

Programming For Problem Solving

Sem 2

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PROGRAMMING FOR PROBLEM SOLVING

Unit 1

Multiple Choice questions MCQ

1. Who is father of C Language? [CLO 1] [PLO1]

A. Bjarne Stroustrup B. Dennis Ritchie

C. James A. Gosling D. Dr. E.F. Codd

2. C Language developed at _____? [CLO 1] [PLO 1]

A. AT & T's Bell Laboratories of USA in 1972

B. AT & T's Bell Laboratories of USA in 1970

C. Sun Microsystems in 1973

D. Cambridge University in 1972

3. For 16-bit compiler allowable range for integer constants is _____ ? [CLO 1] [PLO 2]

A. -3.4e38 to 3.4e38

B. -32767 to 32768

C. -32768 to 32767

D. -32668 to 32667

4. C programs are converted into machine language with the help of [CLO 1] [PLO 1]

A. An Editor

B. A compiler

C. An operating system

D. None of the above

5. A C variable cannot start with [CLO 1] [PLO 1]

A. An alphabet

B. A number

C. A special symbol other than underscore

D. both (b) and (c)

6. Which of the following is allowed in a C Arithmetic Instruction? [CLO 1] [PLO 1]

A. []

B. {}

- C. ()
 - D. None of the above
7. Which of the following shows the correct hierarchy of arithmetic operations in C [CLO 1] [PLO 2]
- A. / + * -
 - B. * - / +
 - C. + - / *
 - D. * / + -
8. Program which is written originally by the programmer is classified as [CLO 1] [PLO 1]
- A. object code
 - B. machine code
 - C. source program
 - D. interactive programs
- [Page 2]
9. Data types are differed on the basis of [CLO 1] [PLO 1]
- A. the way of storage
 - B. the type of operations
 - C. the type of operators used
 - D. both a and b
10. Loop statement which is repeated for some given number of times is classified as [CLO 1]
- A. FOR loop
 - B. GO loop
 - C. REPEAT loop
 - D. GO REPEAT loop
11. Type of statement written in sequence and is repeated until the specific condition met is classified as [CLO 1] [PLO 1]
- A. format
 - B. loop
 - C. case
 - D. condition
12. Size of an array is declared by [CLO 1] [PLO 1]
- A. programmer
 - B. program users
 - C. software
 - D. declared automatically
13. Programming language 'FORTRAN' stands for [CLO 1] [PLO 1]
- A. formula translator
 - B. formula translation
 - C. free translator
 - D. free translation
14. Functions that are used in the programs and are defined by the programmers are called [CLO 1] [PLO 2]
- A. program layout

- B. program procedure
- C. built-in functions
- D. user-defined function

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15. An assembler translates [CLO 1] [PLO 1]

- A. machine code into assembly code
- B. assembly code into machine code
- C. processing time into manual time
- D. routine into subroutine

16. Name given by a programmer to any particular data is classified as [CLO 1] [PLO 1]

- A. identifier
- B. identification
- C. exponent
- D. mantissa

17. When variable used in program is whole number, the variable is stored as [CLO 1] [PLO 2]

- A. fixed string
- B. integers
- C. negative whole numbers
- D. positive whole numbers

18. In programming, programmers use comments to [CLO 1] [PLO 1]

- A. highlight program modules
- B. explain module functions
- C. explain used variables
- D. all of above

19. Variable which uses the same name in whole program and in its all routines thus best classified

as [CLO 1] [PLO 2]

- A. middle variable
- B. default variable
- C. local variable
- D. global variable

20. Statement which is used to make choice between two options and only option is to be performed

is written as [CLO 1] [PLO 2]

- A. if statement
- B. if else statement
- C. then else statement
- D. else one statement

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21. The _____ statement is used to transfer the control to the end of statement block in a loop:

a. Continue [CLO 1] [PLO 2]

b. Break

c. Switch

d. Goto

22. Which of the following is not a valid variable name declaration? [CLO 1] [PLO 2]

a) int _a3;

b) int a_3;

c) int 3_a;

d) int _3a

23. All keywords in C are in [CLO 1] [PLO 2]

a) LowerCase letters

b) UpperCase letters

c) CamelCase letters

d) None of the mentioned

24. Which of the following is true for variable names in C? [CLO 1] [PLO 2]

a) They can contain alphanumeric characters as well as special characters

b) It is not an error to declare a variable to be one of the keywords (like goto, static)

c) Variable names cannot start with a digit

d) Variable can be of any length

25. The format identifier '%i' is also used for _____ data type? [CLO 1] [PLO 2]

a) char

b) int

c) float

d) double

26. Which of the following is a User-defined data type? [CLO 1] [PLO 3]

a) typedef int Boolean;

b) typedef enum {Mon, Tue, Wed, Thu, Fri} Workdays;

c) struct {char name[10], int age};

d) all of the mentioned

27. What is the output of this C code? [CLO 1] [PLO 3]

1. #include <stdio.h>

2. int main()

3. {

4. signed char chr;

5. chr = 128;

6. printf("%d\n", chr);

7. return 0;

8. }

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a) 128

- b) -128
- c) Depends on the compiler
- d) None of the mentioned

28. What is the output of this C code? [CLO 1] PLO 3]

```
1. #include <stdio.h>
2. int main()
3. {
4. j = 10;
5. printf("%d\n", j++);
6. return 0;
7. }
```

- a) 10 b) 11
- c) Compile time error d) 0

Explanation : j is not belongs to any data type

29. The following code 'for(; ;)' represents an infinite loop.

It can be terminated by. [CLO 1] [PLO 3]

- a) break b) exit(0)
- c) abort() d) all of the mentioned

30. The keyword 'break' cannot be simply used within: [CLO 1] [PLO 3]

- a) do-while b) if-else
- c) for d) while

31. Which keyword is used to come out of a loop only for that iteration? [CLO 1] [PLO 3]

- a) break b) continue
- c) return d) none of the mentioned

32. The first step in problem solving is ----- [CLO 1] [PLO 2]

- a) Understand the problem c)Identify the problem
- b) Developing algorithm/Flowchart d)Listing the possible outcome

33. The solutions which has series of actions to solve a problem are called ----- [CLO 1] [PLO 2]

- a) Heuristic solutions
- b) Algorithmic solutions

34. The solution can be reached by completing the actions in steps. These steps are called [CLO 1]

[PLO 2]

- a) Sequence c)Algorithm
- b) Flowchart d)Steps

35. Solutions that cannot be reached through a direct set of steps are called ----- [CLO 1][PLO 2]

- a)Algorithmic solutions b) Heuristic solutions

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36. ----- means the outcome or the completed computer-assisted answer. [CLO 1][PLO 1]

- a) Solution b) Result c) Program

37. The field of computer that deals with the problem of heuristic solution is called-----
[CLO 1]
[PL01]
a) Artificial Intelligence
b) Expert System c) Computer Management
38. ----- means the set of instructions that make up the solution after they have been coded into a programming language. [CLO 1] [PL01]
a) solution b) result c) program d) error
39. ----- are organized facts. [CLO 1] [PL01]
a) Data b) Information
40. A language applied for wide range of application is called as _____language[CLO 1][PLO 2]
a. Special purpose c)General purpose
b. Individual purpose d)Scientific purpose
41. Compiler converts the source code into _____ [CLO 1][PLO 4]
a. C code b. Byte code c. Object code d. Executable code
42. _____ converts source code to machine language one line at a time. [CLO 1][PLO 4]
a. Compiler b. Interpreter c. Assembler d.CPU
43. The program that converts high level language to a machine language is called _____
[CLO 1] [PLO 1]
a. Interpreter b. Linker c. Compiler d. Loader
44. ‘#’ symbol is known to be _____ [PLO 2] [CLO 1]
a. Linker b.Compiler c. Assembler d.Preprocessor directive
45. A _____ is a notational system for describing computations in both machine and human readable form [PLO 1] [CLO 1]
a. Programming language c.Machine language
b. High-level language d.Assembly language
46. ----- program converts assembly language into machine language [PLO 2] [CLO 1]
a. compiler b. interpreter c. assembler d. preprocessor directive
47. which symbol is been used for processing in flowchart? Ans: a [CLO 1] [PLO 3]

a) b) c) d)

48. which symbol is been used for input/output in flowchart? Ans: c [PLO 3] [CLO 1]

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a) b) c) d)

49. which symbol is been used for decision making in flowchart? Ans: d [PLO 3] [CLO 1]

a) b) c) d)

50. which is used as start/stop symbol in flowchart? Ans:b [PLO 3] [CLO 1]

a) b) c) d)

51. ----- is a distinguishing characteristic of human excellence in every area of behavior

[CLO 1] [PLO 1]

a) creativity b) thinking c) visualization d) problem solving

52. .What is the value of x in this C code? [PLO 2] [CLO 1]

```
#include <stdio.h>
```

```
void main()
```

```
{
```

```
int x = 5 * 9 / 3 + 9;
```

```
}
```

(a). 3.75 b) Depends on compiler c) 24 d) 3

53.A character variable can store how many characters at a time? [PLO 2][CLO 1]

(a) 1 character

(b)8 characters

(c)255 character

54.Which of the following is the correct way of writing comments? [PLO 2] [CLO 1]

(a)*/comments/*

(b)/*comment*/

(c)**comment** (

d){comment}

55. C programming language is [PLO 1] [CLO 1]

- (a) object oriented programming language
- (b) Procedure oriented programming language
- (c) function oriented programming language
- (d) None of the above

56. The memory space taken for a char type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes
- (c) 8 bytes
- (d) 1 byte

57. The memory space taken for a int type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes

[Page 8]

- (c) 8 bytes
- (d) 10 bytes

58. The memory space taken for a float type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes
- (c) 8 bytes
- (d) 10 bytes

59. What is the only function all programs must contain ? [PLO 2] [CLO 1]

- (a) start()
- (b) system()
- (c) main()
- (d) program

60. For 16-bit compiler allowable range for integer constants is _____? [PLO 2] [CLO 1]

- (a) -3.4e38 to 3.4e38
- (b) -32767 to 32768
- (c) -32668 to 32667
- (d) -32768 to 32767

61. Every statement in C program is to be terminated by a _____ [PLO 2] [CLO 1]

- (a) dot(.)
- (b) semi-colon(;)
- (c) colon(:)
- (d) Question mark(?)

62. The escape sequence „\b“ is a [PLO 2] [CLO 1]

- (a)back space
- (b)next line
- (c)tab
- (d)none of the above

63.The memory space taken for a long int type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes
- (c) 8 bytes
- (d)10bytes

64. which of the following will not valid expressions in C? [PL02] [CLO 1]

- (a) a=2+(b=5);
- (b) a=11%3
- (c) a=b=c=5
- (d) b+5=2

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65. The ----- operator is true only when both the operands are true. [PLO 2]
[CLO 1]

- a) &&
- b) b) ||
- c) c)!
- d) d) ?:

66. The ----- statement when executed in a switch statement causes
immediate exit
from the structure. [CLO 1] [PLO 2]

- a)goto
- b) default
- c) break
- d) switch

66. The ternary conditional expression using the operator?: could be easily coded using ----

- statement [CLO 1][PLO 2]

- a)Nested if b) if-else
- c) if d) for

4 Marks :

Unit-1 CLO-1

1. Comment "C is mid level language"? [CLO-1][PLO 1]

2. What is problem solving? [CLO-1][PLO 1]
3. What are the six steps of problem solving? [CLO-1][PLO 1]
4. Discuss about how the problems can be solved with computers? [CLO-1][PLO 1]
5. What is a program? [CLO-1][PLO 2]
6. Why is problem analysis important? [CLO-1][PLO 2]
7. What are the tools of problem solving available? [CLO-1][PLO 2]
8. How do problem-solving tools help in leading to a solution? [CLO-1][PLO 2]
9. Why it is important to test a solution before coding it? [CLO-1][PLO 2]
10. What is an algorithm? Give the characteristics of the algorithm. [CLO-1][PLO 2]
11. What is a flowchart? Give the symbols/shapes used in the flowchart.
[CLO-1][PLO 3]
12. Define pseudocode and give its importance with an example. [CLO-1][PLO 2]
13. Discuss the difficulties with problem solving in detail. [CLO-1][PLO 2]

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14. State the use of %d and %f .Write a printf statement in C using the above mentioned symbols? [CLO-1][PLO 4]
15. What is main difference between variable and constant? [CLO-1][PLO 3]
16. Explain bitwise left shift operator? [CLO-1][PLO 4]
17. Explain primary data types used in C? [CLO-1][PLO 3]
18. Difference between formatted & unformatted statement ? [CLO-1][PLO 3]
19. What is mean by storage class of variable? [CLO-1][PLO 3]
20. Explain with example ++i and i++. [CLO-1][PLO 4]
21. Refer all Elab programs. [CLO-1][PLO 3] [PLO 2][PLO 4]

12 Marks:

- 1) Explain the evolution of programming languages. [CLO-1][PLO 1]
- 2) Explain the various steps involved in problem solving with diagram. [CLO-1][PLO 1]
- 3) Draw the flowchart and write the algorithm and c code to find the sum and to reverse the digits of given five digit number. [CLO-1][PLO 1] [PLO2]
- 4) Write an algorithm and draw a flow chart to find the factorial and Fibonacci series of given number. [CLO-1][PLO 2] [PLO 3]
- 5) Write a note on Algorithm, Flow chart and Pseudocode. [CLO-1][PLO 3] [PLO 2]
- 6) Explain the scope, lifetime of variable in C with example. [CLO-1][PLO 2][PLO3]
- 7) Write down the algorithm to find the largest number among three given numbers and outline the steps in the algorithm with the inputs 5, 17, 3. [CLO-1][PLO 2][PLO3]
- 8) Draw flowchart to compute the salary of an employee in a company. Assume that there are two types of employees in the company daily wages and regular. Salary is calculated as number of hours worked* wages per hour for daily wagers and basic pay +(% of DA * basic pay)/100 + HRA + medical allowance for regular employees. Sketch the flow of your design for a regular employee with basic pay = 5000, % of DA = 75 % and HRA = 500. (8)
[CLO-1][PLO 2][PLO3][PLO4]

9) Explain in details about operators with an example [CLO-1][PLO 1] [PLO 2][PLO3]

10) Refer all elab Programs [CLO-1][PLO 3] [PLO 2][PLO 4]

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UNIT - II

PART A

1. The ----- operator is true only when both the operands are true.

e) && b) || c)! d) ?:

Answer: a

2. The ----- statement when executed in a switch statement causes immediate exit

from the structure.

a)goto b) default c) break d) switch

Answer: c

3. The ternary conditional expression using the operator?: could be easily coded using -----

- statement

a)Nested if b) if-else c) if d) for

Answer: b

4. What will be the output when the following segment is executed?

```
Char ch='a';
```

```
Switch(ch)
```

```
{
```

```
case 'a':
```

```
Printf("A");
```

```
case 'b':
```

```
Printf("B");
```

```
Default:
```

```
printf("C");
```

```
}
```

a)A b)B c)C d) a

Answer: a

5. What will be the output of the following segment when executed?

```
int x=10, y=20;
```

```
if((x<y)|| (x+5)>10)
```

```
printf("%d",x);
```

```
else
```

```
printf("%d",y);
```

a)10 b) 20 c) 15 d)5

Answer: a

6. The ----- statement is used to skip a part of the statements in a loop.

a) Continue b) break c) goto d)switch

Answer: a

7. A for loop with no test condition is known as ----- loop

a) Infinite b) time delay c) for d) Incrementing

Answer: a

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8. The sentinel -controlled loop is also known as -----loop

a) Indefinite repetition loop c) time delay

b) Definite repetition loop d) infinite

Answer: a

9. In an exit controlled loop the body of the loop is always executed minimum number of ----

a) 1 time b)2 times c) 3 times d)n times

Answer:a

10. The while is an ----- loop statement.

a) Entry-controlled b)exit-controlled c)indefinite repetition

d)definite repetition

Answer: a

11. The -----specification is used to read or print integers

a)h b)l c)L d)c

Answer: a

12. To print the data left-justified, must use ----- in the field specification

a) - b) + c) / d) *

Answer: a

13. By default, the real numbers are printed with a precision of ----- decimal.

a)6 b)2 c)4 d)0

Answer:a

14. The expression !(x!=y) can be replaced by the expression-----

a) x!=y b) x==y c)x!=y d)!x!=y

Answer:c

15. In a counter controlled loop, variable known as -----is used to count the
loop

operations.

a) Counter b)sentinel c)i d)n

Answer: a

16. Which of the following special symbol allowed in a variable name?

(a) * (asterisk) (b)| (pipeline) (c) - (hyphen) (d) _ (underscore)

17 . Which of the following are invalid variable names?

a) Minimum b) n\$ c) Integer d) float

18. `int a=10;`

`++a;`

`a++;`

`Printf("%d",a);`

a)10 b) 11 c)12 d) 13

19. `int a=11;`

`a=a%2;`

`a=a/2;`

The value of a is,

a) 1, 1 b) 5,1 c) 1,5 d)5

[Page 13]

20. Which of the following is not logical operator?

A. &

B. &&

C. ||

D. !

21 Which of the following cannot be checked in a switch-case statement?

A. Character

B. Integer

C. Float

D. enum

22. What is the output of this C code?

```
int main()
```

```
{
```

```
    int a = 0, i = 0, b;
```

```
    for (i = 0; i < 5; i++)
```

```
    {
```

```
        a++;
```

```
        continue;
```

```
    }
```

```
}
```

A. 2 B. 3

C. 4 D. 5

23. What is the output of this C code?

```

void main()
{
    int i = 0, j = 0;
    for (i = 0; i < 5; i++)
    {
        for (j = 0; j < 4; j++)
        {
            if (i > 1)
                break;
        }
        printf("Hi \n");
    }
}

```

- A. Hi is printed 5 times B. Hi is printed 9 times
 C. Hi is printed 7 times D. Hi is printed 4 times

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24. What is the output of this C code?

```

void main()
{
    int i = 0;
    int j = 0;
    for (i = 0; i < 5; i++)
    {
        for (j = 0; j < 4; j++)
        {
            if (i > 1)
                continue;
            printf("Hi \n");
        }
    }
}

```

- A. Hi is printed 9 times B. Hi is printed 8 times
 C. Hi is printed 7 times D. Hi is printed 6 times

25. What is the output of this C code?

```

void main()
{
    int i = 0;
    for (i = 0; i < 5; i++)

```

```

if (i < 4)
{
printf("Hello");
break;
}
}

```

- A. Hello is printed 5 times B. Hello is printed 4 times
C. Hello D. Hello is printed 3 times

27. What is the output of this C code?

```

int main()
{
int i = 0;
char c = 'a';
while (i < 2){
i++;
switch (c) {
case 'a':
printf("%c ", c);
break;
break;
}
}
}

```

[Page 15]

```

printf("after loop\n");
}

```

- A. a after loop B. a a after loop
C. after loop D. None of the mentioned

28. int main()

```

{
printf("before continue ");
continue;
printf("after continue\n");
}

```

- A. Before continue after continue B. Before continue
C. after continue D. Compile time error

29. What is the output of the code given below?

```

int main()
{
printf("%d ", 1);
goto l1;
printf("%d ", 2);
l1:goto l2;
}

```

```
printf("%d ", 3);
12:printf("%d ", 4);
}
```

- A. 1 4 B. Compilation error
C. 1 2 4 D. 1 3 4

30. What is the output of code given below?

```
int main()
{
printf("%d ", 1);
11:12:
printf("%d ", 2);
printf("%d\n", 3);
}
```

- A. Compilation error B. 1 2 3
C. 1 2 D. 1 3

31. What will happen if in a C program you assign a value to an array element whose subscript

exceeds the size of array?

- A. The element will be set to 0.
B. The compiler would report an error.

[Page 16]

- C. The program may crash if some important data gets overwritten.
D. The array size would appropriately grow.

32 .In C, if you pass an array as an argument to a function, what actually gets passed?

- A. Value of elements in array
B. First element of the array
C. Base address of the array
D. Address of the last element of array

33. Result of a logical or relational expression in C is?

- A. True or False
B. 0 or 1
C. 0 if expression is false and any positive number if expression is true
D. None of the mentioned

34. What will be the value of d in the following program?

```
int main()
{
int a = 10, b = 5, c = 5;
int d;
d = b + c == a;
printf("%d", d);
}
```


}

- A. Syntax error
- B. 1
- C. 5
- D. 10

35. What is the output of this C code?

```
int main()
{
    int a = 10, b = 5, c = 3;
    b != !a;
    c = !!a;
    printf("%d\t%d", b, c);
}
```

[Page 17]

- A. 5 1
- B. 0 3
- C. 5 3
- D. 1 1

36. What is meaning of following declaration ?

```
int arr[20];
```

- a) Array of size 20 that can have integer address
- b) None of the above
- c) Integer array of size 20
- d) Array of size 20

37. In C Programming, If we need to store word "INDIA" then syntax is as below -

- a) `char name[6]={'I','N','D','I','A','\0'}`
- b) `char name[6]={"I","N","D","I","A"}`
- c) `char name[6]={'I','N','D','I','A'}`
- d) `char name[] ; name="INDIA"`

38. what is the way to initialize array?

- a) `int num[6] = {2,4,12,5,45,5};`
- b) `int n{} = {2,4,12,5,45,5};`
- c) `int n{6}={2,4,12};`
- d) `int n(6)={2,4,12,5,45,5};`

39. what will be the output of the program?

- a) 3, 2, 15
- b) 2, 3, 20
- c) 2, 1, 15
- d) 1, 2, 5

Answer: Option A

[Page 18]

Solution:

>> int a[5] = {5, 1, 15, 20, 25}; The variable arr is declared as an integer array with a size of 5 and it is initialized to

a[0] = 5, a[1] = 1, a[2] = 15, a[3] = 20, a[4] = 25.

>> int i, j, m; The variable i, j, m are declared as an integer type.

>> i = ++a[1]; becomes i = ++1; Hence i = 2 and a[1] = 2

>> j = a[1]++; becomes j = 2++; Hence j = 2 and a[1] = 3.

>> m = a[i++]; becomes m = a[2]; Hence m = 15 and i is incremented by 1(i++ means 2++ so i=3)

>> printf("%d, %d, %d", i, j, m); It prints the value of the variables i, j, m

Hence the output of the program is 3, 2, 15.

39. what will be the output of the program?

- a) 5
- b) 6
- c) 9
- d) Error
- e) None of the above

Answer: Option C

Solution:

x[i] is equivalent to *(x + i),

so (buf + 1)[5] is *(buf + 1 + 5), i.e. buf[6].

40) An array elements are always stored in _____memory locations

- a) sequential
- b) Random
- c) Sequential and Random
- d) None of the above

41) Let x be an array. which of the following operations are illegal?

[Page 19]

a) ++x b) x+1 c)x++ d) x*2

options:

- a) I and II
- b) I,II and III
- c) II and III

d) I, III and IV

e) III and IV

Answer: Option D

Solution: `int x[10]; *x` will store the base address of array. `*`

Statement I, III and IV is invalid.

Statement I and III : `++x` and `x++` are throwing an error while compile (lvalue required as increment operand)

Since, `x` is storing in the address of the array which is static value which cannot be change by the operand.

Statement IV : `x*2` is also throw an error while compile (invalid operands to binary `*` (have 'int `*`' and 'int'))

Statement II : `x+1` is throw a warning: assignment makes integer from pointer without a cast [enabled by default]

42) what is the maximum number of dimensions an array in c may have?

a) 2

b) 8

c) 20

d) 50

e) theoretically no limit. the only practical limits are memory size and compilers

43) size of the array need not be specified, when

a) initialized is a part of definition

b) it's a declaration

c) it is a formal parameter

d) All of these

44) what will be the output of the program?

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a) Equal

b) Unequal

c) Error

d) None of these

Answer: Option B

Solution:

Strings are compared using `strcmp()` function defined under `string.h` header file.

45)

Unit-2 PART -B

1. Give the general syntax of conditional operator?
2. Define operator in C. What role an operator plays in C program?
3. Differentiate between relational and logical operators used in C?
4. Give the types of decision making statements
5. What is the general form of switch statement?
6. Differentiate do... while.. and while... loop.
7. Determine the output of the following and justify your answer

```
int main()
{
    int ch; printf("enter a value between 1 to 2:");
    scanf("%d", &ch);
    switch (ch, ch -1){
        case 1: printf("1\n"); break;
        case 2: printf("2"); break;
    } return 0; }
```

8. Explain the limitations of arrays.

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9. Give the steps of looping process.
10. Brief the use of continue, goto, break statement and give the syntax.
11. What are loop control structures? Explain for loop, while loop and do-while loop with their syntax.
12. What do you mean by infinite loop? Give suitable of any infinite loop in a C program.
13. Explain break and continue statements with examples.
14. Determine the output of the following program justify your answer in a few words.

```
int main() {
    float f = 1.0;
    switch(f) {
        case 1.0: printf("one"); break;
        case 2.0: printf("two"); break;
        default: printf("%f", f);
    }
}
```

15. Write a program to print series of number divisible by 3 from 1 to 100 using for loop.
16. Write a C program to print 1, 2, 4, 8, 16, 32, 64N using do...while loop (Read N from user).
17. What are Arrays in C programming? Give the importance of Array in C language
13. What are the rules to declare one dimensional array?

14. What do you mean by compile time initialization? Give suitable example of Compile time initialization of C Array.
15. Describe the array index out of bound error in context of C array program.

PART C

1) Lucy is celebrating her 15th birthday. Her father promised her that he will buy her a new computer on her birthday if she solves the question asked by him. He asks Lucy to find whether the year on which she had born is leap year or not. Help her to solve this puzzle so that she celebrates her

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birthday happily. If her birth year is 2016 and it is a leap year display 2016 is a leap year.? Else

display 2016 is not a leap year and check with other leap year conditions

2) Write a C program to print the multiplication table of an integer n upto m rows using a while loop

sample input and output

INPUT

5

4

OUTPUT

1*5=5

2*5=10

3*5=15

4*5=20

3) Write a program to generate a following @s triangle:

@ @ @ @ @

@ @ @ @

@ @ @

@ @

@

4) Write a program that determines a student's grade. The program will read three scores and determine the grade based on the following rules: score =90% =>grade=A , score >= 70% and <90% => grade=B, score>=50% and <70% =>grade=C, score<50% =>grade=F

5) i)Why switch case is better than else..if... ladder. Justify.

ii)Write a program to read a value (1-7) and print the equivalent day of the week (i.e. 1-SUN, 2-MON.....)

iii) Explain briefly about for loop and Nested for loop with suitable example

iv) Write a program to print the following pattern.

12345

1234
123
12
1

6.a. i) Write a C program to insert an element at a specified position in the array.

7)) Write a program to generate a following patterns

a)

b)

[Page 23]

c)

d)

e)

8) write c Program for Student mark list generation by using control and looping statements

9) write c program for digit rotation. "For any positive integer, we define a digit rotation as either

moving the first digit to the end of the number (left digit rotation), or the last digit to the front of the

number (right digit rotation). For example, the number 12345 could be left digit rotated to 23451, or right

digit rotated to 51234.

10) Write a program to find the sum of positive numbers in an array

11) write c program for palindrome strings , two strings A and B, each consisting of lower case alphabets.

12) Write a arithmetic operation menu driven program using while(1)

13) i) Write short notes on switch case.

ii) Program to create a simple calculator, Performing addition, subtraction, multiplication, division depending the input from user.

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14) i)write a C program to read 10 nos. and reverse it using array

ii)Concatenate two arrays of length minimum 5 numbers.

UNIT III

PART A

[Page 25]

1) Which of the following is not possible statically in C?

- a) Jagged Array
- b) Rectangular Array
- c) Cuboidal Array
- d) Multidimensional Array

2) What is the output of this C code?

```
#include <stdio.h>
void main()
{
int a[2][3] = {1, 2, 3, 4, 5};
int i = 0, j = 0;
for (i = 0; i < 2; i++)
for (j = 0; j < 3; j++)
printf("%d", a[i][j]);
```

- ```
}
```
- a) 1 2 3 4 5 0
  - b) 1 2 3 4 5 junk
  - c) 1 2 3 4 5 5
  - d) Run time error

3) Predict the output of below program:

```
#include <stdio.h>
```

```
int main()
{
 int arr[5];

 // Assume that base address of arr is 2000 and size of integer
 // is 32 bit
 arr++;
 printf("%u", arr);

 return 0;
}
```

- a) 2002
- b) 2004
- c) 2020
- d) lvalue required

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4 ) Predict the output of below program:

```
#include <stdio.h>
```

```
int main()
{
 int arr[5];
 // Assume base address of arr is 2000 and size of integer is 32 bit
 printf("%u %u", arr + 1, &arr + 1);
 return 0;
}
```

- i. 2004 2020
  - ii. 2004 2004
  - iii. 2004 Garbage Value
  - iv. The program fails to compile because address of operator cannot be used with array name
- 5) Size of the array need not be specified, when
- A. Initialization is a part of definition



- B. It is a declaratrion
  - C. It is a formal parameter
  - D. All of these
- 6) While passing an array as an actual argument, the function call must have the array name
- A. with empty brackets
  - B. with its size
  - C. alone
  - D. none of the above
- 7) The parameter passing mechanism for an array is
- A. call by value
  - B. call by value-result
  - C. call by reference
  - D. none of these
- 8) Under which of the following conditions, the size of an one-dimensional array need not be specified?

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- A. when initialization is a part of definition
  - B. when it is a declaration
  - C. when it is a formal parameter and an actual argument
  - D. All of the above
- 9) If a two dimensional array is used as a formal parameter, then
- A. both the subscripts may be left empty
  - B. the first (row) subscript may be left empty
  - C. the first subscript must be left empty
  - D. both the subscripts must be left empty
- 11) Choose the statement that best defines an array
- A. It is a collection of items that share a common name
  - B. It is a collection of items that share a common name and occupy consecutive memory location
  - C. It is a collection of items of the same type and storage class that share a common name and occupy consecutive memory locations
  - D. None of the above
- 12) Choose the correct statements
- A. Array stores data of the same type
  - B. Array can be a part of a structure
  - C. Array of structure is allowed
  - D. All of the above
- 13) What is the output of this C code?
- 1. #include <stdio.h>
  - 2. void main()

```

3. {
4. int a[2][3] = {1, 2, 3, 4, 5};
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
a) 1 2 3 4 5 0
b) 1 2 3 4 5 junk
c) 1 2 3 4 5 5

```

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d) Run time error

Answer: a

14) What is the output of this C code?

```

1. #include <stdio.h>
2. void main()
3. {
4. int a[2][3] = {1, 2, 3, , 4, 5};
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
a) 1 2 3 junk 4 5
b) Compile time error
c) 1 2 3 0 4 5
d) 1 2 3 3 4 5

```

Answer: b

15. What is the output of this C code?

```

1. #include <stdio.h>
2. void f(int a[][3])
3. {
4. a[0][1] = 3;
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
10. void main()
11. {
12. int a[2][3] = {0};
13. f(a);

```

14. }
- a) 0 3 0 0 0 0
  - b) Junk 3 junk junk junk junk
  - c) Compile time error
  - d) All junk values

Answer: a

16. What is the output of this C code?

- 1. #include <stdio.h>
- 2. void f(int a[][])

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- 3. {
  - 4. a[0][1] = 3;
  - 5. int i = 0, j = 0;
  - 6. for (i = 0; i < 2; i++)
  - 7. for (j = 0; j < 3; j++)
  - 8. printf("%d", a[i][j]);
  - 9. }
  - 10. void main()
  - 11. {
  - 12. int a[2][3] = {0};
  - 13. f(a);
  - 14. }
- a) 0 3 0 0 0 0
  - b) Junk 3 junk junk junk junk
  - c) Compile time error
  - d) All junk values

Answer: c

17. What is the output of this C code?

- 1. #include <stdio.h>
- 2. void f(int a[2][3])
- 3. {
- 4. a[0][1] = 3;
- 5. int i = 0, j = 0;
- 6. for (i = 0; i < 2; i++)
- 7. for (j = 0; j < 3; j++)
- 8. printf("%d", a[i][j]);
- 9. }
- 10. void main()
- 11. {
- 12. int a[2][3] = {0};

13. `f(a);`  
14. `}`  
a) `0 3 0 0 0 0`  
b) `Junk 3 junk junk junk junk`  
c) Compile time error  
d) All junk values

Answer: c

18. Comment on the following statement:  
`int (*a)[7];`

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- a) An array "a" of pointers.  
b) A pointer "a" to an array.  
c) A ragged array.  
d) None of the mentioned

Answer: b

19. Comment on the 2 arrays regarding P and Q:

1. `int *a1[8];`  
2. `int *(a3[8]);`  
3. P. Array of pointers  
4. Q. Pointer to an array  
a) a1 is P, a2 is Q  
b) a1 is P, a2 is P  
c) a1 is Q, a2 is P  
d) a1 is Q, a2 is Q

Answer: b

20. Which of the following is not possible statically in C?

- a) Jagged Array  
b) Rectangular Array  
c) Cuboidal Array  
d) Multidimensional Array

Answer: a

21. What is the correct syntax to send a 3-dimensional array as a parameter? (Assuming declaration `int a[5][4][3];`)

- a) `func(a);`

- b) func(&a);
- c) func(\*a);
- d) func(\*\*a);

Answer: a

22. Applications of multidimensional array are?

- a) Matrix-Multiplication
- b) Minimum Spanning Tree
- c) Finding connectivity between nodes

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- d) All of the mentioned

Answer: d

23. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. int ary[2][3];
5. foo(ary);
6. }
7. void foo(int *ary[])
8. {
9. int i = 10, j = 2, k;
10. ary[0] = &i;
11. ary[1] = &j;
12. *ary[0] = 2;
13. for (k = 0; k < 2; k++)
14. printf("%d\n", *ary[k]);
15. }
```

- a) 2 2
- b) Compile time error
- c) Undefined behaviour
- d) 10 2

Answer: a

24. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. int ary[2][3];
5. foo(ary);
6. }
```

```
7. void foo(int (*ary)[3])
8. {
9. int i = 10, j = 2, k;
10. ary[0] = &i;
11. ary[1] = &j;
12. for (k = 0; k < 2; k++)
13. printf("%d\n", *ary[k]);
14. }
a) Compile time error
b) 10 2
c) Undefined behaviour
```

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d) segmentation fault/code crash

Answer: a

25. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. foo(ary);
5. }
6. void foo(int **ary)
7. {
8. int i = 10, k = 10, j = 2;
9. int *ary[2];
10. ary[0] = &i;
11. ary[1] = &j;
12. printf("%d\n", ary[0][1]);
13. }
a) 10
b) 2
c) Compile time error
d) Undefined behaviour
```

Answer: d

26. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. int ary[2][3][4], j = 20;
5. ary[0][0] = &j;
6. printf("%d\n", *ary[0][0]);
```

7. }
- a) Compile time error
  - b) 20
  - c) Address of j
  - d) Undefined behaviour

Answer: a

27. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
```

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```
4. int ary[2][3];
5. ary[][] = {{1, 2, 3}, {4, 5, 6}};
6. printf("%d\n", ary[1][0]);
7. }
```

- a) Compile time error
- b) 4
- c) 1
- d) 2

Answer: a

28. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. void foo();
5. printf("1 ");
6. foo();
7. }
8. void foo()
9. {
10. printf("2 ");
11. }
```

- a) 1 2
- b) Compile time error
- c) 1 2 1 2
- d) Depends on the compiler

Answer: a

29. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. void foo(), f();
5. f();
6. }
7. void foo()
8. {
9. printf("2 ");
10. }
11. void f()
12. {
13. printf("1 ");
```

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```
14. foo();
15. }
```

- a) Compile time error as foo is local to main
- b) 1 2
- c) 2 1
- d) Compile time error due to declaration of functions inside main

Answer: b

30. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. void foo();
5. void f()
6. {
7. foo();
8. }
9. f();
10. }
11. void foo()
12. {
13. printf("2 ");
14. }
a) 2 2
b) 2
c) Compile time error
d) Depends on the compiler
```



Answer: d

Explanation: Even though the answer is 2, this code will compile fine only with gcc. GNU C supports nesting of functions in C as a language extension whereas standard C compiler doesn't.

31. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo();
3. int main()
4. {
5. void foo();
6. foo();
7. return 0;
8. }
9. void foo()
10. {
11. printf("2 ");
12. }
```

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- a) Compile time error
- b) 2
- c) Depends on the compiler
- d) Depends on the standard

Answer: b

32. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo();
3. int main()
4. {
5. void foo(int);
6. foo(1);
7. return 0;
8. }
9. void foo(int i)
10. {
11. printf("2 ");
12. }
```

- a) 2
- b) Compile time error
- c) Depends on the compiler
- d) Depends on the standard

Answer: a

33. What is the output of this C code?

```
1. #include <stdio.h>
```

```
2. void foo();
```

```
3. int main()
```

```
4. {
```

```
5. void foo(int);
```

```
6. foo();
```

```
7. return 0;
```

```
8. }
```

```
9. void foo()
```

```
10. {
```

```
11. printf("2 ");
```

```
12. }
```

a) 2

b) Compile time error

c) Depends on the compiler

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d) Depends on the standard

Answer: b

34. What is the output of this C code?

```
1. include <stdio.h>
```

```
2. void m()
```

```
3. {
```

```
4. printf("hi");
```

```
5. }
```

```
6. void main()
```

```
7. {
```

```
8. m();
```

```
9. }
```

a) hi

b) Run time error

c) Nothing

d) Varies

Answer: a

35. What is the output of this C code?

```
1. #include <stdio.h>
```

```
2. void m();
```

```
3. void n()
```

```
4. {
```

```
5. m();
```

```
6. }
```

```
7. void main()
8. {
9. void m()
10. {
11. printf("hi");
12. }
13. }
a) hi
b) Compile time error
c) Nothing
d) Varies
```

Answer: b

36. What is the return-type of the function sqrt()

a) int  
b) float

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c) double  
d) depends on the data type of the parameter

Answer: c

37. Which of the following function declaration is illegal?

a) double func();  
int main(){  
double func(){  
b) double func(){};  
int main(){  
c) int main()  
{  
double func();  
}  
double func(){//statements}  
d) None of the mentioned

Answer: d

38. What is the output of this code having void return-type function?

```
1. #include <stdio.h>
2. void foo()
3. {
4. return 1;
5. }
```

```
6. void main()
7. {
8. int x = 0;
9. x = foo();
10. printf("%d", x);
11. }
a) 1
b) 0
c) Runtime error
d) Compile time error
```

Answer: d

39. What will be the data type returned for the following function?

```
1. #include <stdio.h>
2. int func()
3. {
4. return (double)(char)5.0;
5. }
```

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a) char  
b) int  
c) double  
d) multiple type-casting in return is illegal

Answer: b

40. What is the problem in the following declarations?

```
int func(int);
double func(int);
int func(float);
```

a) A function with same name cannot have different signatures  
b) A function with same name cannot have different return types  
c) A function with same name cannot have different number of parameters  
d) All of the mentioned

Answer: d

41. The output of the code below is

```
1. #include <stdio.h>
2. void main()
3. {
4. int k = m();
5. printf("%d", k);
```

```
6. }
7. void m()
8. {
9. printf("hello");
10. }
a) hello 5
b) Error
c) Nothing
d) Junk value
```

Answer: a

42. The output of the code below is

```
1. #include <stdio.h>
2. int *m()
3. {
4. int *p = 5;
5. return p;
6. }
```

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```
7. void main()
8. {
9. int *k = m();
10. printf("%d", k);
11. }
a) 5
b) Junk value
c) 0
d) Error
```

Answer: a

43. The output of the code below is

```
1. #include <stdio.h>
2. int *m();
3. void main()
4. {
5. int *k = m();
6. printf("hello ");
7. printf("%d", k[0]);
8. }
9. int *m()
10. {
11. int a[2] = {5, 8};
```

12. return a;
13. }
- a) hello 5 8
- b) hello 5
- c) hello followed by garbage value
- d) Compilation error

Answer: c

## Part B

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1. What is the need for user defined function?
2. Write a multi function program
3. List the Elements of user defined function
4. Give short note on 2-D array processing.
5. Write a C program to find the length of a string.
6. HELLO encode it as IFMMP using array
7. Show the difference between actual and formal parameter in function with piece of code
8. Compare user define function vs System define function
9. List categories of function
10. Explain the concept of function call by reference with a sample program.
11. Explain the concept of function call by value with a sample program
12. Explain about functions with example
13. Discuss Character Arrays
14. Predict output of the following program (CL03)

```
int main(){
 inta[][]={{1,2},{3,4}};
int i,j;
for (i=0;i<2;i++)
for(j=0;j<2;j++)
printf("%d",a[i][j]);
return 0;
}
```
15. Why array index starts from zero?
16. Contrast function declaration vs function definition.
17. List the advantages of functions.

18. Program to Find the 2nd Largest Elements in an Array.
19. Write a program to convert the given string (srm university) Lower to upper case
20. Advantages and limitations of multi dimensional array initialization
21. Explain call by value and call by reference with an example
22. Common programming errors in 2D arrays
12. Explain String functions with example

#### PART C

1. Array construction for student mark list for 100 students. output need to display register number, marks of five subjects , CGPA and PASS/Fail Status
2. Explain in details about String Functions: gets(), puts(), getchar(), putchar(), printf(), with an example programs
3. i)Print the given pattern using 2-D array  
1234567  
23456

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345

4

- ii) Program to divide one array into two arrays
4. i) What are strings in C? Write a C program to read a string in lowercase and convert it to uppercase.
- ii) ) Illustrate call by value and call by reference with example for each.

5. i) Find the factorial of 10 using function recursion.
- ii) What is a string? Explain any 5 string functions.
6. Explain in details about String Functions: atoi, strlen, strcat, strcmp with an example
7. Functions declaration and definition, Types: Call by Value & Call by Reference with example programs
8. Explain and write c program for Function with and without Arguments and no Return Values
9. Explain and write c program for Passing Array to Functions with return type, Recursion Functions
10. Write c program for Matrix Multiplication using Multi-dimensional array
11. Program to Delete duplicate elements from an array

INPUT

1 3 4 5 3

OUTPUT

1 3 4 5

- b. Write a program to perform Matrix addition and Multiplication using 2-D arrays

12. i) State the importance of functions. List out the different types of Function

- ii) Write a swap function using call by value and call by reference
  - iii) Write a C program to concatenate two strings.
  - iv) Write a program to read two strings and compare them and print a message that the first string is equal, less or greater than the second one accordingly.
13. Write c program for Matrix addition and Matrix Transpose using multi dimensional array
14. Given a number , find whether it is a power of 2 or not
15. Write a C program to swap elements in cyclic order using call by reference.

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#### UNIT 4

1. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo(int*);
3. int main()
4. {
5. int i = 10;
6. foo(&i);
7. }
8. void foo(int *p)
9. {
10. printf("%d\n", *p);
11. }
```

a) 10  
b) Some garbage value  
c) Compile time error  
d) Segmentation fault/code crash

Answer: c

2. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo(int*);
3. int main()
```



```
4. {
5. int i = 10, *p = &i;
6. foo(p++);
7. }
8. void foo(int *p)
9. {
10. printf("%d\n", *p);
11. }
a) 10
b) Some garbage value
c) Compile time error
d) Segmentation fault
```

Answer: a

[Page 43]

3. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo(float *);
3. int main()
4. {
5. int i = 10, *p = &i;
6. foo(&i);
7. }
8. void foo(float *p)
9. {
10. printf("%f\n", *p);
11. }
a) 10.000000
b) 0.000000
c) Compile time error
d) Undefined behaviour
```

Answer: b

4. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. int i = 97, *p = &i;
5. foo(&i);
6. printf("%d ", *p);
7. }
8. void foo(int *p)
```

```
9. {
10. int j = 2;
11. p = &j;
12. printf("%d ", *p);
13. }
a) 2 97
b) 2 2
c) Compile time error
d) Segmentation fault/code crash
```

Answer: a

5. What is the output of this C code?

[Page 44]

```
1. #include <stdio.h>
2. int main()
3. {
4. int i = 97, *p = &i;
5. foo(&p);
6. printf("%d ", *p);
7. return 0;
8. }
9. void foo(int **p)
10. {
11. int j = 2;
12. *p = &j;
13. printf("%d ", **p);
14. }
a) 2 2
b) 2 97
c) Undefined behaviour
d) Segmentation fault/code crash
```

Answer: a

6. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. int i = 11;
5. int *p = &i;
6. foo(&p);
7. printf("%d ", *p);
8. }
```

```

9. void foo(int *const *p)
10. {
11. int j = 10;
12. *p = &j;
13. printf("%d ", **p);
14. }
a) Compile time error
b) 10 10
c) Undefined behaviour
d) 10 11

```

[Page 45]

Answer: a

7. What is the output of this C code?

```

1. #include <stdio.h>
2. int main()
3. {
4. int i = 10;
5. int *p = &i;
6. foo(&p);
7. printf("%d ", *p);
8. printf("%d ", *p);
9. }
10. void foo(int **const p)
11. {
12. int j = 11;
13. *p = &j;
14. printf("%d ", **p);
15. }
a) 11 11 11
b) 11 11 Undefined-value
c) Compile time error
d) Segmentation fault/code-crash

```

Answer: b

8. What is the output of the code below?

```

1. #include <stdio.h>
2. int main()
3. {
4. int i = 10;
5. int *const p = &i;
6. foo(&p);

```

```

7. printf("%d\n", *p);
8. }
9. void foo(int **p)
10. {
11. int j = 11;
12. *p = &j;
13. printf("%d\n", **p);
14. }
a) 11 11
b) Undefined behaviour
c) Compile time error

```

[Page 46]

d) Segmentation fault/code-crash

Answer: a

9. Which of the following are correct syntaxes to send an array as a parameter to function:

```

a) func(&array);
b) func(#array);
c) func(*array);
d) func(array[size]);

```

Answer: a.

11. What is the output of this C code?

```

1. #include <stdio.h>
2. void main()
3. {
4. int k = 5;
5. int *p = &k;
6. int **m = &p;
7. printf("%d%d%d\n", k, *p, **m);
8. }
a) 5 5 5
b) 5 5 junk value
c) 5 junk junk
d) Run time error

```

Answer: a

12. What is the output of this C code?

```

1. #include <stdio.h>
2. void main()
3. {

```

```
4. int k = 5;
5. int *p = &k;
6. int **m = &p;
7. printf("%d%d%d\n", k, *p, **p);
8. }
a) 5 5 5
b) 5 5 junk value
c) 5 junk junk
```

[Page 47]

d) Compile time error

Answer: d.

13. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. int k = 5;
5. int *p = &k;
6. int **m = &p;
7. **m = 6;
8. printf("%d\n", k);
9. }
a) 5
b) Compile time error
c) 6
d) Junk
```

Answer: c

14. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. int a[3] = {1, 2, 3};
5. int *p = a;
6. int *r = &p;
7. printf("%d", (**r));
8. }
a) 1
b) Compile time error
c) Address of a
d) Junk value
```

Answer: b

15. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
```

[Page 48]

```
4. int a[3] = {1, 2, 3};
5. int *p = a;
6. int **r = &p;
7. printf("%p %p", *r, a);
8. }
```

a) Different address is printed  
b) 1 2  
c) Same address is printed.  
d) 1 1

View Answer

Answer: c

Explanation: None.

16. How many number of pointer (\*) does C have against a pointer variable declaration?

a) 7  
b) 127  
c) 255  
d) No limits.

Answer: d

17. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. int a = 1, b = 2, c = 3;
5. int *ptr1 = &a, *ptr2 = &b, *ptr3 = &c;
6. int **sptr = &ptr1; //-Ref
7. *sptr = ptr2;
8. }
```

a) ptr1 points to a  
b) ptr1 points to b  
c) sptr points to ptr2  
d) None of the mentioned

Answer: b

18. What is the output of this C code?

```
1. #include <stdio.h>
```

2. void main()

3. {

[Page 49]

4. int a[3] = {1, 2, 3};

5. int \*p = a;

6. int \*\*r = &p;

7. printf("%p %p", \*r, a);

8. }

a) Different address is printed

b) 1 2

c) Same address is printed.

d) 1 1

Answer: c

#### PART B

1) short notes on Passing Array Element to Function with an example

2) Explain Formal and Actual Parameters

3) Give the Advantages of using Functions ,

4) example program for Processor Directives and #define Directives

5) write a program for Pointer Declaration and dereferencing,

6) write a program for pointers, Void Pointers and size of Void Pointers

7. Write algorithm and pseudo code for leap year calculation. [CLO 1]

8. Contrast the declaration of break and continue in while and for loop with justification [CLO 2]

9. Limitations of two dimensional arrays.

#### PART C

1) i) c program to read array elements and print the values with their address

7) Justify and explain the different size of data types using pointer variables with example program.

8) i) write a c program to count vowels and consonants in a string using pointer

ii) Write call by reference and call by value with an example program

ii) what is Pointers and address operator, explain with an example of Size of Pointer Variable and Pointer, Operator

2) Explain in details with an example Pointer Declaration and dereferencing, pointers, Void Pointers and

size of Void Pointers

- 3) write program for Arithmetic Operations, Incrementing Pointers
- 4) what is Constant Pointers, Pointers to array elements and strings
- 5) what is function pointer & Array of Function Pointers with an example
- 6) write the c program for Accessing Array of Function Pointers, Null Pointers

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## UNIT 5

### Part A

1. Which of the following are themselves a collection of different data types?

- a) string
- b) structures
- c) char
- d) all of the mentioned

View Answer

Answer: b

Explanation: None.

2. User-defined data type can be derived by\_\_\_\_\_

- a) struct
- b) enum
- c) typedef
- d) all of the mentioned

Answer: d

3. Which operator connects the structure name to its member name?

- a) -
- b) <-
- c) .
- d) Both <- and .

Answer: c

4. Which of the following cannot be a structure member?

- a) Another structure
- b) Function



c) Array

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d) None of the mentioned

Answer: b

5. Which of the following structure declaration will throw a n error?

a) struct temp{s;

main(){}

b) struct temp{};

struct temp s;

main(){}

c) struct temp s;

struct temp{};

main(){}

d) None of the mentioned

Answer: d

6. What is the output of this C code?

1. #include <stdio.h>

2. struct student

3. {

4. int no;

5. char name[20];

6. }

7. void main()

8. {

9. struct student s;

10. s.no = 8;

11. printf("hello");

12. }

a) Compile time error

b) Nothing

c) hello

d) Varies

Answer: a

7. What is the output of this C code?

1. #include <stdio.h>

2. struct student

3. {

4. int no = 5;

5. char name[20];

```
6. };
7. void main()
```

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```
8. {
9. struct student s;
10. s.no = 8;
11. printf("hello");
12. }
a) Nothing
b) Compile time error
c) hello
d) Varies
```

Answer: b

8. What is the output of this C code?

```
1. #include <stdio.h>
2. struct student
3. {
4. int no;
5. char name[20];
6. };
7. void main()
8. {
9. student s;
10. s.no = 8;
11. printf("hello");
12. }
a) Nothing
b) hello
c) Compile time error
d) Varies
```

Answer: c

9. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. struct student
5. {
6. int no;
7. char name[20];
8. };
```

9. struct student s;

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```
10. s.no = 8;
11. printf("%d", s.no);
12. }
a) Nothing
b) Compile time error
c) Junk
d) 8
```

Answer: d

10. Can the above code be compiled successfully?

```
1. #include <stdio.h>
2. struct p
3. {
4. int k;
5. char c;
6. float f;
7. };
8. int main()
9. {
10. struct p x = {.c = 97, .f = 3, .k = 1};
11. printf("%f\n", x.f);
12. }
a) Yes
b) No
c) Depends on the standard
d) Depends on the platform
```

Answer: c

11. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. struct student
5. {
6. int no;
7. char name[20];
8. };
9. struct student s;
```

[Page 54]

- ```
10. no = 8;
11. printf("%d", no);
12. }
```
- a) Nothing
 - b) Compile time error
 - c) Junk
 - d) 8

Answer: b

12. Number of bytes in memory taken by the below structure is

- ```
1. #include <stdio.h>
2. struct test
3. {
4. int k;
5. char c;
6. };
a) Multiple of integer size
b) integer size+character size
c) Depends on the platform
d) Multiple of word size
```

Answer: a

13. What is the output of this C code?

- ```
1. #include <stdio.h>
2. struct
3. {
4. int k;
5. char c;
6. };
7. int main()
8. {
9. struct p;
10. p.k = 10;
11. printf("%d\n", p.k);
12. }
```
- a) Compile time error
 - b) 10
 - c) Undefined behaviour
 - d) Segmentation fault

[Page 55]

- d) Segmentation fault

14. What is the output of this C code?

```
1. #include <stdio.h>
2. struct
3. {
4. int k;
5. char c;
6. } p;
7. int p = 10;
8. int main()
9. {
10. p.k = 10;
11. printf("%d %d\n", p.k, p);
12. }
```

a) Compile time error
b) 10 10
c) Depends on the standard
d) Depends on the compiler

Answer: a

15. What is the output of this C code?

```
1. #include <stdio.h>
2. struct p
3. {
4. int k;
5. char c;
6. };
7. int p = 10;
8. int main()
9. {
10. struct p x;
11. x.k = 10;
12. printf("%d %d\n", x.k, p);
13. }
```

a) Compile time error
b) 10 10
c) Depends on the standard
d) Depends on the compiler

[Page 56]

Answer: b

16. What is the output of this C code?

```
1. #include <stdio.h>
```

```

2. struct p
3. {
4. int k;
5. char c;
6. float f;
7. };
8. int p = 10;
9. int main()
10. {
11. struct p x = {1, 97};
12. printf("%f %d\n", x.f, p);
13. }

```

a) Compile time error
b) 0.000000 10
c) Somegarbage value 10
d) 0 10

Answer: b

17. What is the output of this C code(according to C99 standard)?

advertisement

```

1. #include <stdio.h>
2. struct p
3. {
4. int k;
5. char c;
6. float f;
7. };
8. int main()
9. {
10. struct p x = {.c = 97, .f = 3, .k = 1};
11. printf("%f\n", x.f);
12. }

```

a) 3.000000
b) Compile time error
c) Undefined behaviour
d) 1.000000

[Page 57]

Answer: a

18. What is the output of this C code(according to C99 standard)?

```

1. #include <stdio.h>
2. struct p

```

```

3. {
4. int k;
5. char c;
6. float f;
7. };
8. int main()
9. {
10. struct p x = {.c = 97, .k = 1, 3};
11. printf("%f \n", x.f);
12. }
a) 3.000000
b) 0.000000
c) Compile time error
d) Undefined behaviour

```

Answer: b

19. What is the output of this C code(according to C99 standard)?

```

1. #include <stdio.h>
2. struct p
3. {
4. int k;
5. char c;
6. float f;
7. };
8. int main()
9. {
10. struct p x = {.c = 97};
11. printf("%f\n", x.f);
12. }
a) 0.000000
b) Somegarbagevalue
c) Compile time error
d) None of the mentioned

```

Answer: a

[Page 58]

PART B

1. Write algorithm and pseudo code for leap year calculation.
2. Contrast the declaration of break and continue in while and for loop with justification
3. Limitations of two dimensional arrays.
4. Explain about nested pre-processor MACRO
5. Categorize the basic operations that can be performed on a file with suitable declarations.
6. Write a c program to demonstrate double pointer for accessing the value of another pointer
7. Explain about file operations and mode with syntax.
- 8) explain shortly about Initializing Structure, Declaring structure, variable,
- 9) give the example of Structure using typedef, Accessing members
- 10) what is Nested structure, Accessing elements in a structure array,
- 11) write c program for Array of structure, Accessing elements in a structure array
12. Explain about file operations and mode with syntax.
13. Give a brief note about preprocessor directives.
14. Differentiate between a union and a structure.
15. Write a program to read a file character by character, and display it simultaneously on the screen
- 16) write a program for Passing Array of structure to function,
- 17) write a program for Array of pointers to structures
18. Explain about nested pre-processor MACRO
19. Categorize the basic operations that can be performed on a file with suitable declarations.
20. Write a c program to demonstrate double pointer for accessing the value of another pointer

PART C

- 1) explain in details about Bit Manipulation to structure and Pointer, to structure, Union Basic and declaration with an example

- 2) explain in details about Accessing Union Members Pointers to Union, Dynamic memory allocation, malloc, realloc, free with an example
- 3) explain in details about Allocating Dynamic Array, Multidimensional array using dynamic memory allocation with an example
- 4) Explain in details about array of structures and accessing elements in a structure array.
- 5) Write a c program to insert a line at the end of text file
- 6) write a program to copy the content from one file to another file
- 7) explain and write the program for library management system using union

[Page 59]

- 14) c program to read array elements and print the values with their address
- 15) Justify and explain the different size of data types using pointer variables with example program.
- 16) write a c program to count vowels and consonants in a string using pointer
- 17) Write call by reference and call by value with an example program
- 18) a) i) Explain in details about array of structures and accessing elements in a structure array.

ii) Write a c program to insert a line at the end of text file
- 19) i) write a program to copy the content from one file to another file
ii) explain and write the program for library management system using union

8) write a c program for file: opening, defining, closing, File Modes, File Types, Writing contents into a

file

9) write a c program for Reading file contents, Appending an existing file

10) Differentiate between gets() and scanf()

(11) Write a program to convert the given string "hello world" to "dlrow olleh".

12. Write a program that passes a pointer to a structure to a function

13 Briefly discuss about file operations in C

32(a) Write a program that passes a pointer to a structure to a function

[CLO 5]

(OR)

32(b) Briefly discuss about file operations in C [CLO 5]

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/CT Papers Compilation
PYQ.url =====

CATEGORY: PYQs

STATUS: url_reference_only

URL: https://drive.google.com/drive/folders/1e00Y2AVToH-FtIpYimh1iE4d0kmW-uUq?usp=drive_link

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/Important Questions.pdf
=====

CATEGORY: PYQs

STATUS: ok

[Page 1]

PPS IMPORTANT TOPIC

Note: Prepare all topics from all the unit, but concentrate more on the following topics

UNIT-1

☐ Compare compiler and interpreter.

☐ What is algorithm? How it is used as a problem-solving tool.

☐ What is pseudo code? How it is used as a problem-solving tool.

☐ Writing algorithms, Pseudo code and Flow Chat for

o Fibonacci Series (OR)

- o Check Number Is Armstrong or Not (OR)
- o Check a Number is Palindrome or Not (OR)
- o Find Factorial of a Number (OR)
- o convert temperature from Fahrenheit to Celsius (OR)
- o convert temperature from Celsius to Fahrenheit (OR)
- o prime number or not (OR)
- o sum of even digits in a number in c (OR)
- o sum of odd digits in a number in c (OR)
- o sum of natural numbers (OR)
- ▣ What is a token? What are different types of tokens available in C language? Explain. (explain the following topic with example Keywords, Identifiers, Constants and Special Characters)
- ▣ Unformatted Input/Output functions.
- ▣ Storage Class (Auto, static, extern, register)
- ▣ Operators (all operators with example)
- ▣ Operator Precedence (They will give some expression, you have to solve expression using operator precedence rules)
- ▣ Write C program to read a radius of a Circle as an input and Calculate its Diameter, Circumference, and Area.
- ▣ Write a C program to swap two numbers without using third variable.
- ▣ Write a C program to enter length and width of a rectangle as an input and calculate its perimeter and its area.
- ▣ Write a C program to swap two numbers using bitwise operator.
- ▣ Read two angle of a triangle as an input and find the third angel. And also read base and height of the triangle as an input from the user and find area of a triangle.
- ▣ Write a C program to convert temperature from Fahrenheit to Celsius.
- ▣ Write a C program to enter basic salary and prepare pay slip using following data. TA = 15% of basic, HRA = 8% of basic, Grade Pay = 9000, and PF = 15% of basic. Calculate Gross salary, PF and Net salary and display them.

UNIT-2

- ▣ Explain in detail about Conditional Control Statements. (Simple if, if...else, else if and nested if, Switch case with example)
- ▣ Explain in detail about Looping Control Statements. (for, While, do.while with example)
- ▣ Write a c programming to
 - o Fibonacci Series (OR)
 - o Check Number Is Armstrong or Not (OR)
 - o Check a Number is Palindrome or Not (OR)
 - o Find Factorial of a Number (OR)
 - o prime number or not (OR)

[Page 2]

- o sum of even digits in a number in c (OR)
- o sum of odd digits in a number in c (OR)

- o sum of natural numbers (OR)
- o Find Factorial of a Number Using Recursion. (OR)
- o They will give some pattern.
- ☐ Write a c program for
 - o Matrix Addition Program
 - o Matrix Subtraction Program
 - o Matrix Multiplication Program
- o Sorting element
- o To Insert An Element Desired or Specific Position In An Array
- o To Delete Element From Array At Desired Or Specific Position
- ☐ Pointer
- ☐ What is dynamic memory allocation? Write and explain the different dynamic memory allocation functions in C

UNIT-3

- ☐ Explain string function with an example. (Prepare all string function with an example).
- ☐ C Program to Count Vowels, Consonants, Digits and Spaces in Given String or Remove all characters in a string except alphabets or Reverse a string using recursion or find index of character.
- ☐ Explain Call by Value and Call by Reference with an example.
- ☐ C function argument and return values. (explain Four types with an example)

UNIT-4

- ☐ Basic Data Types in Python.
- ☐ Explain in detail about Conditional Control Statements in python
- ☐ Explain in detail about Looping Control Statements in python.
- ☐ List function and methods.
- ☐ Tuple function and methods.
- ☐ Set methods.
- ☐ Dictionaries methods.

UNIT-5

- ☐ NumPy ndarray
- ☐ Slicing arrays in NumPy
- ☐ Dealing with Rows and Columns in Pandas.
- ☐ Working with Missing Data
- ☐ Applying Functions to Data frames
- ☐ Comparison between Numpy and Pandas
- ☐ Other Python Libraries

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/PPS CT1 Papers 24-09-2025.pdf =====

CATEGORY: PYQs

STATUS: ok

4
8
9

B) 123 C) "Hello" D) main

Which storage class in C is used to declare variables that are local to a function and retain their values between function calls? A) auto B) register C) static

D) extern

If `int x4,,` result of `x<<I` is: A) 2 B) 8

C) 6

D) 10 Which of the following is a valid single-line comment in C? A) B) /This is a comment

MThis is a comment

This is a comment /l

D) <-This is a comment--> What is the output of the program `kinclude <stdio.h>`

`int main) { int x 2:`

A) One

`switch (x) {`

B) Two

D) Error

A) 0 B) 4

`int i:`

C) 5

`case 1: printf("One"% break:`

C) Three

`case 2: printf("T wo");`

`return 0:`

`break:`

`case 3: printf("Three"): break.`

What will be printed for the code snippet:

`default: printf("Default");`

`for(i -0; ii<5;itt: printf("%d", ):`

D) Garbage value How many times will "C Programming" be printed? `Winclude <stdio.h>`

`int main()`

1

2

2

1

131

1.3.1

13.1

2 2 2 2.1.2

2 1 13.1

22 2 2.1.2

[Page 2]

10

```

int k=2.
do
B)
stop:
printf("C Programmingin"): k-
goto stop: while(k>-1):
if(0
return 0:

```

- A) One time
 Two times C) Three times
 D) Infinite times

Guess the output of the following output: #include <stdio.h> int main) {

```

int a =-5, b =0;
A) True
if(a && b)
printf("True");
else

```

- B) False
 return 0;
 printf("False");
 C) Compilation error D) No output
 2 2 1 L3.1

[Page 3]

OSRM

Test: CT I

Course Code & Title: 21CSSIOLI Programming for Problem Solving Year & Sem: 1 Year /I Sem

11

Answer any three questions out of four:

13

14

SRM Institute of Science and Technology College of Engineering and Technology School of Computing SRM Nagar, Kattankulathur 603203. Chengalpattu District. Tamilnadu Academic Year: 2025-2026 (ODD)

12 A robotics project requires checking whether a number is odd or even

Write a program to illustrate the difference between logical and bitwise operators in C.

Conditions:

using only bitwise operators. Write a C program for the same.

Part-B

(3x5 Marks)

An amusement park has three types of rides based on height. Manya wants to know if she can go on the Adventure Ride. Her height is 140 cm. Write a C program to decide her eligibility.

"Eligible for Giant Ride, if height > 160 cm "Eligible for Adventure Ride, if height > 120 cm and < 160 cm Eligible for Kids Ride, if height < 120 cm

COL

developer is assigned the task of implementing a key feature for a mobile calculator application. The application should provide users with the ability to select an arithmetic operation-such as addition, subtraction, multiplication, or division and perform that operation on two user-input numbers. The developer is required to write a program in C that utilizes a switch statement to construct this simple calculator functionality, enabling users to choose their desired mathematical operation.

CO Distribution

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions

CO2 CO) CO4 CO5

Bloom's level Distribution

5

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Date: 24.09, 2025

Duration i Period Max. Marks: 25

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1.3.1

13.1

13.1

[Page 4]

Course Articulation Matrix

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3

4

Course

Outcome

CO1

CO2

C03

C04

C05

Instructions:

arrays.

POI

Course Outcomes (CO):

A) 2

2

Follow proper dress code. ID card is mandatory

B) 3

2

MCQ sheet to be handed over to the faculty after the first 15 minutes.

MOBILE PHONES/SMART GADGETS STRICTLY NOT PERMITTED.

DO NOT WRITE or MARK anything in the Question Paper (Except Registration Number)

Discussion with your classmates is considered as MALPRACTICE, and it results in 0° Mark.

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Question Paper should be submitted along with Answer Booklet.

C) 1

2

CO1: Solve problems through computer programming. Express the basic data types and variables in C

CO2: Use appropriate data types in simple data processing applications. To create programs using the concept of

D) 0

2

int a2, b3:

2

A) main

B) define

printf("%d", a && b):

C) printf

P02

D) double

3

Predict the output of the following C code snippet:

P03 P04 P05 P06 P07

Which of the following is a keyword in C?

Which is valid C expression?

A) int mynum 100,000:

Question

Part -A

(1x 10 -10 Marks)

Multiple Choice Questions

P05 P06 P010 P011 P012

Marks BL

2

2

CO PO PI

Code

13.1

2.1.2

1.3.1

[Page 5]

4

6

7

9

10

B) int ny num-I00000 C) int ny num 1000,

D) int Smy num

What will be printed after executing the below code sippet

A) 7 B) 6

C) Error

D) Undefined

What will be the output of the folowing C code?

```
int x7: printf("%d",x-.
```

```
Hinclud <stdio.h>
```

```
int main)
```

```
1
```

```
10000:
```

A) Itwill cause a compile-time error B) Itwill cause a run-time error

```
int main 330;
```

```
printf("d", main).
```

```
return 0:
```

C) It will run without any error and prints 3 D) It will experience infinite looping Which of the following is NOT true about pseudocode?

E) It uses natural language mixed with programming constructs F) Itis not executed on

computers G) It helps in designing algorithms H) It requires strict syntax rules like C

The following loop exccutes exactly 5 times: forint i-0; ic5; i++)|/body 71 A) Truc, because it runs from 0 to 4.

B) False, because it actually executes 6 times (0 to 5). C) True, because condition allows only 5 iterations.

D) False, because loop never runs.

C) exit

Fill in the blank so the while loop prints numbers from 0 to 4.

```
int i 0; while )t printf"%d ". i):
```

B) break

keyword is used to exit from a loop immediately? A) continue

D) return Pick the correct statement about the continue statement. A) It skips the remaining statements in the loop and moves to the next iteration B) It terminates the loop immediately C) It is used to exit from a switch-case

D) It ends the program execution

1 2

2

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13.1

1.3.1

2 2.1.2

1 1.3.1

2.1.2

1.3.1

[Page 6]

Test: CTI

12

SRM

Course Code & Title: 21CSSIOLJ Programming for Problem Solving Year & Sem: I Year / L SSem

13

Answer any three questions out of four:

14

SRM Institute of Science and Technology College of Engineering and Technology School of Computing SRM Nagar, Kattankolattur 603203. Chengalpattu District Tamil Nadu Academic Year 2025-2026 (ODD)

In C language, illustrate how storage classes handle the variables, in terms of scope and visibility. along with program code.

COL

Write a pseudocode to convert Celsius to Fahrenheit temperature

Prompt the user to enter the temperature (in Celsius). Explain how pseudocode makes it easy to translate the logic into a C program.

An HR manager wants to automate the employee performance evaluation process in C. The company follows this grading policy: Score >90 Outstanding

Score 75-89 Excellent Score 60-74 Good Score 40-59 Needs Improvement Score < 40 Poor Help the HR manager, by writing a C program using an else-if ladder that accepts an employee's score as input and displays their performance rating.

C0 Distribution

C02

Part-B

(3x5-15 Marks)

Ravi is building a small banking application in C. As part of his project, he needs to implement a feature where the digits of a customer's account number must be reversed for encryption before

storage (Eg: 1234 ' 4321). Ravi decides to use a while loop to solve this. Help Ravi to solve this, using a C program.

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions

COS CO COS

Bloom's level Distribution

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Dates 24.09.2025 Duration 1 Period Max. Marks: 25

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[Page 7]

Tet: CT

Course Articulation Matrix

SRM

Course Code & Title: 21CSSIoL Programming for Problem Solving Year & Sem: Year/ Sem

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C02 C03

Instructions:

COI

COS

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Course COuteomes (CO:

SRM Naga, Kattankulathur 603203, Chenpalpatu Diarict. Tamilnadu

Academic Vear: 202s-2026 (ODD)

DO NOT WRITEor MARK anything in the Question Paper (Except Registration Number)

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arrays.

P02 P03 P04 POS P06 P07 POS

. Question Paper should be submitted along with Answer Booklet.

3

" Discussion with your classmates is considercd as MALPRACTICE, and it results in '0' Mark "

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A) 7 B) 9

co: Solve problems through computer programming. Express the basic data types and variables in C. c0Z: Use appropriate data types in simple data processing aplications. To create programs using the concept of

C) 10 D) 8

int e\$ + 8/4*3-2; Evaluate and find the valuc of e in the expression:

int main()

What will be the output?

int a6;

#include<stdio.he

return 0;

int b a--2;

printf("6d", b):

P09

Part-A

(1x 10-10 Marks) Multiple Choice Questions

Question

Date: 2409.2024

DurationI Period Mas. Marks 25

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SET G

Marks BL CO PO

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1 2

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Code

1.3.1

2.1.2

[Page 8]

4

5

6

B) 3

C) 2 D) -1 Guess the output of the below code:

```
#include<stdio.h> int main() double x=78.999; printf("%2f. x"); return 0;.
```

A) 78.99

B) 79.00

C) 78.90

D) Error

Predict the output of the following code snippet `int p=1; printf("%d %d", p, &p);` A) 02

B) 12 C) 11

D) 01

The sizeof() operator in C returns: A) Number of digits in a value B) Size of memory in bits C) Size of memory in bytes

D) Address of variable

The keyword used to define a constant in C is

A) const

B) #define

C) Both A and B

D) Neither A nor B

What will be the output of the following code snippet?

```
int x=10.
```

```
if(x<5)
```

```
printf("Hello").
```

```
else
```

```
printf("World");
```

A) Hello

B) World

C) HelloWorld D) No output

How many times, value of "i" is checked in the following program

```
#include <stdio.h>
```

```
int main()
```

```
{ int i=0; while (i<3)
```

```
{ printf ("In while loop\n");
```

```
1
```

```
1
```

```
1
```

```
2
```

```
2
```

```
2
```

```
2
```

```
1
```

```
1
```

```
1
```

```
2
```

2
1 13.1
13.1
1.3.1
1.3.1
2 2.12
1 13.1

[Page 9]

Answer any three questions out of four:

11
12
13

Develop an algorithm, pseudocode and flowchart for finding the factorial of the given number.

Write a C program that Swaps two variables without using temporary variables.

Fill the code given and explain how the desired output will be generated.

```
#include <stdio.h>
```

```
int main(){ int num, sum 0: printf("Enter a number: "):
```

```
scanf( / Loop to find divisors for(int i 1;i<
```

```
if
```

```
ifC
```

Part-B

(3x5=15 Marks)

.&num):

Sum m

```
else printf("%d is a Perfect Number", num):
```

```
printf("%d is not a Perfect Number", num):
```

```
return 0:
```

5
5
2
3
3
1
2
13.1
13.1
13.1

[Page 10]

A) 2
B) 1 C) 3
D) 4

Guess the output for following code snippet: int x=3, switch (x) case 1: printf("A"); case 2: printf("B"); case 3: printf("C"); default: printf("D");

A) CD B) C C) ABC D) BCD Write the output for the following code snippet? for (int i = 1; i <= 5; i++) if(i%3 != 0) continue; printf("d. i: ");

A) 12345

B) 12 C) 1245

D) 345

1 2

2

2

2

2 2.1.2

13.1

[Page 11]

Test: CTI

Course Articulation Matrix:

Course Code & Title: 21CS501J -Programming for Problem Solving Year & Sem: I Year /I Sem

S.No Course PO1

Outcome

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Instructions:

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Academic Year: 2025-2026 (ODD)

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PO2

3

SRM Institute of Science and Technology

College of Engineering and Technology School of Computing

of arrays.

Course Outcomes (CO:

. Follow proper dress code. ID card is mandatory

A) static

P03 P04 P05 P06 P07 P08

DO NOT WRITE or MARK anything in the Question Paper (Except Registration Number)

" Question Paper should be submitted along with Answer Booklet.

B) auto

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C) extern

CO1: Solve problems through computer programming. Express the basic data types and variables in C. CO2: Use appropriate data types in simple data processing applications. To create programs using the concept

D) register

What is the default storage class of local variables in C?

```
int shifty;
```

```
#include<stdio.h>
```

```
int main()
```

Question

P09

What is the output of the following C code?

Part -A

(1x 10-10 Marks) Multiple Choice Questions

```
shifty = 0570; shifty = shifty >> 4;
```

Date: 24.09.2025 Duration: I Period

Max. Marks 25

P010 P011 P012

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Marks BL

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CO PO P

Code

13.1

2.1.2

[Page 12]

8

7

4

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3

)

E) The value of shifty is 1500 F) The value of shifty is 4300 G) The value of shifty is 5700 H) The value of shifty is 2700 The following three C language statements is equivalent to which single statement? yy+k

A) 2x y+2: B) 2xt)+(44y): C) z(xt+) +(yt+):

C)

```
shifty shifty <<6;
```

D) $z(xt)+(ty) + 1$: Which of the following identifiers is invalid?

```
printt("The value of shifty is %o, shifty);
```

B) id

A) student 1

C) 9marks D) firstName Which statement is correct about string in C? A) It is a keyword B) it is stored as a character array ending with `\0` C) It can store numbers only

Which of the following produces compile-time error?

D) It has dynamic memory allocation by default

A) 45

B) x+

+x

D) xt-1

What is the output of the following C code? `#include<stdio.h>`

```
int main()
```

```
int k;
```

A) 4 16

B) 4 10 16

```
for(k=0; k< 20; k=k+ 2){ if(k %3 == 1) printf("%d", k);
```

```
printf("%d\n", k);
```

D) | 4 7 10 13 16 19

With respect to following "for" loops in C, pick the best statement

Assume that there is a prior declaration of `i` in all cases

```
for (i = 0; i < 10; i++) { }
```

```
for (i = 0; i < 10; i++) { }
```

1

1

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2

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1

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1

1 1.3.1

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131

2 2.1.2

13.1

[Page 13]

10

A) All the above for" loops would compile successfully Except (iii), the behaviour of all the other "for" loops depend on compiler implementation.

for(=0; i<10;i++) /I/ (H) for (i = 0; t++;i<10) // (iv) for (it++; i =0;i<10)// () for (it*;i <0;i=10) // (v)

B) Only (iii) would compile successfully.

C) Only (ii) and (iv) would compile successfully but behaviour of (iv) would depend on compiler implementation.

D) All the above for" loops would compile successfully.

What is the output of the following C code?

A)

B)

```
#include<stdio.h>
```

```
int main(0
```

```
int index;
```

```
for(index= 1; index <=5; index+)(
```

```
1245
```

```
12345
```

C) 12245

D) 12354

```
printf("%d", index); if (index 3)
```

```
continue;
```

Which of the following statement is correct for switch controlling expression?

A) "switch" control expression can be empty as well.

B) Only int can be used in "switch" control expression.

C) All types i.e. int, char and float can be used in "switch" control expression.

D) Both int and char can be used in "switch control expression.

1

2

2 2

2

2 2.1.2

13.1

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/PPS CT1 Papers ALL SETS 2025.pdf =====

CATEGORY: PYQs

STATUS: ok

[Page 1]

Test: CTI

Course Articulation Matui:

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Course Code & Title: 2CSS0 Prooammno for Problem Solving

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Course Outcomes (CO):

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B) % C)+

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Question Paper should be submitted along with Answer Booklet.

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CO1: Solve problems through computer prouramming. Express the basic data types and variables in C.

CO2:Use appproprhate data types in sinple data processing applications. To create programs using the concept of arrays.

Which operator has the lowest priority?

SRM Ins

r0)

A) 'A'

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Academic Year: 2025-2026 (ODD)

AY int main() { return 0; D) int main() { }

Y main

Which of the following is the correect structure of a C program?

B) main() { return 0;r

B) 123 C) "Hello"

C) void main() { return 0; Which of the following is not avalid constant in C?

A) auto B) register g static D) extern 2

Part -A (1x 10 = 10 Marks) Question

B) 8 C) 6 D) 10

Which storage class in C is used to declare variables that are local to a function and retain their values between function calls?

If `int x=4;`, result of `x<<1` is:

P0 P06 P08

A), `/* This is a comment */` Which of the following is a valid single-line comment in C? BX

I/ This is a comment C) `#This is a comment` D) `<--This is a comment` -

P09

Multiple Choice Questions

Date: 24.09.2025 Duration: I Period Max. Marks: 25

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PI Code 1.3.1

2.1.2

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1.3.1

[Page 2]

10

One

D) Error 0 Three

B) 4

1What will be printed for the code snippet:

D)

`default printfDefault ).`

`for(i=0;i<$;i++){ printf("%d", i); }` D) Garbage value How many times will "C Programming" be

```

printed?
break break
#include <stdio.h> do {
break
int main() {
stop:
int k = 2; k--; printf("C Programmingn"):
A), One time
if(k == 0) goto stop; } while(k>-1);
Two times return 0;
Y True B) False
C) Three times Infinite times Guess the output of the following output: #include <stdio.h>
int main(){
else int a = 5, b=0, if(a && b) printf{"True\n"}: printf{"False\n"}: return 0;
C) Compilation error D) No output
1 2
2
2
1 L3.1
2 2.1.2
1 1.3.1

```

[Page 3]

CSRM

Test: CTI course Code & Title: 21CSST01J -Programming for Problem Solving Year & Sem: 1 Year / I Sem

11 Answer any three questions out of four:

12 13

14

20 15 10

SRM Institute of Science and Technology College of Engineering and Technology School of Computing SRM Nagar, Kattankulathur-603203. Chengalpattu District, Tamilnadu Academic Year: 2025-2026 (ODD)

Conditions:

Write a program to illustrate the difference between logical and bitwise operators in C. A robotics project requires checking whether a number is odd or even using only bitwise operators. Write a C program for the same. An amusement park has three types of rides based on height. Manya wants to know if she can go on the Adventure Ride. Her height is 140 cm. Write a C program to decide her eligibility.

Part -B (3 x5= 15 Marks)

C01

" Eligible for Giant Ride, if height >= 160 cm " Eligible for Adventure Ride. if height >= 120 cm and < 160 cm

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions

" Eligible for Kids Ride, if height < 120 cm

CO Distribution

A developer is assigned the task of implementing a key feature for a mobile calculator application. The application should provide users with the ability to select an arithmetic operation-such as addition, subtraction, multiplication, or division-and perform that operation on two user-input numbers. The developer is required to write a program in C that utilizes a switch statement to construct this simple calculator functionality, enabling users to choose their desired mathematical operation.

C02 C03 C04 C05

Bloom's Level Distribution

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Date: 24.09.2025 Duration: 1 Period Max. Marks: 25

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13.1

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/PYQ Jan 2023.pdf =====

CATEGORY: PYQs

STATUS: ok

[Page 1]

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B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, JANUARY 2023 First Semester

21 CSS101J-PROGRAMMING FOR PROBLEM SOLVING (For the candidates admitted from the academic year 2022-2023) Note:) Part A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed

over to hall invigilator at the end of 40th minute. Part-B and Part -C should be answered in answer booklet. (ii) Max. Marks: 75 Time: 3 Hours Marks BL CO PO PART A (20 x 1 = 20Marks) Answer ALL Questions 1. Find the output for the following statement. printf ("%d", sizeof ("Hello ")); (B) 6 (D) 2 (A) 5 C) 1 2 1 1 2. What is the output for the following expression in C? n =5-2* 7-9: (A) -18 (C) 9 (B) 12 (D)1 2 1 1 3. Determine the correct output. float c = 3.14; printf ("f, c%2); (A) 1.0 C) Compile error (B) 0 (D) 0.14 2 2 4. Find the meaning of following statement: f=(xv)? x :y; (A) fpoints to max of x & y (C) fpoints to addition of x & y (B) fpoints to min of x & y (D) Compile error I 21 5. The 'continue' statement cannot be used with (B) do while (D) switch case (A) while (C) for 12 I 6. Find the correct one to access the 3 element in an array. (A) *a+3 (C)*(a + 3) (B)(a +3) (D) &(a +3) 22 2 7. Find the output for the following C code. for (K 1; K< = 5; K+ +);

printf ("od", K); (A) 1 2 3 45 (C) (B) 1 23 4 (D) 5 6 Page 1 of 4 06JF1/21CSS101J

[Page 2]

5. Size of integer pointer and character pointer in C for 16 bit processor is (A) 2 (C) 4 (B) 2 (D) 4 9. Expected output for the following code is 3 2 2 char str [] = {'g'. T, 'o', b' 10, 'a', T}; printf ("as", str); (A) global (C) globloal (B) glob (D) al T0. The operator used to get value at an address stored in a pointer variable is (A) (C) && 2 3 (B) & (D) 11. The keyword used to transfer the control from a function back to the calling function in C is (A) goto (C) retur
4
(B) back (D) goback
4 12. If both the strings are same, a string function strcmp() returns (A) 1 (C) 2 (B)-1 (D) 13. Keyword used for function in Python language is (A) function (C) fun (B) def (D) define 2 2 14. The following python code will print print (ype (5/2)) print (ype 5//2)) (A) float int float (B) float float (D) int int (C) int 2 2 15. Find the length of the following list in python len (["hello", 2, 4. 6]) (A) 8 (C) 4 (B) 6 (D) 3 1 5 16. Representation of python tuple is (A){} (C) [1,2, 3] (B) {1,2, 3} (D) (1,2, 3) 1 5 17. Number of elements present in the series S1 is SI = pd.series (range (1, 6)) print(S1) (A) 4 (C) 6 (B) (D) 7 18. Numpy is often used along with packages like (A) Node.js (C) SciPy (B) Matplotlib (D) Both (B) and (C) 06JF 1/21CSs101J Page 2 of 4

[Page 3]

19. Size attribute in Numpy is used to find (A) shape (C) objects (B) date and time (D) number of items 20. Library used to create data frame is (A) Sklearn (C) Pandas (B) Data frame (D) Random
Marks L c0 PO PART B (5 x 8 = 40 Marks) Answer ALL Questions 2 21. a. Describe various storage classes in C programming with suitable example. (OR) 3 2 post-increment, pre-decrement, about pre-increment, b. Describe post-decrement, conditional operators, each with suitable example. 2 22. a.i. Write a C program to find the factorial of a given number. 2 2 4 ii. Write the necessary conditions for switch-case in C. 2 (OR) 2 4+4 b. Explain about) (ii) void pointer with an example in C NULL pointer with an exanple in C
8 4 3 2 23. a. Write a C program to find whether the given string is palindrome or not. Explain the string functions used in this program. (OR) 2 4 4 b. Explain the concept of passing the array as an argument to function. Illustrate with C program to find the sum two matrices.
5 2 24. a. Describe any four operations performed on tuples in Python programming. (OR) 5 2 b. Write a Python program to print the following Floyd triangle. 2 3 4 56
4 4 5 2 25. a.i. Explain the significant features of the pandas library
A 5 2 ii. Write a Python program to create a Numpy array filled with all zeros. (OR) A b.i. List out various features of Numpy. 1. Describe about the different types of data structures in pandas.

[Page 4]

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eGrat

CO Marks 91.

PART C (1 x 15 =15 Marks) Answer ANY ONE Question 32 2 26.i. The National Highways

Department announces a scheme for four wheeler riders in the toll gate for its 50th year celebration. If the vehicle number is divisible by both 7 and 3, then they should pay one-third of the fee. If the vehicle number is divisible by 7 or 3, then they should pay half of the fee. Otherwise they have to pay full fees. Write a C program for the given Scenario. 7 3

32 ii. Write a C program to replace all vowels with star (*) and print the output 8 4 2 27.i.

Write a Python program to check whether the given list is in sorted order or

not. If the given list is in sorted order then print true, else print false.

4 5 2 i. Write a python program to create two sets Set1 and Set2 by getting input from user and determine whether the given sets Set1 and Set2 are disjoint or not.

*** *

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/PYQ Jun 2023.pdf =====

CATEGORY: PYQs

STATUS: ok

[Page 1]

Note: (i) Time: 3 Hours

BTech/ M.Tech (Integrated) DGREE ENAMINATION, UNE 2023

Part -A should be answered in OMR sheet within 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40 minutes.

(A) 3 (C) 6

Z1CSST01| -PROGRAMMING FOR POLYMORPHISM (For the candidates admitted from the college)

Part -B And Part C should be answered in answer booklet

Page 1 of 4

Reg. No

programming?

1. The expression $4 + 6/3 * 2 + 7 \% 3$ evaluates to

2. Which of the following is appropriate position for writing comments in C (A) Anywhere in the code (C) Before header file section

(A) && (C) |

A

PART A (20 x 1 = 20 Marks)

(A) char > int > long

'float' & 'double' are not

(C) char < int < double

Answer ALL Questions

3. Which is correct with respect to the size of the data types?

element array[]

(A) Character

(A) (((a + I) + /) + k) + 1) (C) ((a+)+) +k+)

(C) Float

4. Which of the following is not logical oprntor?

include <stdio.h> int main ()

(B) 4 (D) 7

int a switch (a)

(B) Before function (D) Beford NCction

(B) int > char > loat (D) double (B) &R (D) ! 5. What would be the equivalent point expression for referring the array

calling

7. What is the output of the following program?

Variable

6. Which of the following cannot be checked in a switch-case statement? (B) Integer (D) Enum

char> int

the main

(B) (*((*a +) +) +k) +1) (D) (a+i)+j+k*)

Mux, Marks; 75

declaration

[Page 2]

(B) (1, 3, 2, 1,3]

(D) [1,3. 2, 3, 2, 1]

01JEISI.1CSU

(A) (2, 6,4] (C) [1,3,2, 1,3, 2)

15. Suppose list1 is [1,3,2),

lists are mutable (C) Tuples are immutable

while (D) are immutables are mutable while tests

mutable (A)

immutable

14. Choose the correct option with respect to

are

(0) -py (A) -python . Wich of the following

is the correct extension

of the python ie:

(D) p (B) pl

(C) String size is not mentioned

characters (D) String cannot contain special

(A) Character array, array is a string

end (B) array has no null character

at the

char array []="Hello...!"; 12. Choose a correct statement

about

C string.

1 1 3 1

(C) 25 25

(D) Compiler error

(A) 25 1234

(B) 25 0

```
return 0; printf ("%d %d", b, p);
```

```
int *p =b; int b = 25; |// b memory
```

{ #include <stdio.h> 11. What is the output of C program?

1 1 3 I

(C) Recursive function (A) Selffunction 10. A function which calls itself called a (D) Static function (B) Auto function

131

(C) Go back (A) Switch

(D) Return

function is 9. The keyword used to transfer control from a function

(B) Goto

1 131

(C) for (A) do while The keyword 'break cannot be simply used within (D) while (B) if else

(C) No output (A) ABCDEFGH

(D) Compiler (B) EFGH

error

```
printf ("EFGH"); printf ("ABCD");
```

what is list1 * 2? Tuples

Both

tuples and

tuples and lists are (B)

Both lists to python?

location = 1234;

back to the calling

[Page 3]

25. a. Write short notes on numpy array attributes. (OR)

b.

(OR)

4

8

24. a. Write the various data types in python with example.

3

23. a. Differentiate

getchar (),

and b.

8 2 3 I

b. Write

8 I 2 1

(OR)

biggest number among the

b. Write a suitable

C program to explain the pre and post increment operator.

8 2 1 1

(OR)

example. 21. a. List down various storage classes in C. Explain about storage class with

Answer ALL Questions PART-B

(5 x 8= 40 Marks)

Marks BI. CO PO

(C) dtypes (A) Dtypes

(D) datatypes (B) DTypes

of each column

in data frame?

20. Which of the following attribute of data frame is used to display data type

(C) Pandas

(D) Random

(A) Python

(B) Data frame

19. Which library is to be imported for creating data frame?

(C) Numerical programme

(D) Nonlinear python

(A) Numerical python

(B) Natural python

18. What does Numpy stand for?

(C) [5, 6, 7]

(D) (2,4, 6]

1

(A) [2, 3, 4)

(B) [0, 1, 2]

5 1

print (a [2:5]) a = np. a range (10)

import numpy as np

17. What is the output of the following code?

(C) !=

(D) <=

1

(A) & 16. Which of these is not a bit wise operator? (B) &=

1

Write the input and output functions in python with example.

putchar () with a suitable C programming

(OR)

with suitable a example: call

by value Explain

call by reference.

a C program to compute the sum of the first 10 natural numbers.

three using if condition.

22. a. Write a C program to get 3 numbers
as input from the user and print the 8 1 2 1

[Page 4]

flow chart. multiply the last digit and second last digit. Also write the algorithm
and

27. Develop

a C program

to read a three digit

15 2 2 2

draw the flow chart and algorithm.

10% of basic salary. Write a C program

to calculate

his gross salary. Also

35% of basic salary, house rent allowance

is 27% of basic

Answer

ANY ONE Question

PART -C(1x15

=15 Marks)

Marks

BL Co P0

b. Explain with an example numpy indexing.

2 5 1

number

from the user and

salary, and TA is

26. Kannan's

basic salary is given as input through

the keyboard.

His D.A is 15 2 1 2

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/PYQ Nov 2018.pdf =====

CATEGORY: PYQs

STATUS: ok

[Page 1]

31. a.i. List the advantages of uniting function in C programming.

ii. Define preprocessor and its directives with suitable examples.

b.i. Define pointer.

(or)

ii. Write a program to explain the usage of pointer.

32. a.i. Explain dynamic memory allocation in detail.

ii. Explain Structure within Structure.

(Marks)

(8 Marks)

(8 Marks)

(a Marks)

Note i

(i)

(i0

Time: Three Hours

B.Tech. DEGREE EXAMINATION, NOVEMBER 2018

First Semester

I 8CSS I O l J _ PROGRAMMING FOR PROBLEM SOLVING(For the candidates admitted during the academic year zoia-r-"*ara]

Part - A should be answered in oMR sheet within first 45 minutes and oMR sheet should be handedover to hall invigilator at the end of 45fi minute.

Ihrt - B and Part - C should be answered in answer booklet.

b.i.

u.

write aprogram to urite ,Hello worrd,toot}-o, ur".

Discuss all file opening modes in c programming language with respect to file handring.

PART-A(20x1=20Marks)

AnswerALL euestions

Max. Marks: 100

+**,t*

1. Who is the fatherofC langu4ge?

(A) Bjame Stoustrup

(C) Dennis Ritchie @) James A.Gosling

@) Dr.E.F.Codd

which of the following type of operators have the highest

(B) Equality operators

@) Arithmetic opemtors

@) floats and doubles

(D) ints and chars

20M:1/ltcsslou

2. In C programming languagq

precedeirce.

(A) Relational operators

(C) Logical operato,rs

3. Bitwise operators can operate upon?

(A) Doubte and chars

(C) ints and floats

4. What type of value does ,.size of'retum?

(B)

(D)

5. Relational operators cannot be used on
 (A) Structure (B) String (C) String (D) String

6. What is the type of the below assignment expression if x is of type float, y is of type int?
 y = x + y;
 (A) int (B) float (C) There is no type for an assignment (D) double
 expression

7. Which keyword is used to come out of a loop only for that iteration? (A) Break (B) Continue
 (C) Return (D) f"r*inrt"

8. In expression i = g0 + f0, first function called depends on __. (A) Compiler (B) Associativity,
 "f
 () operator (C) Precedence of 0 and + operator (D) f"ttorigtioitdJJrpor"ion
 (A) Char
 (C) Unsigned int
 Short
 Long
 20NFU IECSS101J
 Page 4 of 4
 Page 1 of 1

[Page 2]

9. Set of consecutive memory location is called as

11. If the two strings are identical, then strcmp () function returns '
 (B) 0
 (D) True
 (B) Extracting a substring out of a string
 two (D) Comparing the two strings to define the
 larger one
 (B) Need not start on the first column
 (A) Comes before the first executable statement

14. Can we use a function as a parameter of another function?
 (A) Yes, and we can use the function (B) Yes, but we call the function again to get
 value conveniently the value, not as convenient as in using

PART-B (5 x 4=20 Marks)
 Answer ALL Questions

21. Explain the structure of a C program-

22. Define pseudo code and mention the advantage'

23. What are the differences between L value and R value in expression?

24. What is the difference between a 'break' and 'continue' statement?

25. Differentiate between "call by value" and "call by reference"

26. What is parameter? Mention the difference between formal and actual parameters-

27. Write the difference between union and structure'

PART-C (5x12=60 Marks)
 Answer ALL Questions

28- a.i. write the algorithm and pseudo code that takes minutes as input and displays the total number of hours and minutes.

ii. If the marks obtained by a student in six different subjects are input through the keyboard,

write a C program to find out the aggregate marks and percentage marks obtained by the student.

Pointer

10. Array with last element 'n' will always have array size equal to

(A) Function

(C) Array

(A) n

(C) n-1

(A) 1

(c) -1

12. String concatenation means

(A) Combining two strings

(C) Partitioning the string into strings

13. A preprocessor command

(A) Need not start on a new line

(C) Has # as the first character

(B)

(D)

(B) n+r

(D) n+1

(A) Integer

(C) Array

(A) Nothing

(C) Null

i

20. FILE reserved word is

(A) A structure tag declared in stdio.h

(C) Pointer to the structure defined in stdio.h

Page 2 of 1

Comment on the following pointer declarations? int * ptr, p;

(A) ptr is a pointer to integer, p is not (B) ptr and p, both are pointers to integer
(C) ptr is pointer to integer, p may or may not be (D) ptr and p both are not pointers to integer
may not be

How will you free the allocated memory?

(A) remove (var-name); (B) free (var-name);

(C) delete (var-name); (D) dalloc (var-name);

What is the similarity between a structure, union and enumeration?

(A) All of them let you define new (B) All of them let you define new data types

values

(c) All of them let you define new @) Allofthemletyoudefinewstructure
pointers

(oR)

b.i. Analyze the various storage classes in c with suitable examples.

ii- Write a C program to get two numbers as input from the user, then perform Bitwise-AND and Bitwise-OR and print the results. (4 Marks)

29. a. Analyze the C programming operator precedence and associativity with suitable examples.

(oR)

b. Explain the various decision making statements in C with an example for each'

30. a. write the purpose of following functions with suitable examples

(i) getso

(ii) getcharo

(iii) puts ()

(iv) putchar ()

(") strleno

(vi) strcpy ()

t6.

17.

18.

(C) No, C does not suPPort it

15. Address stored in the pointer variable is of type

-

(D) This case is comPiler

(B) Floating

(D) Character

(B) EOF

(D) Depends on comPiler

(B) One of the basic data tYPes in C

(D) It is a type name defined in stdio.h

(8 Marks)

19. If there is any error while opening a file, fopen will return

(oR)

b. Write about

(i) Recursion function with an example

(ii) Function definition with an example

Pagc 3 of ,l20Nr1/ ltcssl0lJ

20NF1/ r8CESl0lJ

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/Question Bank with
MCQs.pdf =====

CATEGORY: PYQs

STATUS: ok

[Page 1]

PROGRAMMING FOR PROBLEM SOLVING

Unit 1

Multiple Choice questions MCQ

1. Who is father of C Language? [CLO 1] [PLO1]
A. Bjarne Stroustrup B. Dennis Ritchie
C. James A. Gosling D. Dr. E.F. Codd
2. C Language developed at ____? [CLO 1] [PLO 1]
A. AT & T's Bell Laboratories of USA in 1972
B. AT & T's Bell Laboratories of USA in 1970
C. Sun Microsystems in 1973
D. Cambridge University in 1972
3. For 16-bit compiler allowable range for integer constants is _____ ? [CLO 1] [PLO 2]
A. -3.4e38 to 3.4e38
B. -32767 to 32768
C. -32768 to 32767
D. -32668 to 32667
4. C programs are converted into machine language with the help of [CLO 1] [PLO 1]
A. An Editor
B. A compiler
C. An operating system
D. None of the above
5. A C variable cannot start with [CLO 1] [PLO 1]
A. An alphabet
B. A number
C. A special symbol other than underscore
D. both (b) and (c)
6. Which of the following is allowed in a C Arithmetic Instruction? [CLO 1] [PLO 1]
A. []
B. {}
C. ()
D. None of the above
7. Which of the following shows the correct hierarchy of arithmetic operations in C [CLO 1] [PLO 2]
A. / + * -
B. * -/ +
C. + - / *
D. * / + -
8. Program which is written originally by the programmer is classified as [CLO 1] [PLO 1]
A. object code
B. machine code

- C. source program
- D. interactive programs

[Page 2]

9. Data types are differed on the basis of [CLO 1] [PLO 1]
- A. the way of storage
 - B. the type of operations
 - C. the type of operators used
 - D. both a and b
10. Loop statement which is repeated for some given number of times is classified as [CLO 1]
- A. FOR loop
 - B. GO loop
 - C. REPEAT loop
 - D. GO REPEAT loop
11. Type of statement written in sequence and is repeated until the specific condition met is classified as [CLO 1] [PLO 1]
- A. format
 - B. loop
 - C. case
 - D. condition
12. Size of an array is declared by [CLO 1] [PLO 1]
- A. programmer
 - B. program users
 - C. software
 - D. declared automatically
13. Programming language 'FORTRAN' stands for [CLO 1] [PLO 1]
- A. formula translator
 - B. formula translation
 - C. free translator
 - D. free translation
14. Functions that are used in the programs and are defined by the programmers are called [CLO 1] [PLO 2]
- A. program layout
 - B. program procedure
 - C. built-in functions
 - D. user-defined function

[Page 3]

15. An assembler translates [CLO 1] [PLO 1]
- A. machine code into assembly code
 - B. assembly code into machine code
 - C. processing time into manual time
 - D. routine into subroutine

16. Name given by a programmer to any particular data is classified as [CLO 1] [PLO 1]
- A. identifier
 - B. identification
 - C. exponent
 - D. mantissa
17. When variable used in program is whole number, the variable is stored as [CLO 1] [PLO 2]
- A. fixed string
 - B. integers
 - C. negative whole numbers
 - D. positive whole numbers
18. In programming, programmers use comments to [CLO 1] [PLO 1]
- A. highlight program modules
 - B. explain module functions
 - C. explain used variables
 - D. all of above
19. Variable which uses the same name in whole program and in its all routines thus best classified as [CLO 1] [PLO 2]
- A. middle variable
 - B. default variable
 - C. local variable
 - D. global variable
20. Statement which is used to make choice between two options and only option is to be performed is written as [CLO 1] [PLO 2]
- A. if statement
 - B. if else statement
 - C. then else statement
 - D. else one statement

[Page 4]

21. The _____ statement is used to transfer the control to the end of statement block in a loop:
- a. Continue [CLO 1] [PLO 2]
 - b. Break
 - c. Switch
 - d. Goto
22. Which of the following is not a valid variable name declaration? [CLO 1] [PLO 2]
- a) int _a3;
 - b) int a_3;
 - c) int 3_a;
 - d) int _3a
23. All keywords in C are in [CLO 1] [PLO 2]
- a) LowerCase letters

- b) UpperCase letters
- c) CamelCase letters
- d) None of the mentioned

24. Which of the following is true for variable names in C? [CLO 1] [PLO 2]

- a) They can contain alphanumeric characters as well as special characters
- b) It is not an error to declare a variable to be one of the keywords (like goto, static)
- c) Variable names cannot start with a digit
- d) Variable can be of any length

25. The format identifier '%i' is also used for _____ data type? [CLO 1] [PLO 2]

- a) char
- b) int
- c) float
- d) double

26. Which of the following is a User-defined data type? [CLO 1] [PLO 3]

- a) typedef int Boolean;
- b) typedef enum {Mon, Tue, Wed, Thu, Fri} Workdays;
- c) struct {char name[10], int age};
- d) all of the mentioned

27. What is the output of this C code? [CLO 1] [PLO 3]

```
1. #include <stdio.h>
2. int main()
3. {
4. signed char chr;
5. chr = 128;
6. printf("%d\n", chr);
7. return 0;
8. }
```

[Page 5]

- a) 128
- b) -128
- c) Depends on the compiler
- d) None of the mentioned

28. What is the output of this C code? [CLO 1] [PLO 3]

```
1. #include <stdio.h>
2. int main()
3. {
4. j = 10;
5. printf("%d\n", j++);
6. return 0;
7. }
```

- a) 10 b) 11
- c) Compile time error d) 0

Explanation : j is not belongs to any data type

29. The following code 'for(; ;)' represents an infinite loop.

It can be terminated by. [CLO 1] [PLO 3]

- a) break b) exit(0)
- c) abort() d) all of the mentioned

30. The keyword 'break' cannot be simply used within: [CLO 1] [PLO 3]

- a) do-while b) if-else
- c) for d) while

31. Which keyword is used to come out of a loop only for that iteration? [CLO 1] [PLO 3]

- a) break b) continue
- c) return d) none of the mentioned

32. The first step in problem solving is ----- [CLO 1] [PLO 2]

- a) Understand the problem c)Identify the problem
- b) Developing algorithm/Flowchart d)Listing the possible outcome

33. The solutions which has series of actions to solve a problem are called ----- [CLO 1] [PLO 2]

- a) Heuristic solutions
- b) Algorithmic solutions

34. The solution can be reached by completing the actions in steps. These steps are called [CLO 1] [PLO 2]

- a) Sequence c)Algorithm
- b) Flowchart d)Steps

35. Solutions that cannot be reached through a direct set of steps are called ----- [CLO 1][PLO 2]

- a)Algorithmic solutions b) Heuristic solutions

[Page 6]

36. ----- means the outcome or the completed computer-assisted answer. [CLO 1][PLO 1]

- a) Solution b) Result c) Program

37. The field of computer that deals with the problem of heuristic solution is called----- [CLO 1] [PLO1]

- a) Artificial Intelligence
- b) Expert System c) Computer Management

38. ----- means the set of instructions that make up the solution after they have been coded

into a programming language. [CLO 1] [PLO1]

- a) solution b) result c) program d) error

39. ----- are organized facts. [CLO 1] [PLO1]

a) Data b) Information

40. A language applied for wide range of application is called as _____language[CLO 1][PLO 2]

a. Special purpose c)General purpose

b. Individual purpose d)Scientific purpose

41. Compiler converts the source code into _____ [CLO 1][PLO 4]

a. C code b. Byte code c. Object code d. Executable code

42. _____ converts source code to machine language one line at a time. [CLO 1][PLO 4]

a. Compiler b. Interpreter c. Assembler d.CPU

43. The program that converts high level language to a machine language is called _____ [CLO 1] [PLO 1]

a. Interpreter b. Linker c. Compiler d. Loader

44. '#' symbol is known to be _____ [PLO 2] [CLO 1]

a. Linker b.Compiler c. Assembler d.Preprocessor directive

45. A _____ is a notational system for describing computations in both machine and human readable form [PLO 1] [CLO 1]

a. Programming language c.Machine language

b. High-level language d.Assembly language

46. ----- program converts assembly language into machine language [PLO 2] [CLO 1]

a. compiler b. interpreter c. assembler d. preprocessor directive

47. which symbol is been used for processing in flowchart? Ans: a [CLO 1] [PLO 3]

a) b) c) d)

48. which symbol is been used for input/output in flowchart? Ans: c [PLO 3] [CLO 1]

[Page 7]

a) b) c) d)

49. which symbol is been used for decision making in flowchart? Ans: d [PLO 3] [CLO 1]

a) b) c) d)

50. which is used as start/stop symbol in flowchart? Ans:b [PLO 3] [CLO 1]

a) b) c) d)

51. ----- is a distinguishing characteristic of human excellence in every area of behavior

[CLO 1] [PLO 1]

a) creativity b) thinking c) visualization d) problem solving

52. .What is the value of x in this C code? [PLO 2] [CLO 1]

```
#include <stdio.h>
void main()
{
    int x = 5 * 9 / 3 + 9;
}
```

(a). 3.75 b) Depends on compiler c) 24 d) 3

53.A character variable can store how many characters at a time? [PLO 2][CLO 1]

(a) 1 character

(b)8 characters

(c)255 character

54.Which of the following is the correct way of writing comments? [PLO 2] [CLO 1]

(a)*/comments/*

(b)/*comment*/

(c)**comment** (

d){comment}

55. C programming language is [PLO 1] [CLO 1]

(a)object oriented programming language (b)Procedure oriented programming language

(c)function oriented programming language (d)None of the above

56.The memory space taken for a char type data is [PLO 2] [CLO 1]

(a)2 bytes

(b)4 bytes

(c)8 bytes

(d)1bytes

57 .The memory space taken for a int type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes

[Page 8]

- (c) 8 bytes
- (d)10bytes

58.The memory space taken for a float type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes
- (c) 8 bytes
- (d)10bytes

59. What is the only function all programs must contain ? [PLO 2] [CLO 1]

- (a)start()
- (b)system()
- (c)main()
- (d)program

60.For 16-bit compiler allowable range for integer constants is _____? [PLO 2] [CLO 1]

- (a) -3.4e38 to 3.4e38
- (b) -32767 to 32768
- (c) -32668 to 32667
- (d) -32768 to 32767

61.Every statement in C program is to be terminated by a_____ [PLO 2] [CLO 1]

- (a)dot(.)
- (b)semi-colon(;))
- (c)colon(:)
- (d)Question mark(?)

62 The escape sequence „\b“ is a [PLO 2] [CLO 1]

- (a)back space
- (b)next line
- (c)tab
- (d)none of the above

63.The memory space taken for a long int type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes
- (c) 8 bytes
- (d)10bytes

64. which of the following will not valid expressions in C? [PL02] [CLO 1]

- (a) a=2+(b=5);
- (b) a=11%3
- (c) a=b=c=5
- (d) b+5=2

[Page 9]

65. The ----- operator is true only when both the operands are true. [PLO 2]
[CLO 1]

- a) &&
- b) b) ||
- c) c) !
- d) d) ?:

66. The ----- statement when executed in a switch statement causes immediate exit from the structure. [CLO 1] [PLO 2]

- a)goto
- b) default
- c) break
- d) switch

66. The ternary conditional expression using the operator?: could be easily coded using ----

- statement [CLO 1][PLO 2]

- a)Nested if b) if-else
- c) if d) for

4 Marks :

Unit-1 CLO-1

1. Comment "C is mid level language"? [CLO-1][PLO 1]
2. What is problem solving? [CLO-1][PLO 1]
3. What are the six steps of problem solving? [CLO-1][PLO 1]
4. Discuss about how the problems can be solved with computers? [CLO-1][PLO 1]
5. What is a program? [CLO-1][PLO 2]
6. Why is problem analysis important? [CLO-1][PLO 2]
7. What are the tools of problem solving available? [CLO-1][PLO 2]
8. How do problem-solving tools help in leading to a solution? [CLO-1][PLO 2]
9. Why it is important to test a solution before coding it? [CLO-1][PLO 2]
10. What is an algorithm? Give the characteristics of the algorithm. [CLO-1][PLO 2]
11. What is a flowchart? Give the symbols/shapes used in the flowchart.
[CLO-1][PLO 3]

12. Define pseudocode and give its importance with an example. [CLO-1][PLO 2]
13. Discuss the difficulties with problem solving in detail. [CLO-1][PLO 2]

[Page 10]

14. State the use of %d and %f .Write a printf statement in C using the above mentioned symbols? [CLO-1][PLO 4]
15. What is main difference between variable and constant? [CLO-1][PLO 3]
16. Explain bitwise left shift operator? [CLO-1][PLO 4]
17. Explain primary data types used in C? [CLO-1][PLO 3]
18. Difference between formatted & unformatted statement ? [CLO-1][PLO 3]
19. What is mean by storage class of variable? [CLO-1][PLO 3]
20. Explain with example ++i and i++. [CLO-1][PLO 4]
21. Refer all Elab programs. [CLO-1][PLO 3] [PLO 2][PLO 4]

12 Marks:

- 1) Explain the evolution of programming languages. [CLO-1][PLO 1]
2) Explain the various steps involved in problem solving with diagram. [CLO-1][PLO 1]
3) Draw the flowchart and write the algorithm and c code to find the sum and to reverse the digits of given five digit number. [CLO-1][PLO 1] [PLO2]
4) Write an algorithm and draw a flow chart to find the factorial and Fibonacci series of given number. [CLO-1][PLO 2] [PLO 3]
5) Write a note on Algorithm, Flow chart and Pseudocode. [CLO-1][PLO 3] [PLO 2]
6) Explain the scope, lifetime of variable in C with example. [CLO-1][PLO 2][PLO3]
7) Write down the algorithm to find the largest number among three given numbers and outline the steps in the algorithm with the inputs 5, 17, 3. [CLO-1][PLO 2][PLO3]
8) Draw flowchart to compute the salary of an employee in a company. Assume that there are two types of employees in the company daily wages and regular. Salary is calculated as number of hours worked* wages per hour for daily wagers and basic pay +(% of DA * basic pay)/100 + HRA + medical allowance for regular employees. Sketch the flow of your design for a regular employee with basic pay = 5000, % of DA = 75 % and HRA = 500. (8)
[CLO-1][PLO 2][PLO3][PLO4]
9) Explain in details about operators with an example [CLO-1][PLO 1] [PLO 2][PLO3]

10) Refer all elab Programs [CLO-1][PLO 3] [PLO 2][PLO 4]

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UNIT - II

PART A

1. The ----- operator is true only when both the operands are true.

e) && b) || c) ! d) ?:

Answer: a

2. The ----- statement when executed in a switch statement causes immediate exit

from the structure.

a) goto b) default c) break d) switch

Answer: c

3. The ternary conditional expression using the operator?: could be easily coded using -----

- statement

a) Nested if b) if-else c) if d) for

Answer: b

4. What will be the output when the following segment is executed?

```
Char ch='a';
```

```
Switch(ch)
```

```
{
```

```
case 'a':
```

```
Printf("A");
```

```
case 'b':
```

```
Printf("B");
```

```
Default:
```

```
printf("C");
```

```
}
```

a) A b) B c) C d) a

Answer: a

5. What will be the output of the following segment when executed?

```
int x=10, y=20;
```

```
if((x<y)|| (x+5)>10)
```

```
printf("%d",x);
```

```
else
```

```
printf("%d",y);
```

a) 10 b) 20 c) 15 d) 5

Answer: a

6. The ----- statement is used to skip a part of the statements in a loop.

a) Continue b) break c) goto d) switch

Answer: a

7. A for loop with no test condition is known as ----- loop

a) Infinite b) time delay c) for d) Incrementing

Answer: a

8. The sentinel -controlled loop is also known as -----loop

- a) Indefinite repetition loop c) time delay
- b) Definite repetition loop d) infinite

Answer: a

9. In an exit controlled loop the body of the loop is always executed minimum number of ----

- a) 1 time b) 2 times c) 3 times d) n times

Answer: a

10. The while is an ----- loop statement.

- a) Entry-controlled b) exit-controlled c) indefinite repetition
- d) definite repetition

Answer: a

11. The ----- specification is used to read or print integers

- a) h b) l c) L d) c

Answer: a

12. To print the data left-justified, must use ----- in the field specification

- a) - b) + c) / d) *

Answer: a

13. By default, the real numbers are printed with a precision of ----- decimal.

- a) 6 b) 2 c) 4 d) 0

Answer: a

14. The expression $!(x!=y)$ can be replaced by the expression-----

- a) $x!=y$ b) $x==y$ c) $x!=y$ d) $!x!=y$

Answer: c

15. In a counter controlled loop, variable known as ----- is used to count the
loop
operations.

- a) Counter b) sentinel c) i d) n

Answer: a

16. Which of the following special symbol allowed in a variable name?

- (a) * (asterisk) (b) | (pipeline) (c) - (hyphen) (d) _ (underscore)

17 . Which of the following are invalid variable names?

- a) Minimum b) n\$ c) Integer d) float

18. `int a=10;`

`++a;`

`a++;`

`Printf("%d",a);`

- a) 10 b) 11 c) 12 d) 13

19. `int a=11;`

`a=a%2;`

`a=a/2;`

The value of a is,

a) 1, 1 b) 5,1 c) 1,5 d)5

[Page 13]

20. Which of the following is not logical operator?

- A. &
- B. &&
- C. ||
- D. !

21 Which of the following cannot be checked in a switch-case statement?

- A. Character
- B. Integer
- C. Float
- D. enum

22. What is the output of this C code?

```
int main()
{
    int a = 0, i = 0, b;
    for (i = 0; i < 5; i++)
    {
        a++;
        continue;
    }
}
```

- A. 2 B. 3
- C. 4 D. 5

23. What is the output of this C code?

```
void main()
{
    int i = 0, j = 0;
    for (i = 0; i < 5; i++)
    {
        for (j = 0; j < 4; j++)
        {
            if (i > 1)
                break;
        }
    }
}
```

```
printf("Hi \n");  
}  
}
```

- A. Hi is printed 5 times B. Hi is printed 9 times
C. Hi is printed 7 times D. Hi is printed 4 times

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24. What is the output of this C code?

```
void main()  
{  
  int i = 0;  
  int j = 0;  
  for (i = 0; i < 5; i++)  
  {  
    for (j = 0; j < 4; j++)  
    {  
      if (i > 1)  
        continue;  
      printf("Hi \n");  
    }  
  }  
}
```

- A. Hi is printed 9 times B. Hi is printed 8 times
C. Hi is printed 7 times D. Hi is printed 6 times

25. What is the output of this C code?

```
void main()  
{  
  int i = 0;  
  for (i = 0; i < 5; i++)  
  {  
    if (i < 4)  
    {  
      printf("Hello");  
      break;  
    }  
  }  
}
```

- A. Hello is printed 5 times B. Hello is printed 4 times
C. Hello D. Hello is printed 3 times

27. What is the output of this C code?

```
int main()
```

```

{
int i = 0;
char c = 'a';
while (i < 2){
i++;
switch (c) {
case 'a':
printf("%c ", c);
break;
break;
}
}

```

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```

printf("after loop\n");
}

```

- A. a after loop B. a a after loop
C. after loop D. None of the mentioned

28. int main()

```

{
printf("before continue ");
continue;
printf("after continue\n");
}

```

- A. Before continue after continue B. Before continue
C. after continue D. Compile time error

29. What is the output of the code given below?

```

int main()
{
printf("%d ", 1);
goto l1;
printf("%d ", 2);
l1:goto l2;
printf("%d ", 3);
l2:printf("%d ", 4);
}

```

- A. 1 4 B. Compilation error
C. 1 2 4 D. 1 3 4

30. What is the output of code given below?

```

int main()
{
printf("%d ", 1);
l1:l2:
printf("%d ", 2);
}

```

```
printf("%d\n", 3);  
}
```

A. Compilation error B. 1 2 3

C. 1 2 D. 1 3

31. What will happen if in a C program you assign a value to an array element whose subscript

exceeds the size of array?

A. The element will be set to 0.

B. The compiler would report an error.

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C. The program may crash if some important data gets overwritten.

D. The array size would appropriately grow.

32 .In C, if you pass an array as an argument to a function, what actually gets passed?

A. Value of elements in array

B. First element of the array

C. Base address of the array

D. Address of the last element of array

33. Result of a logical or relational expression in C is?

A. True or False

B. 0 or 1

C. 0 if expression is false and any positive number if expression is true

D. None of the mentioned

34. What will be the value of d in the following program?

```
int main()  
{  
    int a = 10, b = 5, c = 5;  
    int d;  
    d = b + c == a;  
    printf("%d", d);  
}
```

A. Syntax error

B. 1

C. 5

D. 10

35. What is the output of this C code?

```
int main()  
{  
    int a = 10, b = 5, c = 3;
```



```

b != !a;
c = !!a;
printf("%d\t%d", b, c);
}

```

[Page 17]

- A. 5 1
- B. 0 3
- C. 5 3
- D. 1 1

36. What is meaning of following declaration ?

```
int arr[20];
```

- a) Array of size 20 that can have integer address
- b) None of the above
- c) Integer array of size 20
- d) Array of size 20

37. In C Programming, If we need to store word "INDIA" then syntax is as below -

- a) `char name[6]={'I','N','D','I','A','\0'}`
- b) `char name[6]={"I","N","D","I","A"}`
- c) `char name[6]={'I','N','D','I','A'}`
- d) `char name[] ; name="INDIA"`

38. what is the way to initialize array?

- a) `int num[6] = {2,4,12,5,45,5};`
- b) `int n{} = {2,4,12,5,45,5};`
- c) `int n{6}={2,4,12};`
- d) `int n(6)={2,4,12,5,45,5};`

39. what will be the output of the program?

- a) 3, 2, 15
- b) 2, 3, 20
- c) 2, 1, 15
- d) 1, 2, 5

Answer: Option A

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Solution:

>> `int a[5] = {5, 1, 15, 20, 25};` The variable arr is declared as an integer array with a size of 5 and it is initialized to

`a[0] = 5, a[1] = 1, a[2] = 15, a[3] = 20, a[4] = 25.`

>> `int i, j, m;` The variable i, j, m are declared as an integer type.

>> `i = ++a[1];` becomes `i = ++1;` Hence `i = 2` and `a[1] = 2`

>> j = a[1]++; becomes j = 2++; Hence j = 2 and a[1] = 3.

>> m = a[i++]; becomes m = a[2]; Hence m = 15 and i is incremented by 1(i++ means 2++ so i=3)

>> printf("%d, %d, %d", i, j, m); It prints the value of the variables i, j, m

Hence the output of the program is 3, 2, 15.

39. what will be the output of the program?

- a) 5
- b) 6
- c) 9
- d) Error
- e) None of the above

Answer: Option C

Solution:

x[i] is equivalent to *(x + i),
so (buf + 1)[5] is *(buf + 1 + 5), i.e. buf[6].

40) An array elements are always stored in _____memory locations

- a) sequential
- b) Random
- c) Sequential and Random
- d) None of the above

41) Let x be an array. which of the following operations are illegal?

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a) ++x b) x+1 c)x++ d) x*2

options:

- a) I and II
- b) I,II and III
- c) II and III
- d) I, III and IV
- e) III and IV

Answer: Option D

Solution: int x[10]; * x will store the base address of array. *

Statement I, III and IV is invalid.

Statement I and III : ++x and x++ are throwing an error while compile (lvalue required as increment operand)

Since, x is storing in the address of the array which is static value which cannot be change by the operand.

Statement IV : `x*2` is also throw an error while compile (invalid operands to binary `*` (have `'int *'` and `'int'`))

Statement II : `x+1` is throw a warning: assignment makes integer from pointer without a cast [enabled by default]

42) what is the maximum number of dimensions an array in c may have?

- a) 2
- b) 8
- c) 20
- d) 50
- e) theoretically no limit. the only practical limits are memory size and compilers

43) size of the array need not be specified, when

- a) initialized is a part of definition
- b) it's a declaration
- c) it is a formal parameter
- d) All of these

44) what will be the output of the program?

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- a) Equal
- b) Unequal
- c) Error
- d) None of these

Answer: Option B

Solution:

Strings are compared using `strcmp()` function defined under `string.h` header file.

45)

Unit-2 PART -B

1. Give the general syntax of conditional operator?
2. Define operator in C. What role an operator plays in C program?
3. Differentiate between relational and logical operators used in C?
4. Give the types of decision making statements
5. What is the general form of switch statement?
6. Differentiate `do... while..` and `while... loop`.
7. Determine the output of the following and justify your answer

```

int main()
{
    int ch; printf("enter a value between 1 to 2:");
    scanf("%d", &ch);
    switch (ch, ch -1){
    case 1: printf("1\n"); break;
    case 2: printf("2"); break;
    } return 0; }

```

8. Explain the limitations of arrays.

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9. Give the steps of looping process.

10. Brief the use of continue, goto, break statement and give the syntax.

11. What are loop control structures? Explain for loop, while loop and do-while loop with their syntax.

12. What do you mean by infinite loop? Give suitable of any infinite loop in a C program.

13. Explain break and continue statements with examples.

14. Determine the output of the following program justify your answer in a few words.

```

int main() {
float f = 1.0;
    switch(f) {
case 1.0: printf("one"); break;
case 2.0: printf("two"); break;
default: printf("%f", f);
    }}

```

15. Write a program to print series of number divisible by 3 from 1 to 100 using for loop.

16. Write a C program to print 1, 2, 4, 8, 16, 32, 64N using do...while loop (Read N from user).

17. What are Arrays in C programming? Give the importance of Array in C language

13. What are the rules to declare one dimensional array?

14. What do you mean by compile time initialization? Give suitable example of Compile time initialization of C Array.

15. Describe the array index out of bound error in context of C array program.

PART C

1) Lucy is celebrating her 15th birthday. Her father promised her that he will buy her a new computer

on her birthday if she solves the question asked by him. He asks Lucy to find whether the year on

which she had born is leap year or not. Help her to solve this puzzle so that she celebrates

her

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birthday happily. If her birth year is 2016 and it is a leap year display 2016 is a leap year.? Else

display 2016 is not a leap year and check with other leap year conditions

2) Write a C program to print the multiplication table of an integer n upto m rows using a while loop

sample input and output

INPUT

5

4

OUTPUT

1*5=5

2*5=10

3*5=15

4*5=20

3) Write a program to generate a following @s triangle:

@ @ @ @ @

@ @ @ @

@ @ @

@ @

@

4) Write a program that determines a student's grade. The program will read three scores and determine the grade based on the following rules: score =90% =>grade=A , score >= 70% and <90% => grade=B, score>=50% and <70% =>grade=C, score<50% =>grade=F

5) i)Why switch case is better than else..if... ladder. Justify.

ii)Write a program to read a value (1-7) and print the equivalent day of the week (i.e. 1-SUN, 2-MON.....)

iii) Explain briefly about for loop and Nested for loop with suitable example

iv) Write a program to print the following pattern.

12345

1234

123

12

1

6.a. i) Write a C program to insert an element at a specified position in the array.

7)) Write a program to generate a following patterns

a)

b)

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c)

d)

e)

8) write c Program for Student mark list generation by using control and looping statements

9) write c program for digit rotation. "For any positive integer, we define a digit rotation as either

moving the first digit to the end of the number (left digit rotation), or the last digit to the front of the

number (right digit rotation). For example, the number 12345 could be left digit rotated to 23451, or right

digit rotated to 51234.

10) Write a program to find the sum of positive numbers in an array

11) write c program for palindrome strings , two strings A and B, each consisting of lower case alphabets.

12) Write a arithmetic operation menu driven program using while(1)

13) i) Write short notes on switch case.

ii) Program to create a simple calculator, Performing addition, subtraction, multiplication, division depending the input from user.

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14) i)write a C program to read 10 nos. and reverse it using array

ii)Concatenate two arrays of length minimum 5 numbers.

UNIT III

PART A

[Page 25]

1) Which of the following is not possible statically in C?

- a) Jagged Array
- b) Rectangular Array
- c) Cuboidal Array
- d) Multidimensional Array

2) What is the output of this C code?

```
#include <stdio.h>
void main()
{
int a[2][3] = {1, 2, 3, 4, 5};
int i = 0, j = 0;
for (i = 0; i < 2; i++)
for (j = 0; j < 3; j++)
printf("%d", a[i][j]);
}
```

- a) 1 2 3 4 5 0
- b) 1 2 3 4 5 junk
- c) 1 2 3 4 5 5
- d) Run time error

3) Predict the output of below program:

```
#include <stdio.h>
```

```
int main()
{
```

```

int arr[5];

// Assume that base address of arr is 2000 and size of integer
// is 32 bit
arr++;
printf("%u", arr);

return 0;
}

```

- a) 2002
- b) 2004
- c) 2020
- d) lvalue required

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4) Predict the output of below program:

```
#include <stdio.h>
```

```

int main()
{
    int arr[5];
// Assume base address of arr is 2000 and size of integer is 32 bit
    printf("%u %u", arr + 1, &arr + 1);
    return 0;
}

```

- i. 2004 2020
 - ii. 2004 2004
 - iii. 2004 Garbage Value
 - iv. The program fails to compile because address of operator cannot be used with array name
- 5) Size of the array need not be specified, when
- A. Initialization is a part of definition
 - B. It is a declaratrion
 - C. It is a formal parameter
 - D. All of these
- 6) While passing an array as an actual argument, the function call must have the array name
- A. with empty brackets
 - B. with its size
 - C. alone
 - D. none of the above
- 7) The parameter passing mechanism for an array is
- A. call by value
 - B. call by value-result

- C. call by reference
- D. none of these

8) Under which of the following conditions, the size of an one-dimensional array need not be specified?

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- A. when initialization is a part of definition
 - B. when it is a declaration
 - C. when it is a formal parameter and an actual argument
 - D. All of the above
- 9) If a two dimensional array is used as a formal parameter, then

- A. both the subscripts may be left empty
- B. the first (row) subscript may be left empty
- C. the first subscript must be left empty
- D. both the subscripts must be left empty

11) Choose the statement that best defines an array

- A. It is a collection of items that share a common name
- B. It is a collection of items that share a common name and occupy consecutive memory location
- C. It is a collection of items of the same type and storage class that share a common name and occupy consecutive memory locations
- D. None of the above

12) Choose the correct statements

- A. Array stores data of the same type
- B. Array can be a part of a structure
- C. Array of structure is allowed
- D. All of the above

13) What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. int a[2][3] = {1, 2, 3, 4, 5};
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
```

- a) 1 2 3 4 5 0
- b) 1 2 3 4 5 junk
- c) 1 2 3 4 5 5

d) Run time error

Answer: a

14) What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. int a[2][3] = {1, 2, 3, , 4, 5};
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
a) 1 2 3 junk 4 5
b) Compile time error
c) 1 2 3 0 4 5
d) 1 2 3 3 4 5
```

Answer: b

15. What is the output of this C code?

```
1. #include <stdio.h>
2. void f(int a[][3])
3. {
4. a[0][1] = 3;
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
10. void main()
11. {
12. int a[2][3] = {0};
13. f(a);
14. }
a) 0 3 0 0 0 0
b) Junk 3 junk junk junk junk
c) Compile time error
d) All junk values
```

Answer: a

16. What is the output of this C code?

```
1. #include <stdio.h>
2. void f(int a[][])
```

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```
3. {
4. a[0][1] = 3;
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
10. void main()
11. {
12. int a[2][3] = {0};
13. f(a);
14. }
a) 0 3 0 0 0 0
b) Junk 3 junk junk junk junk
c) Compile time error
d) All junk values
```

Answer: c

17. What is the output of this C code?

```
1. #include <stdio.h>
2. void f(int a[2][3])
3. {
4. a[0][1] = 3;
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
10. void main()
11. {
12. int a[2][3] = {0};
13. f(a);
14. }
a) 0 3 0 0 0 0
b) Junk 3 junk junk junk junk
c) Compile time error
d) All junk values
```

Answer: c

18. Comment on the following statement:

```
int (*a)[7];
```

[Page 30]

- a) An array "a" of pointers.
- b) A pointer "a" to an array.
- c) A ragged array.
- d) None of the mentioned

Answer: b

19. Comment on the 2 arrays regarding P and Q:

- 1. `int *a1[8];`
 - 2. `int *(a3[8]);`
 - 3. P. Array of pointers
 - 4. Q. Pointer to an array
- a) a1 is P, a2 is Q
 - b) a1 is P, a2 is P
 - c) a1 is Q, a2 is P
 - d) a1 is Q, a2 is Q

Answer: b

20. Which of the following is not possible statically in C?

- a) Jagged Array
- b) Rectangular Array
- c) Cuboidal Array
- d) Multidimensional Array

Answer: a

21. What is the correct syntax to send a 3-dimensional array as a parameter? (Assuming declaration `int a[5][4][3];`)

- a) `func(a);`
- b) `func(&a);`
- c) `func(*a);`
- d) `func(**a);`

Answer: a

22. Applications of multidimensional array are?

- a) Matrix-Multiplication
- b) Minimum Spanning Tree
- c) Finding connectivity between nodes

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d) All of the mentioned

Answer: d

23. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. int ary[2][3];
5. foo(ary);
6. }
7. void foo(int *ary[])
8. {
9. int i = 10, j = 2, k;
10. ary[0] = &i;
11. ary[1] = &j;
12. *ary[0] = 2;
13. for (k = 0; k < 2; k++)
14. printf("%d\n", *ary[k]);
15. }
```

- a) 2 2
- b) Compile time error
- c) Undefined behaviour
- d) 10 2

Answer: a

24. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. int ary[2][3];
5. foo(ary);
6. }
7. void foo(int (*ary)[3])
8. {
9. int i = 10, j = 2, k;
10. ary[0] = &i;
11. ary[1] = &j;
12. for (k = 0; k < 2; k++)
13. printf("%d\n", *ary[k]);
14. }
```

- a) Compile time error
- b) 10 2
- c) Undefined behaviour

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d) segmentation fault/code crash

Answer: a

25. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4.   foo(ary);
5. }
6. void foo(int **ary)
7. {
8.   int i = 10, k = 10, j = 2;
9.   int *ary[2];
10.  ary[0] = &i;
11.  ary[1] = &j;
12.  printf("%d\n", ary[0][1]);
13. }
```

- a) 10
- b) 2
- c) Compile time error
- d) Undefined behaviour

Answer: d

26. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4.   int ary[2][3][4], j = 20;
5.   ary[0][0] = &j;
6.   printf("%d\n", *ary[0][0]);
7. }
```

- a) Compile time error
- b) 20
- c) Address of j
- d) Undefined behaviour

Answer: a

27. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
```

3. {

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4. int ary[2][3];

5. ary[][] = {{1, 2, 3}, {4, 5, 6}};

6. printf("%d\n", ary[1][0]);

7. }

a) Compile time error

b) 4

c) 1

d) 2

Answer: a

28. What is the output of this C code?

1. #include <stdio.h>

2. int main()

3. {

4. void foo();

5. printf("1 ");

6. foo();

7. }

8. void foo()

9. {

10. printf("2 ");

11. }

a) 1 2

b) Compile time error

c) 1 2 1 2

d) Depends on the compiler

Answer: a

29. What is the output of this C code?

1. #include <stdio.h>

2. int main()

3. {

4. void foo(), f();

5. f();

6. }

7. void foo()

8. {

9. printf("2 ");

10. }

```
11. void f()
12. {
13. printf("1 ");
```

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```
14. foo();
15. }
```

- a) Compile time error as foo is local to main
- b) 1 2
- c) 2 1
- d) Compile time error due to declaration of functions inside main

Answer: b

30. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. void foo();
5. void f()
6. {
7. foo();
8. }
9. f();
10. }
11. void foo()
12. {
13. printf("2 ");
14. }
```

- a) 2 2
- b) 2
- c) Compile time error
- d) Depends on the compiler

Answer: d

Explanation: Even though the answer is 2, this code will compile fine only with gcc. GNU C supports nesting of functions in C as a language extension whereas standard C compiler doesn't.

31. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo();
3. int main()
4. {
5. void foo();
6. foo();
```



```
7. return 0;
8. }
9. void foo()
10. {
11. printf("2 ");
12. }
```

[Page 35]

- a) Compile time error
- b) 2
- c) Depends on the compiler
- d) Depends on the standard

Answer: b

32. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo();
3. int main()
4. {
5. void foo(int);
6. foo(1);
7. return 0;
8. }
9. void foo(int i)
10. {
11. printf("2 ");
12. }
```

- a) 2
- b) Compile time error
- c) Depends on the compiler
- d) Depends on the standard

Answer: a

33. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo();
3. int main()
4. {
5. void foo(int);
6. foo();
7. return 0;
8. }
9. void foo()
```

```
10. {
11. printf("2 ");
12. }
a) 2
b) Compile time error
c) Depends on the compiler
```

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d) Depends on the standard

Answer: b

34. What is the output of this C code?

```
1. include <stdio.h>
2. void m()
3. {
4. printf("hi");
5. }
6. void main()
7. {
8. m();
9. }
```

a) hi
b) Run time error
c) Nothing
d) Varies

Answer: a

35. What is the output of this C code?

```
1. #include <stdio.h>
2. void m();
3. void n()
4. {
5. m();
6. }
7. void main()
8. {
9. void m()
10. {
11. printf("hi");
12. }
13. }
```

a) hi
b) Compile time error
c) Nothing
d) Varies

Answer: b

36. What is the return-type of the function sqrt()

- a) int
- b) float

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- c) double
- d) depends on the data type of the parameter

Answer: c

37. Which of the following function declaration is illegal?

- a)

```
double func();
int main(){
    double func();
}
```
- b)

```
double func(){};
int main(){
}
```
- c)

```
int main()
{
    double func();
}
double func(){//statements}
```
- d) None of the mentioned

Answer: d

38. What is the output of this code having void return-type function?

- 1. `#include <stdio.h>`
- 2. `void foo()`
- 3. `{`
- 4. `return 1;`
- 5. `}`
- 6. `void main()`
- 7. `{`
- 8. `int x = 0;`
- 9. `x = foo();`
- 10. `printf("%d", x);`
- 11. `}`
- a) 1
- b) 0
- c) Runtime error
- d) Compile time error

Answer: d

39. What will be the data type returned for the following function?

```
1. #include <stdio.h>
2. int func()
3. {
4.     return (double)(char)5.0;
5. }
```

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- a) char
- b) int
- c) double
- d) multiple type-casting in return is illegal

Answer: b

40. What is the problem in the following declarations?

```
int func(int);
double func(int);
int func(float);
```

- a) A function with same name cannot have different signatures
- b) A function with same name cannot have different return types
- c) A function with same name cannot have different number of parameters
- d) All of the mentioned

Answer: d

41. The output of the code below is

```
1. #include <stdio.h>
2. void main()
3. {
4.     int k = m();
5.     printf("%d", k);
6. }
7. void m()
8. {
9.     printf("hello");
10. }
```

- a) hello 5
- b) Error
- c) Nothing
- d) Junk value

Answer: a

42. The output of the code below is

```
1. #include <stdio.h>
2. int *m()
3. {
4. int *p = 5;
5. return p;
6. }
```

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```
7. void main()
8. {
9. int *k = m();
10. printf("%d", k);
11. }
```

- a) 5
- b) Junk value
- c) 0
- d) Error

Answer: a

43. The output of the code below is

```
1. #include <stdio.h>
2. int *m();
3. void main()
4. {
5. int *k = m();
6. printf("hello ");
7. printf("%d", k[0]);
8. }
9. int *m()
10. {
11. int a[2] = {5, 8};
12. return a;
13. }
```

- a) hello 5 8
- b) hello 5
- c) hello followed by garbage value
- d) Compilation error

Answer: c

Part B

[Page 40]

1. What is the need for user defined function?
2. Write a multi function program
3. List the Elements of user defined function
4. Give short note on 2-D array processing.
5. Write a C program to find the length of a string.
6. HELLO encode it as IFMMP using array
7. Show the difference between actual and formal parameter in function with piece of code
8. Compare user define function vs System define function
9. List categories of function
10. Explain the concept of function call by reference with a sample program.
11. Explain the concept of function call by value with a sample program
12. Explain about functions with example
13. Discuss Character Arrays
14. Predict output of the following program (CL03)

```
int main(){
    inta[][]={{1,2},{3,4}};
int i,j;
for (i=0;i<2;i++)
for(j=0;j<2;j++)
printf("%d",a[i][j]);
return 0;
}
```
15. Why array index starts from zero?
16. Contrast function declaration vs function definition.
17. List the advantages of functions.
18. Program to Find the 2nd Largest Elements in an Array.
19. Write a program to convert the given string (srm university) Lower to upper case
20. Advantages and limitations of multi dimensional array initialization
21. Explain call by value and call by reference with an example
22. Common programming errors in 2D arrays
12. Explain String functions with example

PART C

1. Array construction for student mark list for 100 students. output need to display register number, marks of five subjects , CGPA and PASS/Fail Status

2. Explain in details about String Functions: gets(), puts(), getchar(), putchar(), printf(), with an example programs

3. i) Print the given pattern using 2-D array
1234567
23456

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345

4

ii) Program to divide one array into two arrays

4. i) What are strings in C? Write a C program to read a string in lowercase and convert it to uppercase.

ii)) Illustrate call by value and call by reference with example for each.

5. i) Find the factorial of 10 using function recursion.

ii) What is a string? Explain any 5 string functions.

6. Explain in details about String Functions: atoi, strlen, strcat, strcmp with an example

7. Functions declaration and definition, Types: Call by Value & Call by Reference with example programs

8. Explain and write c program for Function with and without Arguments and no Return Values

9. Explain and write c program for Passing Array to Functions with return type, Recursion Functions

10. Write c program for Matrix Multiplication using Multi-dimensional array

11. Program to Delete duplicate elements from an array

INPUT

1 3 4 5 3

OUTPUT

1 3 4 5

b. Write a program to perform Matrix addition and Multiplication using 2-D arrays

12. i) State the importance of functions. List out the different types of Function

ii) Write a swap function using call by value and call by reference

iii) Write a C program to concatenate two strings.

iv) Write a program to read to strings and compare them and print a message that the first string

is equal, less or greater than the second one accordingly.

13. Write c program for Matrix addition and Matrix Transpose using multi dimensional array

14. Given a number , find whether it is a power of 2 or not

15. Write a C program to swap elements in cyclic order using call by reference.

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UNIT 4

1. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo(int*);
3. int main()
4. {
5.     int i = 10;
6.     foo(&i++);
7. }
8. void foo(int *p)
9. {
10.    printf("%d\n", *p);
11. }
```

a) 10
b) Some garbage value
c) Compile time error
d) Segmentation fault/code crash

Answer: c

2. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo(int*);
3. int main()
4. {
5.     int i = 10, *p = &i;
6.     foo(p++);
7. }
8. void foo(int *p)
9. {
10.    printf("%d\n", *p);
11. }
```

a) 10
b) Some garbage value
c) Compile time error

d) Segmentation fault

Answer: a

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3. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo(float *);
3. int main()
4. {
5.     int i = 10, *p = &i;
6.     foo(&i);
7. }
8. void foo(float *p)
9. {
10.    printf("%f\n", *p);
11. }
```

a) 10.000000
b) 0.000000
c) Compile time error
d) Undefined behaviour

Answer: b

4. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4.     int i = 97, *p = &i;
5.     foo(&i);
6.     printf("%d ", *p);
7. }
8. void foo(int *p)
9. {
10.    int j = 2;
11.    p = &j;
12.    printf("%d ", *p);
13. }
```

a) 2 97
b) 2 2
c) Compile time error
d) Segmentation fault/code crash

Answer: a

5. What is the output of this C code?

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```
1. #include <stdio.h>
2. int main()
3. {
4.     int i = 97, *p = &i;
5.     foo(&p);
6.     printf("%d ", *p);
7.     return 0;
8. }
9. void foo(int **p)
10. {
11.     int j = 2;
12.     *p = &j;
13.     printf("%d ", **p);
14. }
```

a) 2 2
b) 2 97
c) Undefined behaviour
d) Segmentation fault/code crash

Answer: a

6. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4.     int i = 11;
5.     int *p = &i;
6.     foo(&p);
7.     printf("%d ", *p);
8. }
9. void foo(int *const *p)
10. {
11.     int j = 10;
12.     *p = &j;
13.     printf("%d ", **p);
14. }
```

a) Compile time error
b) 10 10
c) Undefined behaviour
d) 10 11

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Answer: a

7. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4.     int i = 10;
5.     int *p = &i;
6.     foo(&p);
7.     printf("%d ", *p);
8.     printf("%d ", *p);
9. }
10. void foo(int **const p)
11. {
12.     int j = 11;
13.     *p = &j;
14.     printf("%d ", **p);
15. }
```

- a) 11 11 11
- b) 11 11 Undefined-value
- c) Compile time error
- d) Segmentation fault/code-crash

Answer: b

8. What is the output of the code below?

```
1. #include <stdio.h>
2. int main()
3. {
4.     int i = 10;
5.     int *const p = &i;
6.     foo(&p);
7.     printf("%d\n", *p);
8. }
9. void foo(int **p)
10. {
11.     int j = 11;
12.     *p = &j;
13.     printf("%d\n", **p);
14. }
```

- a) 11 11
- b) Undefined behaviour
- c) Compile time error

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d) Segmentation fault/code-crash

Answer: a

9. Which of the following are correct syntaxes to send an array as a parameter to function:

- a) func(&array);
- b) func(#array);
- c) func(*array);
- d) func(array[size]);

Answer: a.

11. What is the output of this C code?

- ```
1. #include <stdio.h>
2. void main()
3. {
4. int k = 5;
5. int *p = &k;
6. int **m = &p;
7. printf("%d%d%d\n", k, *p, **m);
8. }
```
- a) 5 5 5
  - b) 5 5 junk value
  - c) 5 junk junk
  - d) Run time error

Answer: a

12. What is the output of this C code?

- ```
1. #include <stdio.h>
2. void main()
3. {
4. int k = 5;
5. int *p = &k;
6. int **m = &p;
7. printf("%d%d%d\n", k, *p, **p);
8. }
```
- a) 5 5 5
 - b) 5 5 junk value
 - c) 5 junk junk

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d) Compile time error

Answer: d.

13. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. int k = 5;
5. int *p = &k;
6. int **m = &p;
7. **m = 6;
8. printf("%d\n", k);
9. }
```

a) 5
b) Compile time error
c) 6
d) Junk

Answer: c

14. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. int a[3] = {1, 2, 3};
5. int *p = a;
6. int *r = &p;
7. printf("%d", (**r));
8. }
```

a) 1
b) Compile time error
c) Address of a
d) Junk value

Answer: b

15. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
```

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```
4. int a[3] = {1, 2, 3};
5. int *p = a;
6. int **r = &p;
7. printf("%p %p", *r, a);
```

8. }

- a) Different address is printed
- b) 1 2
- c) Same address is printed.
- d) 1 1

View Answer

Answer: c

Explanation: None.

16. How many number of pointer (*) does C have against a pointer variable declaration?

- a) 7
- b) 127
- c) 255
- d) No limits.

Answer: d

17. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. int a = 1, b = 2, c = 3;
5. int *ptr1 = &a, *ptr2 = &b, *ptr3 = &c;
6. int **sptr = &ptr1; //-Ref
7. *sptr = ptr2;
8. }
```

- a) ptr1 points to a
- b) ptr1 points to b
- c) sptr points to ptr2
- d) None of the mentioned

Answer: b

18. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
```

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```
4. int a[3] = {1, 2, 3};
5. int *p = a;
6. int **r = &p;
7. printf("%p %p", *r, a);
8. }
```

- a) Different address is printed
- b) 1 2

- c) Same address is printed.
- d) 1 1

Answer: c

PART B

- 1) short notes on Passing Array Element to Function with an example
- 2) Explain Formal and Actual Parameters
- 3) Give the Advantages of using Functions ,
- 4) example program for Processor Directives and #define Directives
- 5) write a program for Pointer Declaration and dereferencing,
- 6) write a program for pointers, Void Pointers and size of Void Pointers
- 7. Write algorithm and pseudo code for leap year calculation. [CLO 1]
- 8. Contrast the declaration of break and continue in while and for loop with justification [CLO 2]
- 9. Limitations of two dimensional arrays.

PART C

- 1) i) c program to read array elements and print the values with their address
- 7) Justify and explain the different size of data types using pointer variables with example program.
- 8) i) write a c program to count vowels and consonants in a string using pointer
- ii) Write call by reference and call by value with an example program
- ii) what is Pointers and address operator, explain with an example of Size of Pointer Variable and Pointer, Operator
- 2) Explain in details with an example Pointer Declaration and dereferencing, pointers, Void Pointers and size of Void Pointers
- 3) write program for Arithmetic Operations, Incrementing Pointers
- 4) what is Constant Pointers, Pointers to array elements and strings
- 5) what is function pointer & Array of Function Pointers with an example
- 6) write the c program for Accessing Array of Function Pointers, Null Pointers

UNIT 5

Part A

1. Which of the following are themselves a collection of different data types?

- a) string
- b) structures
- c) char
- d) all of the mentioned

View Answer

Answer: b

Explanation: None.

2. User-defined data type can be derived by_____

- a) struct
- b) enum
- c) typedef
- d) all of the mentioned

Answer: d

3. Which operator connects the structure name to its member name?

- a) -
- b) <-
- c) .
- d) Both <- and .

Answer: c

4. Which of the following cannot be a structure member?

- a) Another structure
- b) Function
- c) Array

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d) None of the mentioned

Answer: b

5. Which of the following structure declaration will throw an error?

- a)

```
struct temp{}s;
main(){}

```
- b)

```
struct temp{};
```



```
struct temp s;  
main(){}  
c) struct temp s;  
   struct temp{};  
   main(){}  
d) None of the mentioned  
Answer: d
```

6. What is the output of this C code?

```
1. #include <stdio.h>  
2. struct student  
3. {  
4. int no;  
5. char name[20];  
6. }  
7. void main()  
8. {  
9. struct student s;  
10. s.no = 8;  
11. printf("hello");  
12. }  
a) Compile time error  
b) Nothing  
c) hello  
d) Varies
```

Answer: a

7. What is the output of this C code?

```
1. #include <stdio.h>  
2. struct student  
3. {  
4. int no = 5;  
5. char name[20];  
6. };  
7. void main()
```

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```
8. {  
9. struct student s;  
10. s.no = 8;  
11. printf("hello");  
12. }  
a) Nothing  
b) Compile time error
```

- c) hello
- d) Varies

Answer: b

8. What is the output of this C code?

- ```
1. #include <stdio.h>
2. struct student
3. {
4. int no;
5. char name[20];
6. };
7. void main()
8. {
9. student s;
10. s.no = 8;
11. printf("hello");
12. }
```
- a) Nothing
  - b) hello
  - c) Compile time error
  - d) Varies

Answer: c

9. What is the output of this C code?

- ```
1. #include <stdio.h>
2. void main()
3. {
4.     struct student
5.     {
6.         int no;
7.         char name[20];
8.     };
9.     struct student s;
```

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- ```
10. s.no = 8;
11. printf("%d", s.no);
12. }
```
- a) Nothing
  - b) Compile time error
  - c) Junk
  - d) 8

Answer: d

10. Can the above code be compiled successfully?

```
1. #include <stdio.h>
2. struct p
3. {
4. int k;
5. char c;
6. float f;
7. };
8. int main()
9. {
10. struct p x = {.c = 97, .f = 3, .k = 1};
11. printf("%f\n", x.f);
12. }
a) Yes
b) No
c) Depends on the standard
d) Depends on the platform
```

Answer: c

11. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. struct student
5. {
6. int no;
7. char name[20];
8. };
9. struct student s;
```

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```
10. no = 8;
11. printf("%d", no);
12. }
a) Nothing
b) Compile time error
c) Junk
d) 8
```

Answer: b

12. Number of bytes in memory taken by the below structure is

1. #include <stdio.h>
  2. struct test
  3. {
  4. int k;
  5. char c;
  6. };
- a) Multiple of integer size
  - b) integer size+character size
  - c) Depends on the platform
  - d) Multiple of word size

Answer: a

13. What is the output of this C code?

1. #include <stdio.h>
  2. struct
  3. {
  4. int k;
  5. char c;
  6. };
  7. int main()
  8. {
  9. struct p;
  10. p.k = 10;
  11. printf("%d\n", p.k);
  12. }
- a) Compile time error
  - b) 10
  - c) Undefined behaviour

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- d) Segmentation fault

14. What is the output of this C code?

1. #include <stdio.h>
2. struct
3. {
4. int k;
5. char c;
6. } p;
7. int p = 10;
8. int main()
9. {
10. p.k = 10;

```
11. printf("%d %d\n", p.k, p);
12. }
a) Compile time error
b) 10 10
c) Depends on the standard
d) Depends on the compiler
```

Answer: a

15. What is the output of this C code?

```
1. #include <stdio.h>
2. struct p
3. {
4. int k;
5. char c;
6. };
7. int p = 10;
8. int main()
9. {
10. struct p x;
11. x.k = 10;
12. printf("%d %d\n", x.k, p);
13. }
a) Compile time error
b) 10 10
c) Depends on the standard
d) Depends on the compiler
```

[Page 56]

Answer: b

16. What is the output of this C code?

```
1. #include <stdio.h>
2. struct p
3. {
4. int k;
5. char c;
6. float f;
7. };
8. int p = 10;
9. int main()
10. {
11. struct p x = {1, 97};
12. printf("%f %d\n", x.f, p);
```

13. }
- a) Compile time error
  - b) 0.000000 10
  - c) Somegarbage value 10
  - d) 0 10

Answer: b

17. What is the output of this C code(according to C99 standard)?

advertisement

- ```
1. #include <stdio.h>
2. struct p
3. {
4. int k;
5. char c;
6. float f;
7. };
8. int main()
9. {
10. struct p x = {.c = 97, .f = 3, .k = 1};
11. printf("%f\n", x.f);
12. }
```
- a) 3.000000
 - b) Compile time error
 - c) Undefined behaviour
 - d) 1.000000

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Answer: a

18. What is the output of this C code(according to C99 standard)?

- ```
1. #include <stdio.h>
2. struct p
3. {
4. int k;
5. char c;
6. float f;
7. };
8. int main()
9. {
10. struct p x = {.c = 97, .k = 1, 3};
11. printf("%f \n", x.f);
12. }
```
- a) 3.000000

- b) 0.000000
- c) Compile time error
- d) Undefined behaviour

Answer: b

19. What is the output of this C code(according to C99 standard)?

- ```
1. #include <stdio.h>
2. struct p
3. {
4. int k;
5. char c;
6. float f;
7. };
8. int main()
9. {
10. struct p x = {.c = 97};
11. printf("%f\n", x.f);
12. }
```
- a) 0.000000
 - b) Somegarbagevalue
 - c) Compile time error
 - d) None of the mentioned

Answer: a

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PART B

1. Write algorithm and pseudo code for leap year calculation.
2. Contrast the declaration of break and continue in while and for loop with justification
3. Limitations of two dimensional arrays.
4. Explain about nested pre-processor MACRO
5. Categorize the basic operations that can be performed on a file with suitable declarations.
6. Write a c program to demonstrate double pointer for accessing the value of another pointer
7. Explain about file operations and mode with syntax.
- 8) explain shortly about Initializing Structure, Declaring structure, variable,
- 9) give the example of Structure using typedef, Accessing members

- 10) what is Nested structure, Accessing elements in a structure array,
- 11) write c program for Array of structure, Accessing elements in a structure array
12. Explain about file operations and mode with syntax.
13. Give a brief note about preprocessor directives.
14. Differentiate between a union and a structure.
15. Write a program to read a file character by character, and display it simultaneously on the screen
- 16) write a program for Passing Array of structure to function,
- 17) write a program for Array of pointers to structures
18. Explain about nested pre-processor MACRO
19. Categorize the basic operations that can be performed on a file with suitable declarations.
20. Write a c program to demonstrate double pointer for accessing the value of another pointer

PART C

- 1) explain in details about Bit Manipulation to structure and Pointer, to structure, Union Basic and declaration with an example
- 2) explain in details about Accessing Union Members Pointers to Union, Dynamic memory allocation, malloc, realloc, free with an example
- 3) explain in details about Allocating Dynamic Array, Multidimensional array using dynamic memory allocation with an example
- 4) Explain in details about array of structures and accessing elements in a structure array.

- 5) Write a c program to insert a line at the end of text file
- 6) write a program to copy the content from one file to another file
- 7) explain and write the program for library management system using union

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- 14) c program to read array elements and print the values with their address
- 15) Justify and explain the different size of data types using pointer variables with example program.
- 16) write a c program to count vowels and consonants in a string using pointer
- 17) Write call by reference and call by value with an example program
- 18) a) i) Explain in details about array of structures and accessing elements in a structure array.
 - ii) Write a c program to insert a line at the end of text file
- 19) i) write a program to copy the content from one file to another file
 - ii) explain and write the program for library management system using union

- 8) write a c program for file: opening, defining, closing, File Modes, File Types, Writing contents into a file
- 9) write a c program for Reading file contents, Appending an existing file
- 10) Differentiate between gets() and scanf()
- (11) Write a program to convert the given string "hello world" to "dlrow olleh".

12. Write a program that passes a pointer to a structure to a function

13 Briefly discuss about file operations in C

32(a) Write a program that passes a pointer to a structure to a function

[CLO 5]

(OR)

32(b) Briefly discuss about file operations in C [CLO 5]

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SRM Institute of Science and Technology, Ramapuram

Department of Computer Science and Engineering

18CSS101J-Programming for Problem Solving

UNIT 1 MCQ

1. What is the value of variable z? `int x; int y; int z; x=3; y=4; z = ++x * y++;` a) 9 b) 12 c) 16 d)

20 Answer: c

2. To represent two different conditions which of the following is used.

a) Rectangle

b) Diamond

c) Circle

d) Parallelogram

Ans B

3. Give the output of the following program? `#include <stdio.h> main(){ int a=9, b=9; a=b++; b=a++; b=++b; printf("%d %d", a, b); }` a) 9,9 b) 10,10 c) 9,10 d) 10,9

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Answer: b

4. In the below mentioned operators, which of the following has the highest precedence? a)

Unary + b) * c) >= d) == Answer: a

5. What is the output of the given code?

`counter = 2 while counter < 68 puts counter counter**=2 end` a) 2 4 16 64 b) 2 4 16 c) 2 4 16 256 d) None of the mentioned Answer: b

6. Which of the following is a valid assignment operator? a) += b) -= c) *= d) All of the mentioned Answer: d

7. The output for the following C code is? `#include <stdio.h> void main(){ int x = 97; int y = sizeof(x++); printf("x is %d", x); }` a) x is 97 b) x is 98 c) x is 99 d) Run time error Answer: a

8. Give the correct answer for the following C code? `#include <stdio.h> void main()`

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{ int x = 4, y, z; y = --x; z = x--; printf("%d%d%d", x, y, z); } a) 3 2 3 b) 2 2 3 c) 3 2 2 d) 2 3 3
Answer: d

9. What will be the output of the following C code?

```
#include <stdio.h> void main() { int x = 4; int *p = &x; int *k = p++; int r = p - k; printf("%d", r); }
```

a) 4 b) 8 c) 1 d) Run time error
Answer: c

10. Among the following options, which is not a valid representation in bits? a) 8-bit b) 24-bit c) 32-bit d) 64-bit
Answer: b

11. Name the entities whose values can be changed? a) Constants b) Variables c) Modules d) Tokens
Answer: b

12. Which is not a basic data type? a) float b) int c) real

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d) char
Answer: c

13. Name the correct default value of a Boolean type? a) 0 b) 1 c) True d) False e) -1
Answer: Option D

14. Which of the statements given below are correct? 1. We can assign values of any type to variables of type object. 2. When a variable of a value type is converted to object, it is said to be unboxed. 3. When a variable of type object is converted to a value type, it is said to be boxed. 4. Boolean variable cannot have a value of null. 5. When a value type is boxed, an entirely new object must be allocated and constructed. a) 2, 5 b) 1, 5 c) 3, 4 d) 2, 3
Answer: Option B

15. Name the correct ways to set a value as 3.14 in a variable pi such that it cannot be modified? a) float pi = 3.14F; b) #define pi 3.14F; c) const float pi = 3.14F; d) const float pi; pi = 3.14F; e) pi = 3.14F;
Answer: C

16. Why the variable names beginning with underscore is not encouraged? a) It is not standard form b) To avoid conflicts since assemblers and loaders use such names c) To avoid conflicts since library routines use such names d) To avoid conflicts with environment variables of an operating system
Ans: c

17. Mention the variable name which is not valid in C? a) int number; b) float rate; c) int variable_count; d) int \$main;
Ans: d

18. Which among the following options is true for the variable names in C?

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a) They can contain alphanumeric characters as well as special characters b) It is not an error to declare a variable to be one of the keywords (like goto, static) c) Variable names can't start with a digit d) Variable can be of any length
Ans: c

19. Give the output for the following code? #include <stdio.h> int main() { int main = 5; printf("%d", main); return 0; } a) compile-time error b) run-time error c) run without any error and prints 5 d) experience infinite looping
Ans: c

20. Which of the following can never be a variable name in C? a) friend b) true c) volatile d) export
Ans: c

21. Mention the User-defined data type from the following? a) struct { char name[10], int age; } b) typedef enum { Mon, Tue, Wed, Thu, Fri } Workdays; c) typedef int Boolean; d) all of the mentioned
Answer: d

22. What is short int in C programming? a) Basic data type of C b) Qualifier c) short is the qualifier and int is the basic data type d) All of the mentioned
Ans: c

23. What is the output of this C code?#include <stdio.h>int main(){signed char chr;

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chr = 128;printf("%d\n", chr);return 0;} a) 128b) -128c) Depends on the compilerd) None of the mentionedAns:b

24. Which of the data types have size that is in variable?a) intb) structc) floatd) doubleAns:b

25. Mention the User-defined data type?a) typedef int Boolean;b) typedef enum {Mon, Tue, Wed, Thu, Fri} Workdays;c) struct {char name[10], int age};d) All of the mentionedAnswer: Option D

26. Output of this C code is ?

int main(){char chr;chr = 128;printf("%d\n", chr);return 0;} a) 128b) - 128c) Depends on the compilerd) None of the mentionedAnswer: Option B

27. What is the output for the C program ?#include <stdio.h>int main(){ static int i=5;if (-i){printf("%d ",i);

[Page 66]

main();}} a) 4 3 2 1b) 1 2 3 4c) 4 4 4 4d) 0 0 0 0

Answer: A

28. Which of the following is not a storage class in C?a) automatic storage classb) register storage classc) external storage classd) internal storage class

Ans: d

29. Scope of which of these in C is not local to the block in which it is defineda) automatic storage classb) register storage classc) static storage classd) external storage classAns d

30. Single line comments are denoted by _____.a) //b) \\\c) /*d) */Answer: A

31 . How many comment lines are present in the below program -main(){printf("\nHello World in C //Comment \");printf("\nHello World in C \"); //Comment//Comment printf("\nHello World in C \");} a) 1b) 0c) 2d) 3Answer: C

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32. Comments are case Sensitive !a) Trueb) FalseAnswer: B33. What is the use of getchar()?a) The next input character each time it is calledb) EOF when it encounters end of filec) The next input character each time it is called EOFwhen it encounters end of filed) None of the mentionedAnswer: c

34. Name the method which uses a list of well-defined instructions to complete a taska) Algorithmb) Flowchartc) Programsd) FunctionsAns C

35. Sequence logic will not be used whilea) Accepting input from the userb) Giving output to the userc) Adding two numbersd) Comparing two sets of dataAns B

36. Flowchart and algorithms are used for a) Better programmingb) Easy understanding and debuggingc) Efficient codingd) All the aboveAns d

37. What is the default initial value of static storage class?a) 0b) garbage valuec) 1d) -1Ans a

38. Resolving errors in a program is known as...

a) Debuggingb) Refixingc) Error Checkingd) Problem SolvingAns a

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39. Many features of C were derived from an earlier language called _____.a) FORTRANb) BASICc) Bd) PASCALAns C

40. C Programming was created at ?a) MIT Universityb) AT&T Bell Laboratoryc) L&T Laboratoryd) Haward UniversityAns B

41. What will be the output of the following pseudocode?Integer a, bSet a = 15, b = 7a = a mod (a - 3)b = b mod (b - 3)a = a mod 1b = b mod 1Print a + ba) 15b) 7c) 2d) 0[Note-mod finds the remainder after the division of one number by another. For example, the expression "5 mod 2" leaves a quotient of 2 and a remainder of 1]Answer: d

42. What will be the output of the following pseudocode?Integer a, b, cSet b = 5, a = 2, c = 2if(b>a && a>c && c>b)b = a + 1Elsea = b + 1End ifPrint a + b + c[Note-&&: Logical AND - The logical AND operator (&&) returns the Boolean value true (or 1) if both operands--. If (x) gets executed if the value if(), i.e., x is not zero]a) 2b) 13c) 26d) 5Ans: B

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UNIT 2 MCQ

1. Does rational operators in the C language are calculated with the short circuit?a) Trueb) Falsec) Depends on the compilerd) Depends on the standard

Answer: a

2. What is the outcome of relational or logical operation in C?a) True or Falseb) 0 or 1c) 0 if an expression is false and any positive number if an expression is true d) None of the mentioned

Answer: b

3. What will be the output of the following C code?

```
#include <stdio.h>
int main()
{
int a = 20, b = 10, c = 10;
int d;
d = b + c == a;
printf("%d", d);
}
```

a) Syntax errorb) 1c) 5d) 10

Answer: b

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4. What will be the output of the following C program.

```
#include<stdio.h>
int main()
{ int a=-1, b=1, c, d;
c = !a && b;
d = !a || b;
printf("%d %d %d %d",a,b,c,d);
return 0;}
```

a) -1 1 1 0b) -1 1 0 0c) -1 1 1 1d) -1 1 0 1

Ans:- d

5. Which operator of the following is not a logical or relational operator? a) != b) == c) || d) =

Answer: d
6. Relational operators will not be used in _____ a) structure b) long c) strings d) float
Answer: a

7. Which are the operators used to relate the values of operands to create logical value in C language?

- a) Logical operator b) Relational operator c) Assignment operator d) None of the above

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Answer: b

8. In the following operators which one performs processes on data in binary level?

- a) Logical operator b) Bitwise operator c) Additional operators d) None of the above

Answer: b

9. How many number of operands needed to unary operator logical not (!)?

- a) 4
b) 3
c) 2
d) 1

Answer: d

10. An expression comprises relational, arithmetic and assignment operators. If interpolations are not indicated, the order of valuation of the operators will be:

- a) Assignment, arithmetic, relational
b) Relational, assignment, arithmetic
c) Assignment, relational, arithmetic
d) Arithmetic, relational, assignment

Answer: d

11. Which group has higher priority between (>, <=, <, <=) and (==, !=)?

- a) (>, >=, <, <=) has lower priority (==, !=) b) (>, >=, <, <=) has higher priority (==, !=) c) (>, >=, <, <=) has equal priority with (==, !=) d) None of the above

Answer: b

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12. The || and && operators

- A. Relate two Boolean values B. Relate two numeric values C. Combine two Boolean values D. Combine two numeric values

Answer: C

13. Increment operators need L-value Expression What will be the output of the following program?

```
#include <stdio.h>
int main(){ int i = 10; printf("%d", ++(-i)); return 0;}
A) 11 B) 10 C) -9 D) None - l-value required
```

Answer: D
14. What will be the output of the following C program?

```
#include <stdio.h>
int main(){ int a = 3, b = -8, c = 2; printf("%d", a % b / c); return 0;}
A) 1 B) 15 C) -9 D) 12
```

Answer: A

15.What will be the output of the following C code?

```
#include <stdio.h>
```

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```
void main(){ double x = 123828749.66;int y = x;printf("%d\n", y);printf("%lf\n", y);}a) 0, 0.0b) 123828749, 123828749.66c) 12382874, 12382874.0d) 123828749, 0.000000
```

Answer: d16.Which of the following selection statement test onlyequality?a) ifb) switchc) if & switchd) none of the mentionedAnswer: b

17.Which of the following will execute the body of loopcondition is false?

a) do-whileb) whilec) ford) none of the mentioned

Answer: a

18.Which of this following statement is incorrect?

a) switch statement is more efficient than a set ofnested ifsb) two case constants in the same switch can haveidentical valuesc) switch statement can only test for equality, whereasif statement can evaluate any typeof Boolean expressiond) it is possible to create a nested switch statements

Answer: b

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19.The break statement is used to quite from:

a. DO loopb. FOR loopc. SWITCH STATEMENTd. All of above.

Answer: d

20.The advantage of a SWITCH statement over an ELSE-IFstatement:

a.A default condition can be used in the SWITCH.b. It is easier to understand.c. Several conditions can cause one set of statements to be executed in a SWITCH.d. Several diffeent statements can be executed in a SWITCH.

Answer: b

21.The comma (,) operator is primarily used in conjunctionwith:a. FOR statement.b. IF-ELSE statement.c.DO-While statement.d.All of above.

Answer: a

22.The GOTO statement is used:

a.To permit two dissimilar expressions to appearin condition where only one expressionwould ordinarily use.b. to dismiss loops or to exit from a switch.c.an unconditional transfer of control to a namedlabel.d.to carry out a logical test and then take one oftwo possible actions, depending upon theoutcome of a test.

Answer: c23.What is an Array in C language?

A) A group of elements of same data type.B) An array contains more than one elementC) Array elements are stored in memory in continuousor contiguous locations.D) All the above.

Answer :d

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24.An array Index starts with?A) -1B) 0C) 1D) 2

Answer :b

```
25.What is the output of c program?Int main(){Int a[]={5,2,3,4};Int b[4]={10,6,7,8};Printf(“%d,%d”, a[0], b[0]);}A) 5,10B) 2,6C) 0 0D) Compiler error
```

Answer : A

26. Which is correct format for initializing an array in C?

a) `int arr[5] = (1,2,3,4,5);` b) `int arr(3) = {1,2,3,4,5};` c) `int arr[3] = {1,2,3,4,5};` d) `int arr(3) = (1,2,3,4,5);`

Answer: c

27. Which of the following concepts make extensive use of arrays? a) Binary trees b) Scheduling of processes c) Caching d) Spatial locality

Answer: d

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28. What are the disadvantages of arrays?

a) Data structure like queue or stack cannot be implemented b) There are chances of wastage of memory space if elements inserted in an array are lesser than the allocated size c) Index value of an array can be negative d) Elements are sequentially accessed

Answer: b

29. Array is accessed by _____ elements. a) randomly b) sequentially c) exponentially d) logarithmically

Answer: a

30. What is the output of C program with arrays?

```
int main()
{
    char str[] = {"C", "A", "T", "\0"};
    printf("%s", str);
    return 0;
}
```

A) CB) CATC) CAT\0D) Compiler error

Answer: D

31. What is the output of C program with string arrays?

```
int main()
{
    char *p1 = "GOAT";
    char *p2;
    p2 = p1;
    p2 = "ANT";
}
```

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```
printf("%s", p1);
}
```

A) ANTB) GOATC) GDD) A

Answer: B

32. A one-dimensional array contains one-dimensional arrays is called A. Two-dimensional array B. Multi-casting array C. Multi-dimensional array D. Three-dimensional array Answer: A

33. Objects in a sequence that have the same type, is called A. Arrays B. Operators C. Functions D. Stacks Answer: A

34. In an array starting from the beginning and inspecting each element one after other for finding an object, until the object is found is called as A. Linear search algorithm B. Searching C. Inspecting D. All of them Answer: A

35. Which of the following technique is used to find value in an array? A. Binary search algorithm B. Bubble sort

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C. Linear search algorithmD. All of themAnswer: d

36.Programmer's uses _____ for defining their own types.?A. AlgorithmsB. OperatorsC. EnumerationsD. None of themAnswer: c

37.A I-D array is considered as____?A. ComplexB. SequentialC. LinearD. Both C and BAnswer: d

38.Array comprising elements are numbered as 0,1,2,3?These numbers are called asA. Subscripts of the arrayB. Index valuesC. Members of an arrayD. Both A and BAnswer: d

39.Find the output of the code

```
#include<stdio.h>void main(){int m[5]={10,20,30,40,50};int a,b,c;a=++m[1];b=m[1]++;c=m[i++];
```

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```
printf(“%d,%d,%d”,a,b,c);}
```

A. 21,20,30B. 20,21,30C. 22,21,30D. 20,21,40Answer : C

40.What will be the resulting array after rotating arr[]={2,4, 6, 8, 10} by 2?A. 4, 6, 8, 10, 2B. 10, 2, 4, 6, 8C. 8, 10, 2, 4, 6D. 6, 8, 10, 2, 4

Answer : D

41.Which one is used in arrays extensivelyA. SchedulingB. CachingC. Spatial LocalityD. Binary trees

Answer : C

42.What is the size of int arr[10] if the int is of 4bytes?A. 4B. 40C. 10D. 14

Answer : B

43.The resulting array after reversing arr[]={21,23,25,27,29}isA. 29,27,25,23,21B. 25,27,29,21,23C. 27,29,21,23,25D. 23,25,27,29,21

Answer : A

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44.Maximum number of dimensions an array haveA. 8B. 4C. No limitD. 50Answer : C

45.What does int arr[30] mean?A. Array of size 30B. Integer Array of size 30C. Array of size 30 that have integer addressD. None

Answer : A

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UNIT 3 MCQ

1. ____ quotes is used for character array initialization.(A) Any(B) Double(C) No(D)

SingleAnswer: B

2.Data type for array in C belongs to------(A) Primitive(B) User defined(C) Derived(D) BasicAnswer: C

3.Two compare the two strings which of the followingone is used(A) Strcmp ()(B) Strcpy ()(C) Stcp()(D) Stcmp()Answer: A

4.----- Array example is matrix(A) Single(B) 2-D(C) 3-D(D) Multi-dimensionalAnswer: B

5.Array Size should be always -----(A) Positive(B) Negative

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(C) Whole number(D) Real numberAnswer: A

6. The first subscript of 2D array represents the----- size(A) Row(B) Column(C) Object(D) DiagonalAnswer: A

7._____ array is an array in which each element is itself an array(A) 1-D(B) 2-D(C) 4-D(D) 5-DAnswer: B

8.----- character type is used for indicating the terminator of the string.(A) Float(B) Int(C) Null(D) StringAnswer: C

9.How many Array types available?(A) 1(B) 2(C) 3(D) 4Answer: B10.For write function, how many parameters required?

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(A) Two(B) Three(C) Four(D) FiveAnswer: A11.Arranging of elements in array with respect to ascending or descending is called ____ (A) Merging(B) Filtering(C) Sorting(D)

OrderingAnswer: C12. The position of array elements always begins from____ (A) One(B) Two(C) Three(D) ZeroAnswer: D

13.What will be the output of the following C code?

```
#include <stdio.h>
int main()
{
ARRAY1(ary);
}
void ARRAY1(int **ary)
{
int i = 10, k = 10, j = 2;
int *ary[2];
ary[0] = &i;
```

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```
ary[1] = &j;
printf("%d\n", ary[0][1]);
}
```

a.22

b. Compile time error

c. Undefined behavior

d.102

Answer: c

14.What will be the output for the following program?

```
#include <stdio.h>
int main()
{
int ary[2][3];
ary[][] = {{1, 2, 3}, {4, 2, 6}};
printf("%d\n", ary[1][0]);
return 0;
}
```

- a.4
- b. Compile time error
- c.2
- d.1

Answer: a

15. `int (*b)[6];`

- a. An array "b" of pointers

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- b. A pointer "b" to an array
- c. A ragged array
- d. None of the mentioned

Answer: b

15. Statement on the following 2 arrays with respect to M and N.

`int *x1[8];`

`int *(x2[8]);`

- M. Array of pointers
- N. Pointer to an array
- A) x1 is M, x2 is N
- B) x1 is M, x2 is M
- C) x1 is N, x2 is M
- D) x1 is N, x2 is N

Answer: b

16. Which of the following is not possible statically in C?

- a. Jagged array
- b. Rectangular array
- c. Cuboidal array
- d. Multidimensional Array.

Answer: D

17. What will be the output of the following C code?

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```
#include <stdio.h>
```

```
void main()
```

```
{
int a[2][3] = {1, 3, 5, 7, 9};
int i = 0, j = 0;
for (i = 0; i < 2; i++)
for (j = 0; j < 3; j++)
printf("%d", a[i][j]);
}
```

- a. 135790
- b. 13579junk
- c. 135799
- d. Run time error

Answer: a

18. Write the output of the following C code?

```
#include <stdio.h>
int main()
{
char *S1 = "hello, world";
char *S2 = "hello, world";
```

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```
if (strcmp(S1, S2))
printf("equal");
else
printf("unequal");
}
```

- a.equal
- b.Unequal
- c.Compilation error
- d. Depends on the compiler

Answer: b

19. Write the output for the below code.

```
#include <stdio.h>
int main()
{
char *str = "hello, word\n";
printf("%d", strlen(str));
}
```

- a. Compilation error

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- b. Undefined behavior
- c.12
- d.11

Answer: c

20. Always----- function adds null character

- a. strcmp()
- b. strcat()
- c. strcpy()
- d. strrev()

Answer: b

21.----- function is not declared in math.h? a) and() b) pow() c) exp() d) acos()

Answer: a

22. Which one is correct for multidimensional array example.

- a. int[][] code = {{1,2},{3,4,5}};
- b. int[2][] code = {{1,2},{3,4,5}};
- c. int[][] code={1,2,

3,4,5});

d. all

Answer: A

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23. The function function1 is called _____ if it calls the same function function1.

A. indirect recursive B. direct recursive C. Both A and B D. None of the above Answer : B

24. Iteration not requires more system memory than recursion.

A. TRUE B. FALSE C. Can be true or false D. Can not Say Answer : A

25. Write the output of the following C code?

```
#include<stdio.h>
```

```
main()
```

```
{
```

```
int n;
```

```
n=f1(3);
```

```
printf("%d",n);
```

```
}
```

```
f1(int x)
```

```
{
```

```
int b;
```

```
if(x==1)
```

```
return 1;
```

```
else
```

```
b=x*f1(x-1);
```

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```
return b;
```

```
}
```

a.6

b.4

c.12

d.10

Answer: a

26. Find the error of following function give on compilation ?

```
f(int a,int b){int b;a = 20;return a;}
```

a. Missing parentheses in return statement

b. Function should be define as int f(int a, int b)

c. Redeclaration if b

d. No error

Answer:c

27.It is essential to declare the sort of a function within the calling program on the off chance that the function

a.Returns an integer

b.Returns a non-integer value

c.is not defined in the same file

d. None of these.

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Answer: b

28. What is meant by void?

a. Representation of NULL data type b. Representation of void pointer c. Error d. None of above

Answer: A

29. If the two strings are not identical, then strcmp() function returns

a. -1 b. 0 c. 1 d. Yes

Answer: c

30. What is the Format specifier used to print an integer array in C Printf or Scanf function.?

A) %c

B) %CC) %sD) %d

Answer: D

31. Write the output for the below code.

```
int main()
{
char ary[]="SRM IST";
printf("%s",ary);
return 0;
}
```

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A) S

B) SRMC) SRMISTD) SRM IST

Answer : D

32. How do you convert this char array to string.?

```
char str[]={ 'g','o','o','d'};
```

A) str[4] = 0;

B) str[4] = '\0'

C) str[]={ 'g','o','o','d','\0'};

D) All the options

Answer : D

33. Find the output?

```
int main()
{
char str[]={ "G","O","D","\0"};
printf("%s",str);
return 0;
}
```

A) Compiler error

B) GOD

C) GOD\0

D) G

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Answer : A

34.What is the output of C Program with arrays.?

```
int main()
{
char str[100];
scanf("%s", str);
printf("%s",str);
return 0;
}
//input: WEST INDIES
```

- A) WEST
- B) WEST INDIES
- C)W
- D) Compiler error

Answer : D

35. What is the output of C Program with functions.?

```
int main()
{
FUN();
printf("LTD ");
return 0;
}
```

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```
void FUN()
{
printf("HDFC ");
}
```

- A) HDFC LTD
- B) LTD HDFC
- C) LTD
- D) Compiler error

Answer : D

36.In C program, Function return -----number of value at a time.?

- A) Only One Value
- B) Maximum of two values
- C) Maximum of three values
- D) Maximum of 8 values

Answer: A

37.Select the types of function

- A) Library Functions
- B) User Defined Functions
- C) Both Library and User Defined
- D) None

Answer: C

38. What is the constraint for number of functions in a C Program.?

A) 12

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B) 24

C) 56

D) No limit

Answer: D

39. What is the least number of functions to be present in a C Program?

A) 1

B) 3

C) 4

D) 2

Answer: A

40. Each C Program ought to contain which function.?

A) main()

B) show()

C) scanf()

D) printf()

Answer: A

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UNIT 4 MCQ

1. How will you declare the function which is intended to receive an array as an argument.
a. return_type function(type arrayname[])
b. return_type function(type arrayname[SIZE])
c. return_type function(type *arrayname)
d. All of the above

Ans: d

2. To store the array returned from the function, we can define a _____ which points to that array.
a. Structure
b. Array
c. Pointer
d. List

Ans: c

3. The difference between Actual Parameters and Formal Parameters is that Actual Parameters are the values that are _____ the function when it is invoked while Formal Parameters are the variables defined by the function that _____ values when the function is called.
a. Passed to, receives
b. Received by, pass
c. Prints, process
d. Process, prints

Ans: a

4. The advantages of using functions are:
a. Avoid repetition of codes.
b. Increases program readability.
c. Divide a complex problem into simpler ones.
d. All of the above
Ans: d
5. The C preprocessor is a _____ that is used automatically by the C compiler to transform your program before actual compilation.
a. Macro processor
b. Microprocessor
c. Macro controller
d. Microcontroller

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Ans: a

6. Preprocessor directives are executed _____ compilation.
a. Before
b. After
c. During
d. None of the above

Ans: a

7. The _____ preprocessor directive is used to paste code of given file into current file. a. #define b. #include c. #if d. #pragma

Ans: b

8. A macro is a _____ which is replaced by the value of macro. Macro is defined by _____ directive. a. segment of code, #pragma b. file, #define c. segment of code, #defined. d. none of the above

Ans: c

9. _____ is used to undefine a macro definition. a. #undef b. #unfed c. #defund. d. #undef

Ans: d

10. What is the output of the following program? #include <stdio.h> #ifndef __MATH_H #error First include then compile #else void main() { float a; a = sqrt(7); printf("%f", a); } #endif

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a. 2.64575 b. 49 c. 0.7 d. None of the above

Ans: a

11. The _____ directive is used by the compiler to offer machine or operating-system feature. a. #define b. #elif c. #included. d. #pragma

Ans: d

12. Calculate the output of the following program? #include <stdio.h> #define Area(x) x*x #define Costpaint(x,y,z) (z*y + Area(x)) void main() { int A = 8, B = 6, C = 4; clrscr(); printf("The area of square = %d\n", Area(A)); printf("Cost of paint = %d\n", Costpaint(A,B,C)); } a. 64 b. 88 c. 32 d. 76

Ans: b

13. Identify the value that gets printed in the following program

```
#include <stdio.h>
int main() {
    int a = 10; //variable declaration
    int *p; //pointer variable declaration
    p = &a; //store address of variable a in pointer
    printf("Address stored in a variable p is: %x\n", p); //accessing the address
    printf("Value stored in a variable p is: %d\n", *p); //accessing the value
    return 0;
}
```

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a. 10 b. 10, 60 c. 60, 10 d. None of the above

Ans: a

14. _____ operator can be used to evaluate sizeof a variable/pointer in C. a. size() b. eval() c. sizeof() d. None of the above

Ans: c

15. A pointer to void means a _____ pointer that can point to any data type. a. Specific b. Generic c. Exact d. Null

Ans: b

16. Select the possible arithmetic operations applicable on the pointer in C language: a. Increment, Decrement b. Addition, Subtraction c. Comparison d. All of the above

Ans: d

17. What is the output of the following program:

```
#include <stdio.h>
void main() {
    int arr[5] = {1, 2, 3, 4, 5};
    int *p = arr;
    int i;
    printf("printing array elements...\n");
    for(i = 0; i < 5; i++) {
        printf("%d ", *(p+i));
    }
}
```

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} a. 1, 2, 3, 4, 5b. 5, 4, 3, 2, 1c. 1, 2, 3, 5, 8d. None of the above

Ans: a

18. Identify the illegal pointer arithmetic operations: a. Address + Address b. Address * Address c. Address % Address d. All of the above

Ans: d

19. What is the output of the given program:

```
#include<stdio.h> int addition (); int main () { int result; int (*ptr)(); ptr = &addition;
result = (*ptr)(); printf("The sum is %d",result); } int addition() { int a=5, b=2; return
a+b; } a. 10b. 3c. 7d. 25Ans: c20. A _____ pointer in C cannot change the address of the
variable to which it is pointing a. Fixed b. Null c. Void
```

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d. ConstantAns: d21. An array of pointers to strings is an array of character pointers where each pointer points to the _____ of the string or the _____ of the string. a. first character, base address b. last character, last address c. middle character, middle address d. None of the aboveAns: a22. A function pointer points to _____, not _____. a. data, code b. code, data c. type, const d. None of the aboveAns: b23. Identify the correct function pointer declaration: a. int * foo(int) b. int (*foo)(int); c. int (int)(*foo) d. None of the aboveAns: b24. What is the output of the given program: #include <stdio.h> void Hi_function (int times); /* function */ int main() { void (*function_ptr)(int); /* function pointer Declaration */ function_ptr = Hi_function; /* pointer assignment */ function_ptr (3); /* function call */ return 0; } void Hi_function (int times) { int k; for (k = 0; k < times; k++) printf("Hi"); } a. Hi Hi Hi Hi Hi b. Hi Hi Hi Hi Hi c. Hi Hi Hi Hi d. Hi Hi Hi HiAns: d25. What is the output of the given program: #include <stdio.h> void* cube (const void* num); int main() {

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```
int x, cube_int; x = 4; cube_int = cube (&x); printf("%d cubed is %d\n", x, cube_int); return
0; }
void* cube (const void *num) {int result; result = (*(int *)num) * (*(int *)num) * (*
(int*)num); return result; } a. 2 cubed is 9 b. 3 cubed is 28 c. 4 cubed is 64 d. 5 cubed is
26Ans: c26. What is the output of the given program: #include<stdio.h> main () {int a[3] =
{10,20,30}; int *p[3], i; for (i=0; i<3; i++) p[i] = &a[i]; //initializing base address
of array printf ("elements of the array are") for (i=0; i<3; i++) printf ("%d \t", *p[i]);
//printing array of pointers getch(); } a. elements of the array are 10 20 30 b. elements of the
array are 9 19 29 c. elements of the array are 11 21 31 d. None of the aboveAns: a27. What is
the output of the following program: #include<stdio.h> main () {int a = 10; int *p; int **q; p =
&a; q = &p;
```

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```
printf("%d",a); printf("%d", *p); printf("%d", **q); } a. 10 10 11 b. 10 11 12 c. 10 10 10 d. None
of the aboveAns: d28. Choose the correct way of declaring pointer array: a. datatype
pointername [size]; b. datatype *pointername [size]; c. datatype[size] pointername; d. None of
the aboveAns: b29. A _____ is a pointer that does not point to any memory location. a. Null
```

pointerb. Void pointerc. Array pointerd. String pointerAns. a30.The null pointer basically stores the ____ value.a. Voidb. Intc. Chard. NullAns. d31._____ is used to initialize a pointer variablewhen the pointer does not point to avalid memory address.a. Voidb. Intc. Chard. NullAns. d32.What is the output of the following program:#include <stdio.h> int main() { int *ptr; printf("Address: %d", ptr);

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printf("Value: %d", *ptr); return 0; } a. Produces outputb. Program crashesc. Show errord. None of the aboveAns. b33.What is the output of the following program:#include <stdio.h> int main() { int *ptr; ptr=(int*)malloc(4*sizeof(int)); if(ptr==NULL) { printf("Memory is not allocated"); } else { printf("Memory is allocated"); } return 0; } a. Memory is not allocatedb. Memory is allocatedc. Shows compiler errord. Program crashesAns. b34.What is the output of the following program:#include <stdio.h> #include <conio.h> #define NUMBER 0 void main() { #if (NUMBER==0) printf("Value of Number is: %d",NUMBER); #endif getch(); } a. Value of Number is: 10

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b. Value of Number is: 11c. Value of Number is: 0d. Value of Number is 9Ans. c35.The #error preprocessor directive indicates error. The compiler gives fatal errorif #error directive is found and skips further compilationprocess.a. #ifdef, continueb. #error, continuec. #ifdef, skipsd. #error, skipsAns. d36.What is the output of the following program:#include <stdio.h>int main(){char *cities[] = {"Iran", "Iraq"};int i;for(i = 0; i < 2; i++)printf("%s\n", cities[i]);return 0;}a. Iraq, Iranb. India, Iranc. Iran, Iraqd. Iraq, IndiaAns. c37.What is the output of the following program:#include <stdio.h>#include <string.h>void function(char**);int main(){char *str = "Pointer-to-string";int i, j = strlen(str);for(i = 0; i < j; i++)printf("%c", *str++);return 0;}a. gnirts-ot-retnioPb. Pointers to string

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c. Pointers-to-stringd. All of the aboveAns. c38.Choose the correct way thatexpress the Pointer andarray elementsa. a[i]b. i[a]c. *(a + i)d. All of the aboveAns. d

39.What is the output of the following program:#include<stdio.h>int main(){int i = 3;int *j;int **k;j = &i;k = &j;k++;printf("%d ",**k);return 0;}a. Garbage Valueb. Compilation errorc. Runtime errord. Linker errorAns. c

40.What is the output of the following program:#include<stdio.h>int main(){int i = 3;int *j;j = &i;j++;printf("%d ",*j);return 0;}a. Garbage Valueb. Compilation errorc. Runtime errord. Linker errorAns.a

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UNIT 5 MCQ

1. What is the output of the program?#include <stdio.h>struct student{ char *name;};struct student s;struct student fun(void){ s.name = "newton";printf("%s\n", s.name);s.name = "alan";return s;}void main(){struct student m = fun();printf("%s\n", m.name);m.name = "turing";printf("%s\n", s.name);}

a) newtonalanalanb) run tiime errorrc) alanan newtond) no error

Answer: a

2. What will be the output of the program?
`#include <stdio.h>struct student{char *name;};void main(){struct student s, m;s.name = "st";m = s;printf("%s%s", s.name, m.name);}`

a) Compile time error

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b) no errorc) runtime errord) stst

Answer: D

3. Which return-type cannot be used for a function?a) mainb)intc) voidd) none of the mentioned

Answer:D

4. What will be the output of the program?
`#include <stdio.h>struct temp{int a;};voidfunc(struct temp s){s.a = 10;printf("%d\t", s.a);}main(){func(s);printf("%d\t", s.a);}`

a)0b10 20c) 10 0d) (Garbage Value) 10

Answer:C

5. Which of the following is not possible??

a) s1 = s2;b) s1 != s2;c) (*s1).number = 50;d) None of the mentioned

Answer: D

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6. Which operation is illegal in structures?

a) Typecasting of structureb) nonec) Dynamic allocation of memory for structured) All

Answer: A

7. "s.t.b = 10" indicates _____

a) Syntax Errorb) Structuredc) float data typed) variable name

Answer:B

8. What will be the output of theprogram?
`#include <stdio.h>struct student{char *name;};struct student fun(void){struct student s;s.name = "alan";return s;}void main(){struct student m = fun();s.name = "turing";printf("%s", m.name);}`a) alanb) Runtime errorrc) Compile time errord) Noth

Answer:C9. What will be the output of the program?
`#include <stdio.h>struct point`

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`{int x;int y;};int main(){struct point p = {1};struct point p1 = {1};if(p == p1)printf("equal\n");elseprintf("not equal\n");}`

a) Compile time errorb) Runtime errorc) depends on the standardd) equal

Answer: a

10.What will be the output of theprogram?
`#include <stdio.h>struct point{int x;int y;};structnotpoint{int x;int y;};struct point foo();int main(){struct point p = {1};structnotpoint p1 = {2, 3};p1 = foo();printf("%d\n", p1.x);}struct point foo(){struct point temp = {1, 2};return temp;}`

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a) Compile time errorb) 1c) Runtime errord) Undefined behaviour

Answer: a

11. What will be the output of the program? `#include <stdio.h> struct point {int x; int y;}; struct notpoint {int x; int y;}; int main() {struct point p = {1}; struct notpoint p1 = p; printf("%d\n", p1.x);}`

a) Compile time error b) 1 c) 0 d) Runtime error

Answer: a

12. What will be the output of the program? `#include <stdio.h> struct point {int x; int y;}; struct notpoint {int x; int y;};`

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`void foo(struct point); int main() {struct notpoint p1 = {1, 2}; foo(p1);} void foo(struct point p) {printf("%d\n", p.x);}`

a) Compile time error b) 1 c) 0 d) Runtime error

Answer: a

13. What will be the output of the program? `#include <stdio.h> struct point {int x; int y;}; void foo(struct point*); int main() {struct point p1 = {1, 2}; foo(&p1);} void foo(struct point *p) {printf("%d\n", *p.x++);}`

a) Compile time error b) Segmentation fault c) code crash d) Runtime error

Answer: a

14. What will be the output of the program? `#include <stdio.h> struct point`

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`{int x; int y;}; void foo(struct point*); int main() {struct point p1 = {1, 2}; foo(&p1);} void foo(struct point *p) {printf("%d\n", *p->x++);}`

a) Compile time error b) Runtime error c) Segmentation fault d) 5

Answer: a

15. What will be the output of the program? `#include <stdio.h> struct student fun(void) {struct student {char *name;}; struct student s; s.name = "alan"; return s;} void main() {struct student m = fun(); printf("%s", m.name);}`

a) Compile time error b) Runtime error c) Nothing d) code crash

Answer: a

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`#include <stdio.h> struct student {char *name;}; struct student fun(void) {struct student s; s.name = "alan"; return s;} void main() {struct student m = fun(); printf("%s", m.name);}`

a) Compile time error b) alan c) Run time error d) code crash

Answer: b

17. What is the correct syntax to access the member of the ith structure? Assuming: `struct temp {int b;}; s[50];`

a) s.b[i]; b) b[i].s; c) b[i]; d) s[i].b;

Answer: d

18. What is the output of the program. `#include <stdio.h> struct temp {int a; int b; int c;}; main() {struct temp p[] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};`

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}

a) No Compile time error, generates an array of structure of size 3
b) Compile time error, generates an array of structure of size 9
c) no Compile time error, illegal declaration of a multidimensional array
d) illegal assignment to members of structure

Answer: a

19. Which of the following uses structure?
a) Array
b) Binary Tree
c) linked list
d) All

Answer: D

20. What is the correct syntax to declare a function foo() which receives an array of structure in function?
a) void foo(struct *var);
b) void foo(struct.var);
c) void(struct(var));
d) none

Answer: a

21. What is the output of the program (size of int is 4)?
`#include <stdio.h> struct temp { int a; int b; int c; } p[] = {0}; main() { printf("%d", sizeof(p)); }`

a) 8
b) 12
c) 30
d) Can't be estimated

Answer: b

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```
#include <stdio.h> struct student { char *name; }; struct student s[2]; void main() { s[0].name = "alan"; s[1] = s[0]; printf("%s%s", s[0].name, s[1].name); s[1].name = "turing"; printf("%s%s", s[0].name, s[1].name); }
```

a) alanalalananturing
b) alanturingturing
c) alanturingturing
d) run time error

Answer: a

```
23. What is the output of the program? #include <stdio.h> struct student { char *name; }; struct student s[2], r[2]; void main() { s[0].name = "alan"; s[1] = s[0]; r = s; printf("%s%s", r[0].name, r[1].name); }
```

a) Runtime error
b) Compile time error
c) Varies
d) None

Answer: b

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```
{ char *name; }; void main() { struct student s[2], r[2]; s[1] = s[0] = "alan"; printf("%s%s", s[0].name, s[1].name); }
```

a) alanalalb) None
c) Compile time error
d) Runtime error

Answer: c

```
25. What is the output of the program? #include <stdio.h> struct student {}; void main() { struct student s[2]; printf("%d", sizeof(s)); }
```

a) 1
b) 8
c) 10
d) 0

Answer: d

```
26. What is the output of the program? #include <stdio.h> struct point { int x; int y; }; void foo(struct point*); int main() { struct point p1[] = {1, 2, 3, 4}; }
```

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```
foo(p1); } void foo(struct point p[]) { printf("%d\n", p[1].x); }
```

a) Compile time error
b) 3
c) 12
d) 4

Answer: b

27. What is the output of the program?

```
#include <stdio.h> struct point { int x; int y; }; void foo(struct point*); int main() { struct point
```

```
p1[] = {1, 2, 3, 4};foo(p1);}void foo(struct point p[]){printf("%d\n", p->x);}
```

a) 1b) 2c) 4d) Run time error

Answer: a

28.What is the output of theprogram?#include <stdio.h>struct point{int x;int y;

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```
};void foo(struct point*);int main(){struct point p1[] = {1, 2, 3, 4};foo(p1);}void  
foo(struct point p[]){printf("%d %d\n", p->x, ++p->x);}
```

a) 2 4b) 2 2c) Compile time error d) Undefined

Answer: b

29.What is the output of the program?#include <stdio.h>struct point{int x;int y;} p[] = {1,
2, 3, 4, 5};void foo(struct point*);int main(){foo(p);}void foo(struct point p[]){printf("%d
%d\n", p->x, p[2].y);}

a) 1 0b) Run time errorc) garbage valued) Undefined

Answer: a

30.What is the output of the program?#include <stdio.h>struct point

[Page 120]

```
{int x;int y;};void foo(struct point*);int main(){struct point p1[] = {1, 2, 3, 4,  
5};foo(p1);}void foo(struct point p[]){printf("%d %d\n", p->x, p[3].y);}
```

a) Run time errorb) 2 0c) 1 some garbage valued) None

Answer: c

31.What is the output of the program?#include <stdio.h>struct point{int x;int y;};void
foo(struct point*);int main(){struct point p1[] = {1, 2, 3, 4, 5};foo(p1);}void foo(struct
point p[]){printf("%d %d\n", p->x, (p + 2).y);}

a) Compile time errorb) 2 0c) garbagevalued) Undefined

Answer: a

[Page 121]

```
32.What is the output of the program?#include <stdio.h>struct point{int x;int y;};void  
foo(struct point*);int main(){struct point p1[] = {1, 2, 3, 4, 5};foo(p1);}void foo(struct  
point p[]){printf("%d %d\n", p->x, (p + 2)->y);}
```

a) Runtime errorb) 1 0c) garbagevalued) undefined

Answer: b

33.What is the output of the program?#include <stdio.h>struct student{char *c;};void main()
{struct student s[2];printf("%d", sizeof(s));}a) 2b) 3c) 1d) 8

Answer: d34.What is the output of the program?#include <stdio.h>struct p

[Page 122]

```
{int x;char y;};int main(){struct p p1[] = {1, 92, 3, 94, 5, 96};struct p *ptr1 = p1;int x =  
(sizeof(p1) / 3);if (x == sizeof(int) + sizeof(char))printf("%d\n", ptr1-  
>x);elseprintf("falsen");}
```

a) Compile time errorb)Runtime errorc) Undefinedd) false

Answer: d

35.What is the output of the program?#include <stdio.h>struct p{int x;char y;};int main()

```
{struct p p1[] = {1, 92, 3, 94, 5, 96};struct p *ptr1 = p1;int x = (sizeof(p1) / sizeof(ptr1));if (x == 1)printf("%d\n", ptr1->x);elseprintf("false\n");}
```

a) Compile time errorb) Runtime errorc) falsed) Undefined

Answer: c

[Page 123]

36.What is the output of the program?#include <stdio.h>struct p{int x;char y;};typedefstruct p* q*;int main(){struct p p1[] = {1, 92, 3, 94, 5, 96};q ptr1 = p1;printf("%d\n", ptr1->x);}

a) Compile time errorb) Runtime errortc) Undefinedd) Segmentation fault

Answer: a

37.What is the output of the program?#include <stdio.h>struct p{int x;char y;};void foo(struct p*);int main(){typedefstruct p* q;struct p p1[] = {1, 92, 3, 94, 5, 96};foo(p1);}void foo(struct p* p1){ q ptr1 = p1;printf("%d\n", ptr1->x);}

a) Compile time errorb) Runtime errorc) Segmentation fault

[Page 124]

d) Undefined

Answer: a.

38.Which is an incorrect syntax for pointer to structure?(Assuming structtemp{int b;}*my_struct;)a) *my_struct.b = 10;b) (*my_struct).b = 10;c) my_struct->b = 10;d) Both *my_struct.b = 10; and (*my_struct).b = 10;

Answer: a

39.Which is an incorrect syntax to pass by referencea member of a structure?(Assume: structtemp{int a;}s;)a) func(&s.a);b) func(&(s).a);c) func(&(s.a));d) none of the mentioned

Answer: d

40.What is the output of the program?#include <stdio.h>struct temp{int a;} s;void change(struct temp);main(){s.a = 10;change(s);printf("%d\n", s.a);}void change(struct temp s){s.a = 1;}a) Output will be 20b) Output will be 10c) Output variesd) Compile time error

Answer: b

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/pyq txt/Alternate Question Bank with MCQs.txt =====

CATEGORY: PYQs

STATUS: ok

PROGRAMMING FOR PROBLEM SOLVING

Unit 1

Multiple Choice questions MCQ

1. Who is father of C Language? [CLO 1] [PL01]

A. Bjarne Stroustrup B. Dennis Ritchie

C. James A. Gosling D. Dr. E.F. Codd

2. C Language developed at _____? [CLO 1] [PLO 1]

- A. AT & T's Bell Laboratories of USA in 1972
- B. AT & T's Bell Laboratories of USA in 1970
- C. Sun Microsystems in 1973
- D. Cambridge University in 1972

3. For 16-bit compiler allowable range for integer constants is _____ ? [CLO 1] [PLO 2]

- A. -3.4e38 to 3.4e38
- B. -32767 to 32768
- C. -32768 to 32767
- D. -32668 to 32667

4. C programs are converted into machine language with the help of [CLO 1] [PLO 1]

- A. An Editor
- B. A compiler
- C. An operating system
- D. None of the above

5. A C variable cannot start with [CLO 1] [PLO 1]

- A. An alphabet
- B. A number
- C. A special symbol other than underscore
- D. both (b) and (c)

6. Which of the following is allowed in a C Arithmetic Instruction? [CLO 1] [PLO 1]

- A. []
- B. {}
- C. ()
- D. None of the above

7. Which of the following shows the correct hierarchy of arithmetic operations in C [CLO 1] [PLO 2]

- A. / + * -
- B. * - / +
- C. + - / *
- D. * / + -

8. Program which is written originally by the programmer is classified as [CLO 1] [PLO 1]

- A. object code
- B. machine code
- C. source program
- D. interactive programs

9. Data types are differed on the basis of [CLO 1] [PLO 1]

- A. the way of storage
- B. the type of operations
- C. the type of operators used
- D. both a and b

10. Loop statement which is repeated for some given number of times is classified as [CLO 1]

- A. FOR loop
- B. GO loop
- C. REPEAT loop
- D. GO REPEAT loop

11. Type of statement written in sequence and is repeated until the specific condition met is classified as [CLO 1] [PLO 1]

- A. format
- B. loop
- C. case
- D. condition

12. Size of an array is declared by [CLO 1] [PLO 1]

- A. programmer

B. program users

C. software

D. declared automatically

13. Programming language 'FORTRAN' stands for [CLO 1] [PLO 1]

A. formula translator

B. formula translation

C. free translator

D. free translation

14. Functions that are used in the programs and are defined by the programmers are called [CLO 1] [PLO 2]

A. program layout

B. program procedure

C. built-in functions

D. user-defined function

15. An assembler translates [CLO 1] [PLO 1]

A. machine code into assembly code

B. assembly code into machine code

C. processing time into manual time

D. routine into subroutine

16. Name given by a programmer to any particular data is classified as [CLO 1] [PLO 1]

A. identifier

B. identification

C. exponent

D. mantissa

17. When variable used in program is whole number, the variable is stored as [CLO 1] [PLO 2]

A. fixed string

B. integers

C. negative whole numbers

D. positive whole numbers

18. In programming, programmers use comments to [CLO 1] [PLO 1]

A. highlight program modules

B. explain module functions

C. explain used variables

D. all of above

19. Variable which uses the same name in whole program and in its all routines thus best classified as [CLO 1] [PLO 2]

A. middle variable

B. default variable

C. local variable

D. global variable

20. Statement which is used to make choice between two options and only option is to be performed is written as [CLO 1] [PLO 2]

A. if statement

B. if else statement

C. then else statement

D. else one statement

21. The_____ statement is used to transfer the control to the end of statement block in a loop:

a. Continue [CLO 1] [PLO 2]

b. Break

c. Switch

d. Goto

22. Which of the following is not a valid variable name declaration? [CLO 1] [PLO 2]

a) int _a3;

b) int a_3;

c) int 3_a;

d) int _3a

23. All keywords in C are in [CLO 1] [PLO 2]

a) LowerCase letters

b) UpperCase letters

c) CamelCase letters

d) None of the mentioned

24. Which of the following is true for variable names in C? [CLO 1] [PLO 2]

a) They can contain alphanumeric characters as well as special characters

b) It is not an error to declare a variable to be one of the keywords(like goto, static)

c) Variable names cannot start with a digit

d) Variable can be of any length

25. The format identifier '%i' is also used for _____ data type? [CLO 1] [PLO 2]

a) char

b) int

c) float

d) double

26. Which of the following is a User-defined data type? [CLO 1] [PLO 3]

a) typedef int Boolean;

- b) typedef enum {Mon, Tue, Wed, Thu, Fri} Workdays;
- c) struct {char name[10], int age};
- d) all of the mentioned

27. What is the output of this C code? [CLO 1] [PLO 3]

```
1. #include <stdio.h>
2. int main()
3. {
4. signed char chr;
5. chr = 128;
6. printf("%d\n", chr);
7. return 0;
8. }
```

- a) 128
- b) -128
- c) Depends on the compiler
- d) None of the mentioned

28. What is the output of this C code? [CLO 1] PLO 3]

```
1. #include <stdio.h>
2. int main()
3. {
4. j = 10;
5. printf("%d\n", j++);
6. return 0;
7. }
```

- a) 10 b) 11
- c) Compile time error d) 0

Explanation : j is not belongs to any data type

29. The following code 'for(; ;)' represents an infinite loop.

It can be terminated by. [CLO 1] [PLO 3]

- a) break b) exit(0)
- c) abort() d) all of the mentioned

30. The keyword 'break' cannot be simply used within: [CLO 1] [PLO 3]

- a) do-while b) if-else
- c) for d) while

31. Which keyword is used to come out of a loop only for that iteration? [CLO 1] [PLO 3]

- a) break b) continue
- c) return d) none of the mentioned

32. The first step in problem solving is ----- [CLO 1] [PLO 2]

- a) Understand the problem c) Identify the problem
- b) Developing algorithm/Flowchart d) Listing the possible outcome

33. The solutions which has series of actions to solve a problem are called ----- [CLO 1] [PLO 2]

- a) Heuristic solutions
- b) Algorithmic solutions

34. The solution can be reached by completing the actions in steps. These steps are called [CLO 1] [PLO 2]

- a) Sequence c) Algorithm
- b) Flowchart d) Steps

35. Solutions that cannot be reached through a direct set of steps are called ----- [CLO 1] [PLO 2]

- a) Algorithmic solutions b) Heuristic solutions

36. ----- means the outcome or the completed computer-assisted answer. [CLO 1] [PLO 1]

- a) Solution b) Result c) Program

37. The field of computer that deals with the problem of heuristic solution is called----- [CLO 1] [PLO 1]

- a) Artificial Intelligence
- b) Expert System c) Computer Management

38. ----- means the set of instructions that make up the solution after they have been coded into a programming language. [CLO 1] [PLO1]

a) solution b) result c) program d) error

39. ----- are organized facts. [CLO 1] [PLO1]

a) Data b) Information

40. A language applied for wide range of application is called as _____language[CLO 1][PLO 2]

a. Special purpose c)General purpose
b. Individual purpose d)Scientific purpose

41. Compiler converts the source code into _____ [CLO 1][PLO 4]

a. C code b. Byte code c. Object code d. Executable code

42. _____ converts source code to machine language one line at a time. [CLO 1][PLO 4]

a. Compiler b. Interpreter c. Assembler d.CPU

43. The program that converts high level language to a machine language is called _____

[CLO 1] [PLO 1]

a. Interpreter b. Linker c. Compiler d. Loader

44. ‘#’ symbol is known to be _____ [PLO 2] [CLO 1]

a. Linker b.Compiler c. Assembler d.Preprocessor directive

45. A _____ is a notational system for describing computations in both machine and human readable form [PLO 1] [CLO 1]

a. Programming language c.Machine language
b. High-level language d.Assembly language

46. ----- program converts assembly language into machine language [PLO 2] [CLO

1]

a. compiler b. interpreter c. assembler d. preprocessor directive

47. which symbol is been used for processing in flowchart? Ans: a [CLO 1] [PLO 3]

a) b) c) d)

48. which symbol is been used for input/output in flowchart? Ans: c [PLO 3] [CLO 1]

a) b) c) d)

49. which symbol is been used for decision making in flowchart? Ans: d [PLO 3] [CLO 1]

a) b) c) d)

50. which is used as start/stop symbol in flowchart? Ans:b [PLO 3] [CLO 1]

a) b) c) d)

51. ----- is a distinguishing characteristic of human excellence in every area of behavior

[CLO 1] [PLO 1]

a) creativity b) thinking c) visualization d) problem solving

52. .What is the value of x in this C code? [PLO 2] [CLO 1]

```
#include <stdio.h>
void main()
{
int x = 5 * 9 / 3 + 9;
}
```

(a). 3.75 b) Depends on compiler c) 24 d) 3

53.A character variable can store how many characters at a time? [PLO 2][CLO 1]

- (a) 1 character
- (b) 8 characters
- (c) 255 character

54. Which of the following is the correct way of writing comments? [PLO 2] [CLO 1]

- (a) `*/comments/*`
- (b) `/*comment*/`
- (c) `**comment**` (
- (d) `{comment}`

55. C programming language is [PLO 1] [CLO 1]

- (a) object oriented programming language
- (b) Procedure oriented programming language
- (c) function oriented programming language
- (d) None of the above

56. The memory space taken for a char type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes
- (c) 8 bytes
- (d) 1 byte

57. The memory space taken for a int type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes
- (c) 8 bytes
- (d) 10 bytes

58. The memory space taken for a float type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes
- (c) 8 bytes
- (d) 10 bytes

59. What is the only function all programs must contain ? [PLO 2] [CLO 1]

- (a)start()
- (b)system()
- (c)main()
- (d)program

60.For 16-bit compiler allowable range for integer constants is _____? [PLO 2] [CLO 1]

- (a) -3.4e38 to 3.4e38
- (b) -32767 to 32768
- (c) -32668 to 32667
- (d) -32768 to 32767

61.Every statement in C program is to be terminated by a_____ [PLO 2] [CLO 1]

- (a)dot(.)
- (b)semi-colon(;;)
- (c)colon(:)
- (d)Question mark(?)

62 The escape sequence „\b“ is a [PLO 2] [CLO 1]

- (a)back space
- (b)next line
- (c)tab
- (d)none of the above

63.The memory space taken for a long int type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes
- (c) 8 bytes
- (d)10bytes

64. which of the following will not valid expressions in C? [PLO2] [CLO 1]

- (a) a=2+(b=5);
- (b) a=11%3

- (c) a=b=c=5
- (d) b+5=2

65. The ----- operator is true only when both the operands are true. [PLO 2]
[CLO 1]

- a) &&
- b) ||
- c) !
- d) ?:

66. The ----- statement when executed in a switch statement causes immediate exit

from the structure. [CLO 1] [PLO 2]

- a) goto
- b) default
- c) break
- d) switch

66. The ternary conditional expression using the operator?: could be easily coded using ----

- statement [CLO 1][PLO 2]

- a) Nested if b) if-else
- c) if d) for

4 Marks :

Unit-1 CLO-1

1. Comment "C is mid level language"? [CLO-1][PLO 1]

2. What is problem solving? [CLO-1][PLO 1]

3. What are the six steps of problem solving? [CLO-1][PLO 1]

4. Discuss about how the problems can be solved with computers? [CLO-1][PLO 1]
5. What is a program? [CLO-1][PLO 2]
6. Why is problem analysis important? [CLO-1][PLO 2]
7. What are the tools of problem solving available? [CLO-1][PLO 2]
8. How do problem-solving tools help in leading to a solution? [CLO-1][PLO 2]
9. Why it is important to test a solution before coding it? [CLO-1][PLO 2]
10. What is an algorithm? Give the characteristics of the algorithm. [CLO-1][PLO 2]
11. What is a flowchart? Give the symbols/shapes used in the flowchart.
[CLO-1][PLO 3]
12. Define pseudocode and give its importance with an example. [CLO-1][PLO 2]
13. Discuss the difficulties with problem solving in detail. [CLO-1][PLO 2]
14. State the use of %d and %f .Write a printf statement in C using the above mentioned symbols? [CLO-1][PLO 4]
15. What is main difference between variable and constant? [CLO-1][PLO 3]
16. Explain bitwise left shift operator? [CLO-1][PLO 4]
17. Explain primary data types used in C? [CLO-1][PLO 3]
18. Difference between formatted & unformatted statement ? [CLO-1][PLO 3]
19. What is mean by storage class of variable? [CLO-1][PLO 3]
20. Explain with example ++i and i++. [CLO-1][PLO 4]
21. Refer all Elab programs. [CLO-1][PLO 3] [PLO 2][PLO 4]

12 Marks:

- 1) Explain the evolution of programming languages. [CLO-1][PLO 1]
- 2) Explain the various steps involved in problem solving with diagram. [CLO-1][PLO 1]
- 3) Draw the flowchart and write the algorithm and c code to find the sum and to reverse the digits of given five digit number. [CLO-1][PLO 1] [PL02]
- 4) Write an algorithm and draw a flow chart to find the factorial and Fibonacci series of given number. [CLO-1][PLO 2] [PLO 3]
- 5) Write a note on Algorithm, Flow chart and Pseudocode. [CLO-1][PLO 3] [PLO 2]
- 6) Explain the scope, lifetime of variable in C with example. [CLO-1][PLO 2][PL03]
- 7) Write down the algorithm to find the largest number among three given numbers and outline the steps in the algorithm with the inputs 5, 17, 3. [CLO-1][PLO 2][PL03]
- 8) Draw flowchart to compute the salary of an employee in a company. Assume that there are two types of employees in the company daily wages and regular. Salary is calculated as number of hours worked* wages per hour for daily wagers and basic pay +(% of DA * basic pay)/100 + HRA + medical allowance for regular employees. Sketch the flow of your design for a regular employee with basic pay = 5000, % of DA = 75 % and HRA = 500. (8)
[CLO-1][PLO 2][PL03][PL04]
- 9) Explain in details about operators with an example [CLO-1][PLO 1] [PLO 2][PL03]
- 10) Refer all elab Programs [CLO-1][PLO 3] [PLO 2][PLO 4]

UNIT - II

PART A

1. The ----- operator is true only when both the operands are true.

e) && b) || c)! d) ?:

Answer: a

2. The ----- statement when executed in a switch statement causes immediate exit

from the structure.

a)goto b) default c) break d) switch

Answer: c

3. The ternary conditional expression using the operator?: could be easily coded using -----

- statement

a)Nested if b) if-else c) if d) for

Answer: b

4. What will be the output when the following segment is executed?

```
Char ch='a';
```

```
Switch(ch)
```

```
{
```

```
case 'a':
```

```
Printf("A");
```

```
case 'b':
```

```
Printf("B");
```

```
Default:
```

```
printf("C");
```

```
}
```

a)A b)B c)C d) a

Answer: a

5. What will be the output of the following segment when executed?

```
int x=10, y=20;
```

```
if((x<y)|| (x+5)>10)
```

```
printf("%d",x);
```

```
else
```

```
printf("%d",y);
```

a)10 b) 20 c) 15 d)5

Answer: a

6. The ----- statement is used to skip a part of the statements in a loop.

- a) Continue b) break c) goto d) switch

Answer: a

7. A for loop with no test condition is known as ----- loop

- a) Infinite b) time delay c) for d) Incrementing

Answer: a

8. The sentinel -controlled loop is also known as -----loop

- a) Indefinite repetition loop c) time delay

- b) Definite repetition loop d) infinite

Answer: a

9. In an exit controlled loop the body of the loop is always executed minimum number of ----

- a) 1 time b) 2 times c) 3 times d) n times

Answer: a

10. The while is an ----- loop statement.

- a) Entry-controlled b) exit-controlled c) indefinite repetition

- d) definite repetition

Answer: a

11. The ----- specification is used to read or print integers

- a) h b) l c) L d) c

Answer: a

12. To print the data left-justified, must use ----- in the field specification

- a) - b) + c) / d) *

Answer: a

13. By default, the real numbers are printed with a precision of ----- decimal.

- a) 6 b) 2 c) 4 d) 0

Answer: a

14. The expression $!(x!=y)$ can be replaced by the expression-----

- a) $x!=y$ b) $x==y$ c) $x=!y$ d) $!x=!y$

Answer: c

15. In a counter controlled loop, variable known as ----- is used to count the loop

operations.

a) Counter b) sentinel c) i d) n

Answer: a

16. Which of the following special symbol allowed in a variable name?

(a) * (asterisk) (b) | (pipeline) (c) - (hyphen) (d) _ (underscore)

17 . Which of the following are invalid variable names?

a) Minimum b) n\$ c) Integer d) float

18. `int a=10;`

`++a;`

`a++;`

`Printf("%d",a);`

a) 10 b) 11 c) 12 d) 13

19. `int a=11;`

`a=a%2;`

`a=a/2;`

The value of a is,

a) 1, 1 b) 5, 1 c) 1, 5 d) 5

20. Which of the following is not logical operator?

A. &

B. &&

C. ||

D. !

21 Which of the following cannot be checked in a switch-case statement?

A. Character

B. Integer

C. Float

D. enum

22. What is the output of this C code?

```
int main()
{
    int a = 0, i = 0, b;
    for (i = 0; i < 5; i++)
    {
        a++;
        continue;
    }
}
```

A. 2 B. 3

C. 4 D. 5

23. What is the output of this C code?

```
void main()
{
    int i = 0, j = 0;
    for (i = 0; i < 5; i++)
    {
        for (j = 0; j < 4; j++)
        {
            if (i > 1)
                break;
        }
        printf("Hi \n");
    }
}
```

A. Hi is printed 5 times B. Hi is printed 9 times

C. Hi is printed 7 times D. Hi is printed 4 times

24. What is the output of this C code?

```
void main()
{
    int i = 0;
    int j = 0;
    for (i = 0; i < 5; i++)
    {
        for (j = 0; j < 4; j++)
        {
            if (i > 1)
                continue;
            printf("Hi \n");
        }
    }
}
```

- A. Hi is printed 9 times B. Hi is printed 8 times
C. Hi is printed 7 times D. Hi is printed 6 times

25. What is the output of this C code?

```
void main()
{
    int i = 0;
    for (i = 0; i < 5; i++)
    {
        if (i < 4)
        {
            printf("Hello");
            break;
        }
    }
}
```

- A. Hello is printed 5 times B. Hello is printed 4 times
C. Hello D. Hello is printed 3 times

27. What is the output of this C code?

```
int main()
```

```

{
int i = 0;
char c = 'a';
while (i < 2){
i++;
switch (c) {
case 'a':
printf("%c ", c);
break;
break;
}
}

```

```

printf("after loop\n");
}

```

A. a after loop B. a a after loop

C. after loop D. None of the mentioned

28. int main()

```

{
printf("before continue ");
continue;
printf("after continue\n");
}

```

A. Before continue after continue B. Before continue

C. after continue D. Compile time error

29. What is the output of the code given below?

```

int main()
{
printf("%d ", 1);
goto l1;
printf("%d ", 2);
l1:goto l2;
printf("%d ", 3);
l2:printf("%d ", 4);
}

```

A. 1 4 B. Compilation error

C. 1 2 4 D. 1 3 4

30. What is the output of code given below?

```
int main()
{
printf("%d ", 1);
11:12:
printf("%d ", 2);
printf("%d\n", 3);
}
```

A. Compilation error B. 1 2 3

C. 1 2 D. 1 3

31. What will happen if in a C program you assign a value to an array element whose subscript exceeds the size of array?

A. The element will be set to 0.

B. The compiler would report an error.

C. The program may crash if some important data gets overwritten.

D. The array size would appropriately grow.

32 .In C, if you pass an array as an argument to a function, what actually gets passed?

A. Value of elements in array

B. First element of the array

C. Base address of the array

D. Address of the last element of array

33. Result of a logical or relational expression in C is?

A. True or False

B. 0 or 1

C. 0 if expression is false and any positive number if expression is true

D. None of the mentioned

34. What will be the value of d in the following program?

```
int main()
{
    int a = 10, b = 5, c = 5;
    int d;
    d = b + c == a;
    printf("%d", d);
}
```

A. Syntax error

B. 1

C. 5

D. 10

35. What is the output of this C code?

```
int main()
{
    int a = 10, b = 5, c = 3;
    b != !a;
    c = !!a;
    printf("%d\t%d", b, c);
}
```

A. 5 1

B. 0 3

C. 5 3

D. 1 1

36. What is meaning of following declaration ?

```
int arr[20];
```

- a) Array of size 20 that can have integer address
- b) None of the above
- c) Integer array of size 20
- d) Array of size 20

37. In C Programming, If we need to store word "INDIA" then syntax is as below -

- a) `char name[6]={'I','N','D','I','A','\0'}`
- b) `char name[6]={"I","N","D","I","A"}`
- c) `char name[6]={'I','N','D','I','A'}`
- d) `char name[] ; name="INDIA"`

38. what is the way to initialize array?

- a) `int num[6] = {2,4,12,5,45,5};`
- b) `int n{} = {2,4,12,5,45,5};`
- c) `int n{6}={2,4,12};`
- d) `int n(6)={2,4,12,5,45,5};`

39. what will be the output of the program?

- a) 3, 2,15
- b) 2, 3, 20
- c) 2, 1, 15

d) 1, 2, 5

Answer: Option A

Solution:

>> int a[5] = {5, 1, 15, 20, 25}; The variable arr is declared as an integer array with a size of 5 and it is initialized to

a[0] = 5, a[1] = 1, a[2] = 15, a[3] = 20, a[4] = 25.

>> int i, j, m; The variable i, j, m are declared as an integer type.

>> i = ++a[1]; becomes i = ++1; Hence i = 2 and a[1] = 2

>> j = a[1]++; becomes j = 2++; Hence j = 2 and a[1] = 3.

>> m = a[i++]; becomes m = a[2]; Hence m = 15 and i is incremented by 1(i++ means 2++ so i=3)

>> printf("%d, %d, %d", i, j, m); It prints the value of the variables i, j, m

Hence the output of the program is 3, 2, 15.

39. what will be the output of the program?

a) 5

b) 6

c) 9

d) Error

e) None of the above

Answer: Option C

Solution:

x[i] is equivalent to *(x + i),

so (buf + 1)[5] is *(buf + 1 + 5), i.e. buf[6].

40) An array elements are always stored in _____memory locations

- a) sequential
- b) Random
- c) Sequential and Random
- d) None of the above

41) Let x be an array. which of the following operations are illegal?

- a) ++x b) x+1 c)x++ d) x*2

options:

- a) I and II
- b) I,II and III
- c) II and III
- d) I, III and IV
- e) III and IV

Answer: Option D

Solution: `int x[10]; * x` will store the base address of array. *
Statement I, III and IV is invalid.

Statement I and III : `++x` and `x++` are throwing an error while compile (lvalue required as increment operand)

Since, x is storing in the address of the array which is static value which cannot be change by the operand.

Statement IV : `x*2` is also throw an error while compile (invalid operands to binary * (have 'int *' and 'int'))

Statement II : `x+1` is throw a warning: assignment makes integer from pointer without a cast [enabled by

default]

42) what is the maximum number of dimensions an array in c may have?

- a) 2
- b) 8
- c) 20
- d) 50
- e) theoretically no limit. the only practical limits are memory size and compilers

43) size of the array need not be specified, when

- a) initialized is a part of definition
- b) it's a declaration
- c) it is a formal parameter
- d) All of these

44) what will be the output of the program?

- a) Equal
- b) Unequal

c) Error

d) None of these

Answer: Option B

Solution:

Strings are compared using strcmp() function defined under string.h header file.

45)

Unit-2 PART -B

1. Give the general syntax of conditional operator?
2. Define operator in C. What role an operator plays in C program?
3. Differentiate between relational and logical operators used in C?
4. Give the types of decision making statements
5. What is the general form of switch statement?
6. Differentiate do... while.. and while... loop.
7. Determine the output of the following and justify your answer

```
int main()
{
    int ch; printf("enter a value between 1 to 2:");
    scanf("%d", &ch);
    switch (ch, ch -1){
        case 1: printf("1\n"); break;
        case 2: printf("2"); break;
    } return 0; }
```

8. Explain the limitations of arrays.
9. Give the steps of looping process.
10. Brief the use of continue, goto, break statement and give the syntax.

11. What are loop control structures? Explain for loop, while loop and do-while loop with their syntax.
12. What do you mean by infinite loop? Give suitable of any infinite loop in a C program.
13. Explain break and continue statements with examples.
14. Determine the output of the following program justify your answer in a few words.

```
int main() {  
float f = 1.0;  
    switch(f) {  
  
case 1.0: printf("one"); break;  
case 2.0: printf("two"); break;  
default: printf("%f", f);  
  
}}
```

15. Write a program to print series of number divisible by 3 from 1 to 100 using for loop.
16. Write a C program to print 1, 2, 4, 8, 16, 32, 64N using do...while loop (Read N from user).
17. What are Arrays in C programming? Give the importance of Array in C language
13. What are the rules to declare one dimensional array?
14. What do you mean by compile time initialization? Give suitable example of Compile time initialization of C Array.
15. Describe the array index out of bound error in context of C array program.

1) Lucy is celebrating her 15th birthday. Her father promised her that he will buy her a new computer on her birthday if she solves the question asked by him. He asks Lucy to find whether the year on which she had born is leap year or not. Help her to solve this puzzle so that she celebrates her

birthday happily. If her birth year is 2016 and it is a leap year display 2016 is a leap year.? Else display 2016 is not a leap year and check with other leap year conditions

2) Write a C program to print the multiplication table of an integer n upto m rows using a while loop

sample input and output

INPUT

5

4

OUTPUT

1*5=5

2*5=10

3*5=15

4*5=20

3) Write a program to generate a following @s triangle:

@ @ @ @ @

@ @ @ @

@ @ @

@ @

@

4) Write a program that determines a student's grade. The program will read three scores and determine the grade based on the following rules: score =90% =>grade=A , score >= 70% and <90% => grade=B, score>=50% and <70% =>grade=C, score<50% =>grade=F

5) i)Why switch case is better than else..if... ladder. Justify.

- ii) Write a program to read a value (1-7) and print the equivalent day of the week (i.e. 1-SUN, 2-MON.....)
- iii) Explain briefly about for loop and Nested for loop with suitable example

iv) Write a program to print the following pattern.

```
12345
1234
123
12
1
```

6.a. i) Write a C program to insert an element at a specified position in the array.

7)) Write a program to generate a following patterns

a)

b)

c)

d)

e)

8) write c Program for Student mark list generation by using control and looping statements

9) write c program for digit rotation. "For any positive integer, we define a digit rotation as either moving the first digit to the end of the number (left digit rotation), or the last digit to the front of the number (right digit rotation). For example, the number 12345 could be left digit rotated to 23451, or right digit rotated to 51234.

10) Write a program to find the sum of positive numbers in an array

11) write c program for palindrome strings , two strings A and B, each consisting of lower case alphabets.

12) Write a arithmetic operation menu driven program using while(1)

13) i) Write short notes on switch case.

ii) Program to create a simple calculator, Performing addition, subtraction, multiplication, division depending the input from user.

14) i)write a C program to read 10 nos. and reverse it using array

ii)Concatenate two arrays of length minimum 5 numbers.

UNIT III

PART A

1) Which of the following is not possible statically in C?

- a) Jagged Array
- b) Rectangular Array
- c) Cuboidal Array
- d) Multidimensional Array

2) What is the output of this C code?

```
#include <stdio.h>
void main()

{
int a[2][3] = {1, 2, 3, 4, 5};
int i = 0, j = 0;
for (i = 0; i < 2; i++)
for (j = 0; j < 3; j++)
printf("%d", a[i][j]);
}
```

- a) 1 2 3 4 5 0
- b) 1 2 3 4 5 junk
- c) 1 2 3 4 5 5
- d) Run time error

3) Predict the output of below program:

```
#include <stdio.h>

int main()
{
    int arr[5];

    // Assume that base address of arr is 2000 and size of integer
    // is 32 bit
    arr++;
    printf("%u", arr);
}
```

```
    return 0;
}
```

- a) 2002
- b) 2004
- c) 2020
- d) lvalue required

4) Predict the output of below program:

```
#include <stdio.h>
```

```
int main()
{
    int arr[5];
    // Assume base address of arr is 2000 and size of integer is 32 bit
    printf("%u %u", arr + 1, &arr + 1);
    return 0;
}
```

- i. 2004 2020
- ii. 2004 2004
- iii. 2004 Garbage Value
- iv. The program fails to compile because address of operator cannot be used with array name

5) Size of the array need not be specified, when

- A. Initialization is a part of definition
- B. It is a declaratrion
- C. It is a formal parameter
- D. All of these

6) While passing an array as an actual argument, the function call must have the array name

A. with empty brackets

B. with its size

C. alone

D. none of the above

7) The parameter passing mechanism for an array is

A. call by value

B. call by value-result

C. call by reference

D. none of these

8) Under which of the following conditions, the size of an one-dimensional array need not be specified?

A. when initialization is a part of definition

B. when it is a declaration

C. when it is a formal parameter and an actual argument

D. All of the above

9) If a two dimensional array is used as a formal parameter, then

A. both the subscripts may be left empty

B. the first (row) subscript may be left empty

C. the first subscript must be left empty

D. both the subscripts must be left empty

11) Choose the statement that best defines an array

A. It is a collection of items that share a common name

B. It is a collection of items that share a common name and occupy consecutive memory location

C. It is a collection of items of the same type and storage class that share a common name and occupy consecutive memory locations

D. None of the above

12) Choose the correct statements

A. Array stores data of the same type

B. Array can be a part of a structure

C. Array of structure is allowed

D. All of the above

13) What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4.   int a[2][3] = {1, 2, 3, 4, 5};
5.   int i = 0, j = 0;
6.   for (i = 0; i < 2; i++)
7.     for (j = 0; j < 3; j++)
8.       printf("%d", a[i][j]);
9. }
```

a) 1 2 3 4 5 0

b) 1 2 3 4 5 junk

c) 1 2 3 4 5 5

d) Run time error

Answer: a

14) What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. int a[2][3] = {1, 2, 3, , 4, 5};
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
```

a) 1 2 3 junk 4 5

b) Compile time error

c) 1 2 3 0 4 5

d) 1 2 3 3 4 5

Answer: b

15. What is the output of this C code?

```
1. #include <stdio.h>
2. void f(int a[][3])
3. {
4. a[0][1] = 3;
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
10. void main()
11. {
12. int a[2][3] = {0};
13. f(a);
14. }
```

- a) 0 3 0 0 0 0
- b) Junk 3 junk junk junk junk
- c) Compile time error
- d) All junk values

Answer: a

16. What is the output of this C code?

```
1. #include <stdio.h>
2. void f(int a[][])

3. {
4. a[0][1] = 3;
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }

10. void main()
11. {
12. int a[2][3] = {0};
13. f(a);
14. }
```

- a) 0 3 0 0 0 0
- b) Junk 3 junk junk junk junk
- c) Compile time error
- d) All junk values

Answer: c

17. What is the output of this C code?

```
1. #include <stdio.h>
2. void f(int a[2][])
3. {
4. a[0][1] = 3;
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
10. void main()
11. {
12. int a[2][3] = {0};
13. f(a);
14. }
```

- a) 0 3 0 0 0 0
- b) Junk 3 junk junk junk junk
- c) Compile time error
- d) All junk values

Answer: c

18. Comment on the following statement:

```
int (*a)[7];
```

- a) An array "a" of pointers.
- b) A pointer "a" to an array.
- c) A ragged array.
- d) None of the mentioned

Answer: b

19. Comment on the 2 arrays regarding P and Q:

1. `int *a1[8];`
2. `int *(a3[8]);`
3. P. Array of pointers
4. Q. Pointer to an array

- a) a1 is P, a2 is Q
- b) a1 is P, a2 is P
- c) a1 is Q, a2 is P
- d) a1 is Q, a2 is Q

Answer: b

20. Which of the following is not possible statically in C?

- a) Jagged Array
- b) Rectangular Array
- c) Cuboidal Array
- d) Multidimensional Array

Answer: a

21. What is the correct syntax to send a 3-dimensional array as a parameter? (Assuming declaration `int a[5][4][3];`)

- a) func(a);
- b) func(&a);
- c) func(*a);
- d) func(**a);

Answer: a

22. Applications of multidimensional array are?

- a) Matrix-Multiplication
- b) Minimum Spanning Tree
- c) Finding connectivity between nodes
- d) All of the mentioned

Answer: d

23. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4.     int ary[2][3];
5.     foo(ary);
6. }
7. void foo(int *ary[])
8. {
9.     int i = 10, j = 2, k;
10.    ary[0] = &i;
11.    ary[1] = &j;
12.    *ary[0] = 2;
13.    for (k = 0; k < 2; k++)
14.        printf("%d\n", *ary[k]);
15. }
```

- a) 2 2
- b) Compile time error
- c) Undefined behaviour
- d) 10 2

Answer: a

24. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4.     int ary[2][3];
5.     foo(ary);
6. }
7. void foo(int (*ary)[3])
8. {
9.     int i = 10, j = 2, k;
10.    ary[0] = &i;
11.    ary[1] = &j;
12.    for (k = 0; k < 2; k++)
13.        printf("%d\n", *ary[k]);
14. }
```

- a) Compile time error
- b) 10 2
- c) Undefined behaviour
- d) segmentation fault/code crash

Answer: a

25. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4.     foo(ary);
5. }
6. void foo(int **ary)
7. {
8.     int i = 10, k = 10, j = 2;
9.     int *ary[2];
10.    ary[0] = &i;
11.    ary[1] = &j;
12.    printf("%d\n", ary[0][1]);
13. }
```

- a) 10
- b) 2
- c) Compile time error
- d) Undefined behaviour

Answer: d

26. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4.     int ary[2][3][4], j = 20;
5.     ary[0][0] = &j;
6.     printf("%d\n", *ary[0][0]);
7. }
```

- a) Compile time error
- b) 20
- c) Address of j
- d) Undefined behaviour

Answer: a

27. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {

4. int ary[2][3];
5. ary[][] = {{1, 2, 3}, {4, 5, 6}};
6. printf("%d\n", ary[1][0]);
7. }
```

a) Compile time error

b) 4

c) 1

d) 2

Answer: a

28. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. void foo();
5. printf("1 ");
6. foo();
7. }
8. void foo()
9. {
10. printf("2 ");
11. }
```

- a) 1 2
- b) Compile time error
- c) 1 2 1 2
- d) Depends on the compiler

Answer: a

29. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. void foo(), f();
5. f();
6. }
7. void foo()
8. {
9. printf("2 ");
10. }
11. void f()
12. {
13. printf("1 ");
```

```
14. foo();
15. }
```

- a) Compile time error as foo is local to main
- b) 1 2
- c) 2 1
- d) Compile time error due to declaration of functions inside main

Answer: b

30. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. void foo();
5. void f()
6. {
7. foo();
8. }
9. f();
10. }
11. void foo()
12. {
13. printf("2 ");
14. }
```

- a) 2 2
- b) 2
- c) Compile time error
- d) Depends on the compiler

Answer: d

Explanation: Even though the answer is 2, this code will compile fine only with gcc. GNU C supports nesting of functions in C as a language extension whereas standard C compiler doesn't.

31. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo();
3. int main()
4. {
5. void foo();
6. foo();
7. return 0;
8. }
9. void foo()
```

```
10. {  
11. printf("2 ");  
12. }
```

- a) Compile time error
- b) 2
- c) Depends on the compiler
- d) Depends on the standard

Answer: b

32. What is the output of this C code?

```
1. #include <stdio.h>  
2. void foo();  
3. int main()  
4. {  
5. void foo(int);  
6. foo(1);  
7. return 0;  
8. }  
9. void foo(int i)  
10. {  
11. printf("2 ");  
12. }
```

- a) 2
- b) Compile time error
- c) Depends on the compiler
- d) Depends on the standard

Answer: a

33. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo();
3. int main()
4. {
5. void foo(int);
6. foo();
7. return 0;
8. }
9. void foo()
10. {
11. printf("2 ");
12. }
```

- a) 2
- b) Compile time error
- c) Depends on the compiler
- d) Depends on the standard

Answer: b

34. What is the output of this C code?

```
1. include <stdio.h>
2. void m()
3. {
4. printf("hi");
5. }
6. void main()
7. {
8. m();
9. }
```

- a) hi
- b) Run time error

c) Nothing

d) Varies

Answer: a

35. What is the output of this C code?

```
1. #include <stdio.h>
2. void m();
3. void n()
4. {
5. m();
6. }
7. void main()
8. {
9. void m()
10. {
11. printf("hi");
12. }
13. }
```

a) hi

b) Compile time error

c) Nothing

d) Varies

Answer: b

36. What is the return-type of the function sqrt()

a) int

b) float

c) double

d) depends on the data type of the parameter

Answer: c

37. Which of the following function declaration is illegal?

- a)

```
double func();
int main(){
    double func();
}
```
- b)

```
double func(){};
int main(){
}
```
- c)

```
int main()
{
    double func();
}
double func(){//statements}
```
- d) None of the mentioned

Answer: d

38. What is the output of this code having void return-type function?

```
1. #include <stdio.h>
2. void foo()
3. {
4.     return 1;
5. }
6. void main()
7. {
8.     int x = 0;
9.     x = foo();
10.    printf("%d", x);
11. }
```

- a) 1
- b) 0
- c) Runtime error

d) Compile time error

Answer: d

39. What will be the data type returned for the following function?

```
1. #include <stdio.h>
2. int func()
3. {
4.     return (double)(char)5.0;
5. }
```

a) char

b) int

c) double

d) multiple type-casting in return is illegal

Answer: b

40. What is the problem in the following declarations?

```
int func(int);
double func(int);
int func(float);
```

a) A function with same name cannot have different signatures

b) A function with same name cannot have different return types

c) A function with same name cannot have different number of parameters

d) All of the mentioned

Answer: d

41. The output of the code below is

```
1. #include <stdio.h>
2. void main()
3. {
4. int k = m();
5. printf("%d", k);
6. }
7. void m()
8. {
9. printf("hello");
10. }
```

- a) hello 5
- b) Error
- c) Nothing
- d) Junk value

Answer: a

42. The output of the code below is

```
1. #include <stdio.h>
2. int *m()
3. {
4. int *p = 5;
5. return p;
6. }

7. void main()
8. {
9. int *k = m();
10. printf("%d", k);
11. }
```

- a) 5
- b) Junk value
- c) 0
- d) Error

Answer: a

43. The output of the code below is

```
1. #include <stdio.h>
2. int *m();
3. void main()
4. {
5.     int *k = m();
6.     printf("hello ");
7.     printf("%d", k[0]);
8. }
9. int *m()
10. {
11.     int a[2] = {5, 8};
12.     return a;
13. }
```

- a) hello 5 8
- b) hello 5
- c) hello followed by garbage value
- d) Compilation error

Answer: c

Part B

1. What is the need for user defined function?
2. Write a multi function program
3. List the Elements of user defined function
4. Give short note on 2-D array processing.
5. Write a C program to find the length of a string.
6. HELLO encode it as IFMMP using array
7. Show the difference between actual and formal parameter in function with piece of code
8. Compare user define function vs System define function
9. List categories of function
10. Explain the concept of function call by reference with a sample program.
11. Explain the concept of function call by value with a sample program
12. Explain about functions with example
13. Discuss Character Arrays
14. Predict output of the following program (CL03)

```
int main(){
    inta[][]={{1,2},{3,4}};

    int i,j;
    for (i=0;i<2;i++)
    for(j=0;j<2;j++)
    printf(“%d”,a[i][j]);
    return 0;
```

}

15. Why array index starts from zero?

16. Contrast function declaration vs function definition.

17. List the advantages of functions.

nd

18. Program to Find the 2 Largest Elements in an Array.

19. Write a program to convert the given string (srm university) Lower to upper case

20. Advantages and limitations of multi dimensional array initialization

21. Explain call by value and call by reference with an example

22. Common programming errors in 2D arrays

12. Explain String functions with example

PART C

1. Array construction for student mark list for 100 students. output need to display register number, marks of

five subjects , CGPA and PASS/Fail Status

2. Explain in details about String Functions: gets(), puts(), getchar(), putchar(), printf(), with an example

programs

3. i)Print the given pattern using 2-D array

1234567

23456

345

4

ii) Program to divide one array into two arrays

4. i) What are strings in C? Write a C program to read a string in lowercase and convert it

to
uppercase.

ii)) Illustrate call by value and call by reference with example for each.

5. i) Find the factorial of 10 using function recursion.

ii) What is a string? Explain any 5 string functions.

6. Explain in details about String Functions: atoi, strlen, strcat, strcmp with an example

7. Functions declaration and definition, Types: Call by Value & Call by Reference with example programs

8. Explain and write c program for Function with and without Arguments and no Return Values

9. Explain and write c program for Passing Array to Functions with return type, Recursion Functions

10. Write c program for Matrix Multiplication using Multi-dimensional array

11. Program to Delete duplicate elements from an array

INPUT

1 3 4 5 3

OUTPUT

1 3 4 5

b. Write a program to perform Matrix addition and Multiplication using 2-D arrays

12. i) State the importance of functions. List out the different types of Function

ii) Write a swap function using call by value and call by reference

iii) Write a C program to concatenate two strings.

iv) Write a program to read two strings and compare them and print a message that the first string

is equal, less or greater than the second one accordingly.

13. Write c program for Matrix addition and Matrix Transpose using multi dimensional array

14. Given a number , find whether it is a power of 2 or not

15. Write a C program to swap elements in cyclic order using call by reference.

UNIT 4

1. What is the output of this C code?

```
1. #include <stdio.h>
```

```
2. void foo(int*);
```

```
3. int main()
```

```
4. {
```

```
5. int i = 10;
```

```
6. foo(&i++);
```

```
7. }
```

```
8. void foo(int *p)
```

```
9. {
```

```
10. printf("%d\n", *p);
```

```
11. }
```

a) 10

b) Some garbage value

- c) Compile time error
- d) Segmentation fault/code crash

Answer: c

2. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo(int*);

3. int main()
4. {

5.     int i = 10, *p = &i;
6.     foo(p++);

7. }
8. void foo(int *p)

9. {
10.     printf("%d\n", *p);

11. }
```

- a) 10
- b) Some garbage value
- c) Compile time error
- d) Segmentation fault

Answer: a

3. What is the output of this C code?

```
1. #include <stdio.h>
```

```

2. void foo(float *);

3. int main()
4. {

5. int i = 10, *p = &i;
6. foo(&i);

7. }
8. void foo(float *p)

9. {
10. printf("%f\n", *p);

11. }

```

- a) 10.000000
- b) 0.000000
- c) Compile time error
- d) Undefined behaviour

Answer: b

4. What is the output of this C code?

```

1. #include <stdio.h>
2. int main()

3. {
4. int i = 97, *p = &i;

5. foo(&i);
6. printf("%d ", *p);

7. }
8. void foo(int *p)

9. {
10. int j = 2;

```

```
11. p = &j;  
12. printf("%d ", *p);  
  
13. }
```

- a) 2 97
- b) 2 2
- c) Compile time error
- d) Segmentation fault/code crash

Answer: a

5. What is the output of this C code?

```
1. #include <stdio.h>  
2. int main()  
  
3. {  
4. int i = 97, *p = &i;  
  
5. foo(&p);  
6. printf("%d ", *p);  
  
7. return 0;  
8. }  
  
9. void foo(int **p)  
10. {  
  
11. int j = 2;  
12. *p = &j;  
  
13. printf("%d ", **p);  
14. }
```

- a) 2 2

- b) 2 97
- c) Undefined behaviour
- d) Segmentation fault/code crash

Answer: a

6. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4.     int i = 11;

5.     int *p = &i;
6.     foo(&p);

7.     printf("%d ", *p);
8. }

9. void foo(int *const *p)
10. {

11.     int j = 10;
12.     *p = &j;

13.     printf("%d ", **p);
14. }
```

- a) Compile time error
- b) 10 10
- c) Undefined behaviour
- d) 10 11

Answer: a

7. What is the output of this C code?

```
1. #include <stdio.h>

2. int main()
3. {

4. int i = 10;
5. int *p = &i;

6. foo(&p);
7. printf("%d ", *p);

8. printf("%d ", *p);
9. }

10. void foo(int **const p)
11. {

12. int j = 11;
13. *p = &j;

14. printf("%d ", **p);
15. }
```

- a) 11 11 11
- b) 11 11 Undefined-value
- c) Compile time error
- d) Segmentation fault/code-crash

Answer: b

8. What is the output of the code below?

```
1. #include <stdio.h>
2. int main()
```

```

3. {
4. int i = 10;

5. int *const p = &i;
6. foo(&p);

7. printf("%d\n", *p);
8. }

9. void foo(int **p)
10. {

11. int j = 11;
12. *p = &j;

13. printf("%d\n", **p);
14. }

```

- a) 11 11
- b) Undefined behaviour
- c) Compile time error
- d) Segmentation fault/code-crash

Answer: a

9. Which of the following are correct syntaxes to send an array as a parameter to function:

- a) func(&array);
- b) func(#array);
- c) func(*array);
- d) func(array[size]);

Answer: a.

11. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. int k = 5;
5. int *p = &k;
6. int **m = &p;
7. printf("%d%d%d\n", k, *p, **m);
8. }
```

- a) 5 5 5
- b) 5 5 junk value
- c) 5 junk junk
- d) Run time error

Answer: a

12. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. int k = 5;
5. int *p = &k;
6. int **m = &p;
7. printf("%d%d%d\n", k, *p, **p);
8. }
```

- a) 5 5 5

- b) 5 5 junk value
- c) 5 junk junk
- d) Compile time error

Answer: d.

13. What is the output of this C code?

```
1. #include <stdio.h>

2. void main()
3. {

4. int k = 5;
5. int *p = &k;

6. int **m = &p;
7. **m = 6;

8. printf("%d\n", k);
9. }
```

- a) 5
- b) Compile time error
- c) 6
- d) Junk

Answer: c

14. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
```

```
3. {  
4. int a[3] = {1, 2, 3};  
  
5. int *p = a;  
6. int *r = &p;  
  
7. printf("%d", (**r));  
8. }
```

- a) 1
- b) Compile time error
- c) Address of a
- d) Junk value

Answer: b

15. What is the output of this C code?

```
1. #include <stdio.h>  
  
2. void main()  
3. {  
  
4. int a[3] = {1, 2, 3};  
5. int *p = a;  
  
6. int **r = &p;  
7. printf("%p %p", *r, a);  
  
8. }
```

- a) Different address is printed
- b) 1 2
- c) Same address is printed.

d) 1 1

View Answer

Answer: c

Explanation: None.

16. How many number of pointer (*) does C have against a pointer variable declaration?

a) 7

b) 127

c) 255

d) No limits.

Answer: d

17. What is the output of this C code?

```
1. #include <stdio.h>
```

```
2. int main()
```

```
3. {
```

```
4. int a = 1, b = 2, c = 3;
```

```
5. int *ptr1 = &a, *ptr2 = &b, *ptr3 = &c;
```

```
6. int **sptr = &ptr1; //-Ref
```

```
7. *sptr = ptr2;
```

```
8. }
```

a) ptr1 points to a

b) ptr1 points to b

c) sptr points to ptr2

d) None of the mentioned

Answer: b

18. What is the output of this C code?

```
1. #include <stdio.h>

2. void main()
3. {

4. int a[3] = {1, 2, 3};
5. int *p = a;

6. int **r = &p;
7. printf("%p %p", *r, a);

8. }
```

- a) Different address is printed
- b) 1 2
- c) Same address is printed.
- d) 1 1

Answer: c

PART B

- 1) short notes on Passing Array Element to Function with an example
- 2) Explain Formal and Actual Parameters
- 3) Give the Advantages of using Functions ,
- 4) example program for Processor Directives and #define Directives
- 5) write a program for Pointer Declaration and dereferencing,
- 6) write a program for pointers, Void Pointers and size of Void Pointers
7. Write algorithm and pseudo code for leap year calculation. [CLO 1]
8. Contrast the declaration of break and continue in while and for loop with justification

[CLO 2]

9. Limitations of two dimensional arrays.

PART C

- 1) i) c program to read array elements and print the values with their address
ii) what is Pointers and address operator, explain with an example of Size of Pointer Variable and
Pointer, Operator
- 2) Explain in details with an example Pointer Declaration and dereferencing, pointers, Void Pointers and
size of Void Pointers
- 3) write program for Arithmetic Operations, Incrementing Pointers
- 4) what is Constant Pointers, Pointers to array elements and strings
- 5) what is function pointer & Array of Function Pointers with an example
- 6) write the c program for Accessing Array of Function Pointers, Null Pointers
- 7) Justify and explain the different size of data types using pointer variables with example program.
- 8) i) write a c program to count vowels and consonants in a string using pointer
ii) Write call by reference and call by value with an example program

UNIT 5

Part A

1. Which of the following are themselves a collection of different data types?

- a) string
- b) structures
- c) char
- d) all of the mentioned

View Answer

Answer: b

Explanation: None.

2. User-defined data type can be derived by_____

- a) struct
- b) enum
- c) typedef
- d) all of the mentioned

Answer: d

3. Which operator connects the structure name to its member name?

- a) -
- b) <-
- c) .

d) Both <- and .

Answer: c

4. Which of the following cannot be a structure member?

a) Another structure

b) Function

c) Array

d) None of the mentioned

Answer: b

5. Which of the following structure declaration will throw an error?

a) struct temp{s;
main(){}

b) struct temp{};
struct temp s;
main(){}

c) struct temp s;
struct temp{};
main(){}

d) None of the mentioned

Answer: d

6. What is the output of this C code?

1. #include <stdio.h>

2. struct student

3. {

```
4. int no;
5. char name[20];

6. }
7. void main()

8. {
9. struct student s;

10. s.no = 8;
11. printf("hello");

12. }
```

- a) Compile time error
- b) Nothing
- c) hello
- d) Varies

Answer: a

7. What is the output of this C code?

```
1. #include <stdio.h>

2. struct student
3. {

4. int no = 5;
5. char name[20];

6. };
7. void main()

8. {
9. struct student s;

10. s.no = 8;
```


11. printf("hello");

12. }

a) Nothing

b) Compile time error

c) hello

d) Varies

Answer: b

8. What is the output of this C code?

1. #include <stdio.h>

2. struct student

3. {

4. int no;

5. char name[20];

6. };

7. void main()

8. {

9. student s;

10. s.no = 8;

11. printf("hello");

12. }

a) Nothing

b) hello

c) Compile time error

d) Varies

Answer: c

9. What is the output of this C code?

```
1. #include <stdio.h>

2. void main()
3. {

4. struct student
5. {

6. int no;
7. char name[20];

8. };
9. struct student s;

10. s.no = 8;
11. printf("%d", s.no);

12. }
```

- a) Nothing
- b) Compile time error
- c) Junk
- d) 8

Answer: d

10. Can the above code be compiled successfully?

```
1. #include <stdio.h>

2. struct p
3. {
```

```
4. int k;  
5. char c;  
  
6. float f;  
7. };  
  
8. int main()  
9. {  
  
10. struct p x = {.c = 97, .f = 3, .k = 1};  
11. printf("%f\n", x.f);  
  
12. }
```

- a) Yes
- b) No
- c) Depends on the standard
- d) Depends on the platform

Answer: c

11. What is the output of this C code?

```
1. #include <stdio.h>  
  
2. void main()  
3. {  
  
4. struct student  
5. {  
  
6. int no;  
7. char name[20];  
  
8. };  
9. struct student s;
```

```
10. no = 8;  
11. printf("%d", no);  
  
12. }
```

- a) Nothing
- b) Compile time error
- c) Junk
- d) 8

Answer: b

12. Number of bytes in memory taken by the below structure is

```
1. #include <stdio.h>  
  
2. struct test  
3. {  
  
4. int k;  
5. char c;  
  
6. };
```

- a) Multiple of integer size
- b) integer size+character size
- c) Depends on the platform
- d) Multiple of word size

Answer: a

13. What is the output of this C code?

```
1. #include <stdio.h>
2. struct
3. {
4. int k;
5. char c;
6. };
7. int main()
8. {
9. struct p;
10. p.k = 10;
11. printf("%d\n", p.k);
12. }
```

- a) Compile time error
- b) 10
- c) Undefined behaviour
- d) Segmentation fault

14. What is the output of this C code?

```
1. #include <stdio.h>
2. struct
3. {
4. int k;
5. char c;
6. } p;
7. int p = 10;
8. int main()
9. {
```

```
10. p.k = 10;
11. printf("%d %d\n", p.k, p);

12. }
```

- a) Compile time error
- b) 10 10
- c) Depends on the standard
- d) Depends on the compiler

Answer: a

15. What is the output of this C code?

```
1. #include <stdio.h>

2. struct p
3. {

4. int k;
5. char c;

6. };
7. int p = 10;

8. int main()
9. {

10. struct p x;
11. x.k = 10;

12. printf("%d %d\n", x.k, p);
13. }
```

- a) Compile time error
- b) 10 10

- c) Depends on the standard
- d) Depends on the compiler

Answer: b

16. What is the output of this C code?

```
1. #include <stdio.h>
2. struct p
3. {
4. int k;
5. char c;
6. float f;
7. };
8. int p = 10;
9. int main()
10. {
11. struct p x = {1, 97};
12. printf("%f %d\n", x.f, p);
13. }
```

- a) Compile time error
- b) 0.000000 10
- c) Somegarbage value 10
- d) 0 10

Answer: b

17. What is the output of this C code(according to C99 standard)?

advertisement

```
1. #include <stdio.h>
2. struct p
3. {
4. int k;

5. char c;
6. float f;

7. };
8. int main()

9. {
10. struct p x = {.c = 97, .f = 3, .k = 1};

11. printf("%f\n", x.f);
12. }
```

- a) 3.000000
- b) Compile time error
- c) Undefined behaviour
- d) 1.000000

Answer: a

18. What is the output of this C code(according to C99 standard)?

```
1. #include <stdio.h>
2. struct p
3. {
4. int k;

5. char c;
```



```

6. float f;

7. };

8. int main()

9. {
10. struct p x = {.c = 97, .k = 1, 3};

11. printf("%f \n", x.f);
12. }

```

- a) 3.000000
- b) 0.000000
- c) Compile time error
- d) Undefined behaviour

Answer: b

19. What is the output of this C code(according to C99 standard)?

```

1. #include <stdio.h>
2. struct p
3. {
4. int k;

5. char c;
6. float f;

7. };

8. int main()

9. {
10. struct p x = {.c = 97};

11. printf("%f\n", x.f);
12. }

```

- a) 0.000000

- b) Somegarbagevalue
- c) Compile time error
- d) None of the mentioned

Answer: a

PART B

1. Write algorithm and pseudo code for leap year calculation.
2. Contrast the declaration of break and continue in while and for loop with justification
3. Limitations of two dimensional arrays.
4. Explain about nested pre-processor MACRO
5. Categorize the basic operations that can be performed on a file with suitable declarations.
6. Write a c program to demonstrate double pointer for accessing the value of another pointer
7. Explain about file operations and mode with syntax.
- 8) explain shortly about Initializing Structure, Declaring structure, variable,
- 9) give the example of Structure using typedef, Accessing members
- 10) what is Nested structure, Accessing elements in a structure array,
- 11) write c program for Array of structure, Accessing elements in a structure array
12. Explain about file operations and mode with syntax.
13. Give a brief note about preprocessor directives.
14. Differentiate between a union and a structure.
15. Write a program to read a file character by character, and display it simultaneously on the screen

- 16) write a program for Passing Array of structure to function,
- 17) write a program for Array of pointers to structures
18. Explain about nested pre-processor MACRO
19. Categorize the basic operations that can be performed on a file with suitable declarations.
20. Write a c program to demonstrate double pointer for accessing the value of another pointer

PART C

- 1) explain in details about Bit Manipulation to structure and Pointer, to structure, Union Basic and declaration with an example
- 2) explain in details about Accessing Union Members Pointers to Union, Dynamic memory allocation, malloc, realloc, free with an example
- 3) explain in details about Allocating Dynamic Array, Multidimensional array using dynamic memory allocation with an example
- 4) Explain in details about array of structures and accessing elements in a structure array.
- 5) Write a c program to insert a line at the end of text file
- 6) write a program to copy the content from one file to another file
- 7) explain and write the program for library management system using union
- 8) write a c program for file: opening, defining, closing, File Modes, File Types, Writing contents into a file
- 9) write a c program for Reading file contents, Appending an existing file
- 10) Differentiate between gets() and scanf()
- (11) Write a program to convert the given string "hello world" to "dlrow olleh".
12. Write a program that passes a pointer to a structure to a function

13 Briefly discuss about file operations in C

14) c program to read array elements and print the values with their address

32(a) Write a program that passes a pointer to a structure to a function

15) Justify and explain the different size of data types using pointer variables with example program.

[CLO 5]

(OR)

16) write a c program to count vowels and consonants in a string using pointer

32(b) Briefly discuss about file operations in C [CLO 5]

17) Write call by reference and call by value with an example program

18) a) i) Explain in details about array of structures and accessing elements in a structure array.

ii) Write a c program to insert a line at the end of text file

19) i) write a program to copy the content from one file to another file

ii) explain and write the program for library management system using union

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/pyq txt/Important Questions.txt =====

CATEGORY: PYQs

STATUS: ok

PPS IMPORTANT TOPIC

Note: Prepare all topics from all the unit, but concentrate more on the following topics

UNIT-1

☐ Compare compiler and interpreter.

☐ What is algorithm? How it is used as a problem-solving tool.

☐ What is pseudo code? How it is used as a problem-solving tool.

☐ Writing algorithms, Pseudo code and Flow Chart for

o Fibonacci Series (OR)

- o Check Number Is Armstrong or Not (OR)
- o Check a Number is Palindrome or Not (OR)
- o Find Factorial of a Number (OR)
- o convert temperature from Fahrenheit to Celsius (OR)
- o convert temperature from Celsius to Fahrenheit (OR)
- o prime number or not (OR)
- o sum of even digits in a number in c (OR)
- o sum of odd digits in a number in c (OR)
- o sum of natural numbers (OR)

▣ What is a token? What are different types of tokens available in C language? Explain. (explain the following topic with example Keywords, Identifiers, Constants and Special Characters)

▣ Unformatted Input/Output functions.

▣ Storage Class (Auto, static, extern, register)

▣ Operators (all operators with example)

▣ Operator Precedence (They will give some expression, you have to solve expression using operator precedence rules)

▣ Write C program to read a radius of a Circle as an input and Calculate its Diameter, Circumference, and Area.

▣ Write a C program to swap two numbers without using third variable.

▣ Write a C program to enter length and width of a rectangle as an input and calculate its perimeter and its area.

▣ Write a C program to swap two numbers using bitwise operator.

▣ Read two angle of a triangle as an input and find the third angel. And also read base and height of the triangle as an input from the user and find area of a triangle.

▣ Write a C program to convert temperature from Fahrenheit to Celsius.

▣ Write a C program to enter basic salary and prepare pay slip using following data. TA = 15%

of basic, HRA = 8% of basic, Grade Pay = 9000, and PF = 15% of basic. Calculate Gross salary, PF and Net salary and display them.

UNIT-2

▣ Explain in detail about Conditional Control Statements. (Simple if, if...else, else if and nested if, Switch case with example)

▣ Explain in detail about Looping Control Statements. (for, While, do.while with example)

❑ Write a c programming to

- o Fibonacci Series (OR)
- o Check Number Is Armstrong or Not (OR)
- o Check a Number is Palindrome or Not (OR)
- o Find Factorial of a Number (OR)
- o prime number or not (OR)

- o sum of even digits in a number in c (OR)
- o sum of odd digits in a number in c (OR)
- o sum of natural numbers (OR)
- o Find Factorial of a Number Using Recursion. (OR)
- o They will give some pattern.

❑ Write a c program for

- o Matrix Addition Program
- o Matrix Subtraction Program
- o Matrix Multiplication Program
- o Sorting element
- o To Insert An Element Desired or Specific Position In An Array
- o To Delete Element From Array At Desired Or Specific Position

❑ Pointer

❑ What is dynamic memory allocation? Write and explain the different

dynamic memory allocation functions in C

UNIT-3

- ❑ Explain string function with an example. (Prepare all string function with an example).
- ❑ C Program to Count Vowels, Consonants, Digits and Spaces in Given String or

Remove all characters in a string except alphabets or Reverse a string using recursion or find index of character.

- ❑ Explain Call by Value and Call by Reference with an example.
- ❑ C function argument and return values. (explain Four types with an example)

UNIT-4

- ❑ Basic Data Types in Python.
- ❑ Explain in detail about Conditional Control Statements in python
- ❑ Explain in detail about Looping Control Statements in python.

- ☐ List function and methods.
- ☐ Tuple function and methods.
- ☐ Set methods.
- ☐ Dictionaries methods.

UNIT-5

- ☐ NumPy ndarray
- ☐ Slicing arrays in NumPy
- ☐ Dealing with Rows and Columns in Pandas.
- ☐ Working with Missing Data
- ☐ Applying Functions to Data frames
- ☐ Comparison between Numpy and Pandas
- ☐ Other Python Libraries

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/pyq txt/PPS CT1 Papers
24-09-2025.txt =====
CATEGORY: PYQs
STATUS: ok

- B) 123
- C) "Hello"
- D) main

4 Which storage class in C is used to declare variables that are local to a function and retain their values between function calls? 131

- A) auto
- B) register
- C) static
- D) extern

If int x4,, result of x<<I is: 1.3.1

- A) 2
- B) 8
- C) 6
- D) 10

Which of the following is a valid single-line comment in C? 1 13.1

- A) This is a comment /1
- B) /This is a comment

MThis is a comment

D) <- This is a comment-->

What is the output of the program 2 2 2 2.1.2

```
kinclude <stdio.h>
```

```
int main) {
```

```
int x 2:
```

```
switch (x) {
```

```
case 1: printf("One"%
```

```
break:
```

```
case 2: printf("T wo"):
```

```
break:
```

```
case 3: printf("Three"):
```

```
break.
```

```
default: printf("Default"):
```

```
return 0:
```

A) One

B) Two

C) Three

D) Error

8 What will be printed for the code snippet: 2 2 1 13.1

```
int i:
```

```
for(i =0; ii<5;itt:
```

```
printf("%d", ):
```

A) 0

B) 4

C) 5

D) Garbage value

9 How many times will "C Programming" be printed? 1 22 2 2.1.2

```
Winclude <stdio.h>
```

```
int main()
```

```
int k=2.
```

```
do
```



```
printf("C Programmingin");
```

```
k-
```

```
if(0
```

```
goto stop;
```

```
while(k>-1):
```

```
stop:
```

```
retun 0:
```

A) One time

B) Two times

C) Three times

D) Infinite times

10 Guess the output of the following output: 2 2 1 L3.1

```
#include <stdio.h>
```

```
int main) {
```

```
int a - 5, b 0;
```

```
ifla && b)
```

```
printf("Trucn");
```

```
else
```

```
printf("Falscin"): 
```

```
return 0:
```

A) True

B) False

C) Compilation error

D) No output

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OSRM Science and Technology

College of Engineering and Teehnology SET C

School of Computing

SRM Nagar, Kattankulathr 603203. Chengalpattu Disarict. Tamilnadu

Acndemie Year: 2025-2026 (ODD)

Test: CT I

Course Code & Title: 21CSSIOLI Date: 24.09, 2025

Year & Sem: 1Year /I Sem Programming for Problem Solving Duration: 1 Period
Max. Marks: 25

Part- B
(3x15 Marks)

Answer any three questions out of four:

11 Write a program to illustrate the difference between logical and bitwise operators in C.

12 An Robotics project requires checking whether a number is odd or even using only bitwise operators. Write a C program for the same.

13 An amusement park has three types of rides based on height. Many wants to know if she can go on the Adventure Ride. Her height is 140 cm. Write a C program to decide her eligibility.

Conditions:

"Eligible for Giant Ride, if height > 160 cm
"Eligible for Adventure Ride, if height > 120 cm and < 160 cm
Eligible for Kids Ride, if height < 120 cm

14 A developer is assigned the task of implementing a key feature for a mobile calculator application. The application should provide users with the ability to select an arithmetic operation-such as addition, subtraction, multiplication, or division and perform that operation on two user-input numbers. The developer is required to write a program in C that utilizes a switch statement to construct this simple calculator functionality, enabling users to choose their desired mathematical operation.

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions

CO Distribution Bloom's level Distribution

L

CO1 CO2 CO3 CO4 CO5

Course Articulation Matrix

SNo Course POI P02 P03 P04 POS P06 P07 POS POP P010 P0IL P0I2

Outcone

COI 2

2 C02 2

3 C03 2

4 C04 2 3

COS 2

Instructions:

MCQ sheet to be handed over to the faculty after the first 15 minutes.

MOBILE PHONES/SMART GADGETS STRICTLY NOT PERMITTED.

DO NOT WRITE or MARK anything in the Question Paper (Except Registration Number)

Discussion with your classmates is considered as MALPRACTICE, and it results in 0° Mark.

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Question Paper should be submitted along with Answer Booklet.

Course Outcomes (CO):

C01: Solve problems through computer programming. Express the basic data types and variables in C

C02: Use appropriate data types in simple data processing applications. To create programs using the concept of

arrays.

Part - A

(1x 10 - 10 Marks)

Multiple Choice QQuestions

Marks BL CO PO PI

Question

Q. Code

No 13.1

Predict the output of the following C code snippet:

```
int a2, b3;  
printf("%d", a && b):
```

- A) 2
- B) 3
- C) 1
- D) 0 2 2.1.2

2 Which of the following is a keyword in C?

- A) main
- B) define
- C) printf
- D) double 2 1.3.1

Which is valid C expression?

3

A) int mynum 100,000:

B) int ny num- I00000

C) int ny num 1000,

D) int Smy num 10000:

4 What will be printed after executing the below code snippet 2 131

```
int x7;  
printf("%d",x-.
```

- A) 7
- B) 6
- C) Error
- D) Undefined

What will be the output of the following C code? 13.1

```
#include <stdio.h>  
int main)
```

```
int main 330;  
printf("d", main).  
return 0:
```

1

- A) It will cause a compile-time error
- B) It will cause a run-time error
- C) It will run without any error and prints 3
- D) It will experience infinite looping

6 Which of the following is NOT true about pseudocode? 1.3.1

- E) It uses natural language mixed with programming constructs
- F) It is not executed on computers
- G) It helps in designing algorithms
- H) It requires strict syntax rules like C

7 The following loop executes exactly 5 times: 2 2 2 2.1.2

```
for(int i=0; i<5; i++) {  
    //body  
}
```

- A) True, because it runs from 0 to 4.
- B) False, because it actually executes 6 times (0 to 5).
- C) True, because condition allows only 5 iterations.
- D) False, because loop never runs.

Fill in the blank so the while loop prints numbers from 0 to 4. 1 2 2 1 1.3.1

```
int i = 0;  
while ( ) {  
  
    printf("%d ", i);  
}
```

9 keyword is used to exit from a loop immediately? 2 2 2 2.1.2

- A) continue
- B) break
- C) exit
- D) return

10 Pick the correct statement about the continue statement. 2 2 1.3.1

- A) It skips the remaining statements in the loop and moves to the next iteration
- B) It terminates the loop immediately
- C) It is used to exit from a switch-case
- D) It ends the program execution

SRM

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College of Engineering and Technology SETE
School of Computing

SRM Nagar, Kattankolattur 603203, Chengalpattu District Tamil Nadu
Academic Year 2025-2026 (ODD)

Test: CTI

Course Code & Title: 21CSSIOLJ Dates 24.09.2025

Year & Sem: I Year / LSSem Programming for Problem Solving Duration 1 Period

Max. Marks: 25

Part-B

(3x5=15 Marks)

Answer any three questions out of four:

In C language, illustrate how storage classes handle the variables, in 5 marks.
terms of scope and visibility. along with program code.

12 Write a pseudocode to convert Celsius to Fahrenheit temperature 3 marks.
Prompt the user to enter the temperature (in Celsius). Explain how
pseudocode makes it easy to translate the logic into a C program.

13 An HR manager wants to automate the employee performance 3 marks.
evaluation process in C. The company follows this grading policy:

Score >90 Outstanding

Score 75-89 Excellent

Score 60-74 Good

Score 40-59 Needs Improvement

Score < 40 Poor

Help the HR manager, by writing a C program using an else-if ladder
that accepts an employee's score as input and displays their
performance rating.

14 Ravi is building a small banking application in C. As part of his 5 marks.
project, he needs to implement a feature where the digits of a

customer's account number must be reversed for encryption before
storage (Eg: 1234 → 4321). Ravi decides to use a while loop to solve

this. Help Ravi to solve this, using a C program.

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions

Bloom's level Distribution

CO Distribution

BLS

COL C02 C05 C0 C05

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Course Code & Title: 21CSSIoL Programming for

Year & Sem: tYear/ Sem Problem Solving Duration/ Period

Max. Marks 25

Course Articulation Matrix

S.No Course PO1 PO2 PO3 PO4 PO5 PO6 PO7 PO8 PO9 PO10 PO11 PO12

Outcome

C01

C02 3

C03

C04

C05

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Course COuteomes (CO:

co: Solve problems through computer programming. Express the basic data types and variables in C.

coZ: Use appropriate data types in simple data processing applications. To create programs using the concept of

arrays.

Part- A

(1x 10- 10 Marks)

Multiple Choice Questions

Q. Question Marks BL CO PO PI

No Code

Evaluate and find the valuc of e in the expression: 1 1.3.1
 inte\$+ 8/4*3- 2;

- A) 7
- B) 9
- C) 10
- D) 8

2 What will be the output? 2 1 2 2.1.2

```
#include<stdio.he
int main()
```

```
int a6;
int b a--2;
printf("6d", b):
return 0;
```

- B) 3
- C) 2
- D) -1

Guess the output of the below code: 1 2 1 1 13.1

```
#include<stdio.h>
int main0
```



```
double x=78.999;
printf("%.2f", x);
return 0;
```

- A) 78.99
- B) 79.00
- C) 78.90
- D) Error

4 Predict the output of the following code snippet

```
int p=13;
printf("%d", p);
```

- A) 02
- B) 12
- C) 11
- D) 01

5 The sizeof() operator in C returns:

- A) Number of digits in a value
- B) Size of memory in bits
- C) Size of memory in bytes
- D) Address of variable

6 The keyword used to define a constant in C is

- A) const
- B) #define
- C) Both A and B
- D) Neither A nor B

What will be the output of the following code snippet?

```
int x=10;
if(x<5)
printf("Hello");
```

else

```
printf("World");
```

- A) Hello
- B) World
- C) HelloWorld
- D) No output

How many times, value of "i is checked in the following program 2 2 1 13.1

```
#include <stdio.h>
```

```
int main(
```

```
int i=0:
```

```
while (i<3)
```

```
printf ("ln while loop\n");
```

Part- B

(3x5=15 Marks)

Answer any three questions out of four:

11 Develop an algorithm, pseudocode and flowchart for finding the 2 13.1

factorial of the given number.

12 Write a C program that Swaps two variables without using temporary 5 3 1 L3.1

variables.

13 Fill the code given and explain how the desired output will be 5 3 2 13.1

generated.

```
#include <stdio.h>
```

```
int main(){
```

```
int num, sum = 0;
```

```
printf("Enter a number: ");
```

```
scanf( "%d",&num);
```

```
/*Loop to find divisors
```

```
for(int i = 1;i<
```

```
if
```

```
Sum = 0
```

```
if
```

```
printf("%d is a Perfect Number", num);
```

```
else
```

```
printf("%d is not a Perfect Number", num);
```

```
return 0;
```

- A) 2
- B) 1
- C) 3
- D) 4

Guess the output for following code snippet: 1 2 2 2 2.1.2

```
int x=3,  
switch ()
```

```
case 1: printf("A")%  
case 2: printf("B"):  
case 3: printf("C");  
default: printf("D"):
```

- A) CD
- B) C
- C) ABC
- D) BCD

Write the output for the following code snippet? 2 2 13.1

```
for (int i = 1; i <= 5; i++)  
if(i%3 == 0) continue;  
printf("%d. ", i);
```

- A) 12345
- B) 12
- C) 1245
- D) 345

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College of Engineering and Technology
School of Computing

SRM Nagar, Kattankulathur-603203, Chennai District, Tamil Nadu
Academic Year: 2025-2026 (ODD)

Test: CTI Date: 24.09.2025

Course Code & Title: 21CS501J - Programming for Problem Solving Duration: 1 Period
Year & Sem: I Year / I Sem Max. Marks 25

Course Articulation Matrik:

S.No Cours P01 P02 P03 P04 POS P06 P07 POS P09 P010 POIT P012

Outceme

COI 2

CO2

CO3

CO4 3

COs

Instructions:

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Course Outeomes (CO:

CO1: Solve problems through computer programming. Express the basic data types and variables in C.

CO2: Use appropriate data types in simple data processing applications. To create programs using the concept

of arrays.

Part - A

(1x 10- 10 Marks)

Multiple Choice Questions

Question Marks BL CO PO P

Q. Code

N

What is the default storage class of local variables in C? 13.1

A) static

B) auto

C) extern

D) register 2.1.2

2 What is the output of the following C code? 2

```
#include<stdio.h>
int main()

int shifty;
shifty = 0570;
shifty = shifty >> 4;

shifty shifty <<6;
printf("The value of shifty is %d, shifty);
```

E) The value of shifty is 1500

F) The value of shifty is 4300

G) The value of shifty is 700

H) The value of shifty is 2700

3 The following three C language statements is equivalent to which single statement? to which
yy+k

A) 2x y+2:

B) 2xt)+(44y):

C) z(xt+) +(yt+):

D) z(xt)+(ty) + 1:

4 Which of the following identifiers is invalid?

A) student 1 2 131

B) id

C) 9marks

D) firstName

5 Which statement is correct about string in C? 2 1 13.1

A) It is a keyword

B) it is stored as a character array ending with 0'

C) It can store numbers only

D) It has dynamic memory allocation by default

Which of the following produces compile-time error? 1 1 131

A) 45

B) x+

+x

D) xt-1

7 What is the output of the following C code? 1 2 2 2 2.1.2

```
#include<stdio.h>
```

```
int main0
```

```
int k;
```

```
for(k=0; k< 20; k=k+ 2)(
```

```
if(k %3 = 1)
```

```
printf("%d", k):
```

```
)
```

A) 41 6

B) 4 10 16

C) 0612 18

D) |47 10 13 16 19

2 2 13.1

8 With respect to following "for" loops in C, pick the best statement

Assume that there is a prior declaration of '' in all cases

```
for (1 < 10; i=0; i++)//)
```

```
for (i<1 0; it+;i= 0) //( )
```

```
for (=0;i < 10;i++) /I/ (H)
```

```
for (i = 0; t+;i<10) // (iv)
```

```
for (it+; i =0;i<10)// ( )
```

```
for (it*;i<0;i=10) //( v)
```

A) All the above for" loops would compile successfully

Except (iii), the behaviour of all the other "for" loops depend on compiler implementation.

B) Only (iii) would compile successfully.

C) Only (ii) and (iv) would compile successfully but

behaviour of (iv) would depend on compiler implementation.

D) All the above for" loops would compile successfully.

2 2 2.1.2

wing C code? 1 2

What is the output of the follo

```
#include<stdio.h>
```

```
int main(0
```

```
int index;
```

```
for(index= 1; index <=5; index+)(
```

```
printf("%d", index);
```

```
if (index 3)
```

```
continue:
```

A) 1245

B) 12345

C) 12245

D) 12354 2 2 13.1

ollowing statement is correct for switch controlling

10 Which of the f

expression?

can be empty as well.

A) "switch" control expression

l expression.

B) Only int can be used in "switch" contro

in "switch"

C) All types i.e. int, char and float can be used

control expression.

in "switch control

D) Both int and char can be used

expression.

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/pyq txt/PPS CT1 Papers
ALL SETS 2025.txt =====
CATEGORY: PYQs
STATUS: ok

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District.

SRM Chengalpattu
603203.

Kaltkulathur DD) Date: 24.09.2025

SRINgar. 2025- 2026 (0
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Academic Year: Dura

Max. Mlarks: 25
Solving

for Problem
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Test: CTI 2CSS0

Course Code & Title:
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Year &S em: 1 ear /|S em
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P09

Course Articulation Matui: P08
P06 P0

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Outcome

2

CO)

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after the first 15 min

Instructions:

NOT PERMITTED.

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SMARTADGLIS (Registration Number)

MOBILE PHONE Question Paper (Except Reg
Mark.

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Question Paper should be subm

types and variables in C.

Course Outcomes (CO): concept of arrays.

ramming. Express the basic data
ams using the co

gh computer prou

CO1: Solve problems throu ng applications. To create progr

ta types in sinple data processi

CO2:Use appoprhate da Multiple Choice Questions

Marks BL CO PO PI

Part- A (1x 10 = 10 Marks) Code

Question

Q. 1 1.3.1

1 1

No

Which operator has the lowest priority?

A) +t

B) %

C)+ 1 2.1.2

1 2

tructure of a C program?

Which of the following is the correect s

AY int main() { return 0;

B) main() { return 0;r

C) void main() { return 0;

int main() {} 2 1.3.1

D)

lid constant in C?

Which of the following is not ava

A) 'A'

B) 123

C) "Hello"

1 1 1.3.1

Y main

Which storage class in C is used to declare variables that are local to

a function and retain their values between function calls?

A) auto

B) register

g static

D) extern 2 1 1.3.1

5 If int x=4;, result of x<<| is:

2

B) 8

C) 6

D) 10

6 ht in C? 1 1 1 1 1.3.1

Which of the following is a valid single-line comment

A), /* This is a comment */

BX I/ This is a comment

C) #This is a comment

D) <-- This is a comment -

break

break

break

default printfDefault).

One

0 Three

D) Error

1What will be printed for the code snippet: 1 1.3.1

```
for(i=0;i< $;i++){
printf("%d", i):
```

B) 4

D) Garbage value

How many times will "C Programming" be printed? 1 2 2 2 2.1.2

```
#include <stdio.h>
int main() {

int k = 2;
do {

printf("C Programmingn"):
k--;
if(k == 0)

goto stop;
}while(k>-1);

stop:
return 0;;
```

A), One time

Two times

C) Three times

D) Infinite times

10 Guess the output of the following output: 2 1 1.3.1

```
#include <stdio.h>
int main(){

int a = 5, b=0,
if(a&& b)

printf{"True\n"):
else

printf{"False\n"):
return 0:
```

YTrue

- B) False
- C) Compilation error
- D) No output

SRM Institute of Science and Technology SET C

CSRMR College of Engineering and Technology
School of Computing

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Academic Year: 2025-2026 (ODD)

Test: CTI Date: 24.09.2025

Course Code & Title: 21CSST01J -Programming for Problem Solving Duration: 1Period
Year & Sem: 1 Max. Marks: 25

Year / I Sem

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12 A robotics project requires checking whether a number is odd or even using only bitwise operators. Write a C program for the same. 5 3 1 1 13.1

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14 A developer is assigned the task of implementing a key feature for a mobile calculator application. The application should provide users with the ability to select an arithmetic operation-such as addition, subtraction, multiplication, or division-and perform that operation on 3 2 13.1

two user-input numbers. The developer is required to write a program in C that utilizes a switch statement to construct this simple calculator functionality, enabling users to choose their desired mathematical operation.

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions

CO Distribution Bloom's Level Distribution

20

15 S BL1

BL2

10

0 BL3

0 BL4

BL5

C01 C02 C03 C04 C05

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/pyq txt/PYQ Jan 2023.txt
=====

CATEGORY: PYQs

STATUS: ok

Reg. No. o1oa| o o5|

B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, JANUARY 2023

First Semester

21CSS101J-PROGRAMMING FOR PROBLEM SOLVING

(For the candidates admitted from the academic year 2022-2023)

Note:

) Part A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed

over to hall invigilator at the end of 40th minute.

Part- B and Part - C should be answered in answer booklet.

(ii)

Max. Marks: 75

Time: 3 Hours

Marks BL CO PO

PART A (20 x 1 = 20Marks)

Answer ALL Questions

1. Find the output for the following statement.

```
printf ("%d", sizeof ("Hello "));
```

(A) 5 (B) 6

C) 1 (D) 2

2 1 1

2. What is the output for the following expression in C?

```
n =5-2*7-9;
```

(A) -18 (B) 12

(C) 9 (D)1

2 1 1

3. Determine the correct output.

```
float c = 3.14;
```

```
printf ("f, c%2);
```

(A) 1.0 (B) 0

C) Compile error (D) 0.14

2 2

4. Find the meaning of following statement:

```
f=(xv)? x :y;
```

(A) fpoints to max of x & y (B) fpoints to min ofx &y

(C) fpoints to addition of x & y (D) Compile error

5. The 'continue' statement cannot be used with I 21

(A) while (B) do while

(C) for (D) switch case

6. Find the correct one to access the 3 element in an array. 12 I

- (A) *a+3 (B)(a +3)
(C)*(*a +3) (D) &(a +3)

7. Find the output for the following C code. 22 2

```
for( K 1; K< = 5; K+ +);
```

```
printf ("od", K);
```

- (A) 1 2 3 45 (B) 1 23 4

- (C) 6 (D) 5

Page 1 of 4 06JF1/21CSS101J

5. Size of integer pointer and character pointer in C for 16 bit processor is

- (A) 2 (B) 2

- (C) 4 (D) 4

9. Expected output for the following code is 2 3 2

```
charstr [] ={'g'. T, 'o', b' 10, 'a', T};
```

```
printf ("as", str);:
```

- (A) global (B) glob

- (C) globloal (D) al

T0. The operator used to get value at an address stored in a pointer variable is 2 3

- (A) (B) &

- (C) && (D)

11. The keyword used to transfer the control from a function back to the 4

calling function in C is

- (A) goto (B) back

- (C) retum (D) goback

12. If both the strings are same, a string function strcmp()returns 4

- (A) 1 (B)-1

- (C) 2 (D)

13. Keyword used for function in Python language is

- (A) function (B) def

(C) fun (D) define

14. The following python code will print 2 2

```
print (type (5/2))  
print (type 5//2))
```

(A) float int (B) float float
(C) int float (D) int int

2 2

15. Find the length of the following list in python

```
len (["hello", 2, 4. 6])  
(A) 8 (B) 6  
(C) 4 (D) 3
```

1 5

16. Representation of python tuple is

(A){} (B) {1,2, 3}
(C) [1,2, 3] (D) (1,2, 3)

1 5

17. Number of elements present in the series S1 is

```
SI = pd.series (range (1, 6))
```

```
print(S1)  
(A) 4 (B)  
(C) 6 (D) 7
```

18. Numpy is often used along with packages like

(A) Node.js (B) Matplotlib

(D) Both (B) and (C)
(C) SciPy

06JF1/21CSs101J

Page 2 of 4

19. Size attribute in Numpy is used to find

(A) shape (B) date and time

(C) objects (D) number of items

20. Library used to create data frame is

- (A) Sklearn (B) Data frame
- (C) Pandas (D) Random

PART B (5 x 8 =40 Marks) Marks L c0 P0

Answer ALL Questions

21. a. Describe various storage classes in C programming with suitable example. 2

(OR)

b. Describe about 3 2

pre-increment, post-increment, pre-decrement,
post-decrement, conditional operators, each with suitable example.

22. a.i. Write a C program to find the factorial of a given number. 2

ii. Write the necessary conditions for switch-case in C. 4 2 2

(OR)

b. Explain about 4+4 2 2

) void pointer with an example in C

(ii) NULL pointer with an example in C

23. a. Write a C program to find whether the given string is palindrome or not. 8 4 3 2

Explain the string functions used in this program.

(OR)

b. Explain the concept of passing the array as an argument to function. 4 4 2

Illustrate with C program to find the sum two matrices.

24. a. Describe any four operations performed on tuples in Python programming. 5 2

(OR)

b. Write a Python program to print the following Floyd triangle. 5 2

2 3

4 56

25. a.i. Explain the significant features of the pandas library 4 4 5 2

ii. Write a Python program to create a Numpy array filled with all zeros. A 5 2

(OR)

b.i. List out various features of Numpy. A

1. Describe about the different types of data structures in pandas.

Page 3 of 44 06JF1/21CSSi01J

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PART C (1 x 15 =15 Marks)

Answer ANY ONE Question

32 2

26.i. The National Highways Department announces a scheme for four wheeler riders in the toll gate for its 50th year celebration. If the vehicle number is divisible by both 7 and 3, then they should pay one-third of the fee. If the vehicle number is divisible by 7 or 3, then they should pay half of the fee.

Otherwise they have to pay full fees. Write a C program for the given

Scenario.

7 3 32

ii. Write a C program to replace all vowels with star (*) and print the output
8 4 2

27.i. Write a Python program to check whether the given list is in sorted order or not. If the given list is in sorted order then print true, else print false.

4 5 2

i. Write a python program to create two sets Set1 and Set2 by getting input from user and determine whether the given sets Set1 and Set2 are disjoint or not.

*** *

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/pyq txt/PYQ Jun 2023.txt

=====

CATEGORY: PYQs

STATUS: ok

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() Part - Nshould tbe AAWered in OMR heet witlbin iet 40 mimes and OMI shee should be handed

(ii) over to hall invigilator at the ond of 4oh nimte.

Part -B And PartC should be anawered in anawer booklet

Mux, Marks; 75

Time: 3 Hours

PARTA (20 x1 20Mur1w)

Annwer ALL Queations

I The CxprcNsjon $4 + 6/3 * 2 * 2 + 7 \% 3$ ovaluates t)

(A) 3 (B) 4

(C) 6 (D) 7

2. Which of the following is appropriate position for writing comments in C

programming? the main

(B) Before calling

(A) Anywhere in the code function

(D) Beford Variable declaration

(C) Before hender file scction

NCction

3. Which is correct with respect to the size of the data types?

(B) $\text{int} > \text{char} > \text{long}$

(A) $\text{char} > \text{int} > \text{long}$ (D) $\text{double} > \text{char} > \text{int}$

(C) $\text{char} < \text{int} < \text{double}$

4. Which of the following is not logical operator?

(B) $\&$

(A) $\&\&$ (D) $!$

(C) $|$

5. What would be the equivalent point expression for referring the array element `arr[i][j]`

(B) $*(\&\text{arr} + i * \text{sizeof}(\text{arr}[0]) + j * \text{sizeof}(\text{arr}[0][0]))$

(A) $((\&\text{arr} + i) + j) * \text{sizeof}(\text{arr}[0][0])$

(C) $((\&\text{arr} + i) + j) * \text{sizeof}(\text{arr}[0])$ (D) $(\&\text{arr} + i) + j * \text{sizeof}(\text{arr}[0])$

6. Which of the following cannot be checked in a switch-case statement?

(A) Character (B) Integer

(D) Enum

(C) Float

7. What is the output of the following program?

```
#include <stdio.h>
```

```
int main ()
```

```
{
```

```
    int a;
```

```
    switch (a)
```

15.

14. .

6) 12.

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===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/pyq txt/PYQ Nov 2018.txt

=====

CATEGORY: PYQs

STATUS: ok

31. a.i. List the advantages of uniting function in C programming.

ii. Define preprocessor and its directives with suitable examples.

(or) B.Tech. DEGREE EXAMINATION,

b.i. NOVEMBER

Define pointer. 2018

(Marks) First Semester

ii. Write a program to explain the usage of pointer. (8 I 8CSS

Marks) I O I J _ PROGRAMMING FOR PROBLEM SOLVING

(For the candidates admitted during the academic year

32. a.i. Explain dynamic memory allocation in Note: zoia-r-"*araJ

detail. (8 Marks) (i) Part - A should be answered in oMR sheet within first 45 minutes and over oMR

ii. Explain Structure within to hall

Structure. invigilator sheet should be handed

at the end of 45 minutes.

(a Marks) (i) Part - B and Part - C should be answered in answer booklet.

Time:

write Three

b.i. Hours

a program to write ,Hello world, toot}-o, ur". Max. Marks: 100

u. Discuss all file opening modes in c programming PART-A(20x1=20Marks)

language with respect to file handling. Answer ALL questions

1. Who is the father of C language?

***,t* (A) Bjame Stoustrup

(C) @) James A.Gosling

Dennis Ritchie @) Dr.E.F.Codd

2. In C programming language which of the following type of operators

precedence. have the highest

(A) Relational operators (B)

(C) Equality operators

Logical operators (D) Arithmetic operators

3. Bitwise operators can operate upon?

(A) Doubles and chars

(C) floats and doubles

ints and floats (D) ints and chars

4. What type of value does sizeof return?

(A) Char (B)

(C) Short

Unsigned int (D) Long

5. Relational operators cannot be used on

(A) Structures

(C) Strings

6. What is the type of the below assignment expression

if (x < y) x = y; is of type float, y is of type int?

(A) int

(C) float

(D) double

There is no type for an assignment (D) double

expression

7. Which keyword is used to come out of a loop

(A) Break (B) Continue (C) Return (D) Exit

1. Which keyword is used to come out of a loop

(A) Break (B) Continue (C) Return (D) Exit

8. In expression i = g0 + f0, first function called depends on __.

(A) Compiler (B)

(C) Associativity (D)

Precedence of + operator (D) operator

10. Which keyword is used to come out of a loop

20M:1/ltcsslou

PART-B (5 x 4=20 Marks)

9. Set of consecutive memory location is called as

Answer: Array

(A) Questions

Function (B)

(C) Array (D) Pointer

21. Explain the structure of a C program-

10. Array with last element 'n' will always have array size equal to

(A) n. Define pseudo code and mention the advantage'

n (B) n+1

(C) n-1 (D) n+1

23. What are the differences between L value and R value in expression?

11. If the two strings are identical, then strcmp () function returns'

(B) "continue" statement?

(A) 0 24. what is the difference between a "break" and

(c) -1 (D) True

25. Differentiate between "call by value" and "call by reference"

12. String concatenation means

(A) Combining two strings (B) Extracting a substring out of a string 26. What is parameter?

Mention the difference between formal and actual parameter

(C) Partitioning the string into two (D) Comparing the two strings to define the larger one 27. Write the difference between union and structure'

strings

PART-C (5x12=60 Marks)

13. A preprocessor command

(A) art on a new line (B) Need not start on the first column Answer ALL Questions
Need not st

(C) Has # as the first character (D) Comes before the first executable statement

28- a.i. write the algorithm and pseudo code that takes minutes as input and displays the total

14. a parameter of another function? number of hours and minutes.

Can we use a function as

(A) Yes, and we can use the function (B) Yes, but we call the function again to get

ent subjects are input through the keyboard,

value conveniently the value, not as convenient as in using ii. If the marks obtained by a student in six differ

write a C program to find out the aggregate marks and percentage marks obtained by the

(C) No, C does not support it (D) This case is compiler

(or)

15. Address stored in the pointer variable is of type (8

(A) (B) b.i. Analyze the various storage classes in C with suitable examples. Marks)
Integer Floating

(C) Array (D) Character

ii- Write a C program to get two numbers as input from the user, then perform Bitwise-AND

16. pointer declarations? `int * ptr, p;` and Bitwise-OR and print the results. (4 Marks)

Comment on the following

(A) ptr is a pointer to integer, p is not (B) ptr and p, both are pointers to integer

ptr and p both are not pointers to integer 29. a. Analyze the C programming operator precedence and associativity with suitable examples.

if `i*` is pointer to integer, p may or (D)

may not be (or)

17. How will you free the allocated memory?

name); (B) b. Explain the various decision making statements in C with an example for each'

(A) remove (var- free (var-name));

(C) delete (var-name); (D) dalloc (var-name);

30. a. write the purpose of following functions with suitable examples

(i) getso

18. What is the similarity between a structure, union and enumeration?

(A) All of them let you define new (B) All of them let you define new data types (ii) getchar

(iii) puts (
 values)

(c) All of them let you define new (d) All of them let you define new structure (iv) putchar ()
 (") strlen

pointers (vi) strcpy (

19. If there is any error while opening a file, fopen will return)

(A) Nothing (B) EOF

(C) Null (D) (or)

Depends on compiler

i b. Write about

20. FILE reserved word is (i) Recursion function with an example

(A) A structure tag declared in stdio.h (B) One of the basic data types in C (ii) Function
 definition with an example

(C) Pointer to the structure defined in (D) It is a type name defined in stdio.h

stdio.h 20NF1/ r8CES101J

20Nr1/ ltcssl01J Page 3 of ,1

Page 2 of ,1

===== SOURCE_FILE: nexus sem 2/Programming For Problem Solving/PYQs/pyq txt/Question Bank
 with MCQs.txt =====

CATEGORY: PYQs

STATUS: ok

PROGRAMMING FOR PROBLEM SOLVING

Unit 1

Multiple Choice questions MCQ

1. Who is father of C Language? [CLO 1] [PL01]

A. Bjarne Stroustrup B. Dennis Ritchie

C. James A. Gosling D. Dr. E.F. Codd

2. C Language developed at _____? [CLO 1] [PLO 1]

- A. AT & T's Bell Laboratories of USA in 1972
- B. AT & T's Bell Laboratories of USA in 1970
- C. Sun Microsystems in 1973
- D. Cambridge University in 1972

3. For 16-bit compiler allowable range for integer constants is _____ ? [CLO 1] [PLO 2]

- A. -3.4e38 to 3.4e38
- B. -32767 to 32768
- C. -32768 to 32767
- D. -32668 to 32667

4. C programs are converted into machine language with the help of [CLO 1] [PLO 1]

- A. An Editor
- B. A compiler
- C. An operating system
- D. None of the above

5. A C variable cannot start with [CLO 1] [PLO 1]

- A. An alphabet
- B. A number
- C. A special symbol other than underscore
- D. both (b) and (c)

6. Which of the following is allowed in a C Arithmetic Instruction? [CLO 1] [PLO 1]

- A. []
- B. {}
- C. ()
- D. None of the above

7. Which of the following shows the correct hierarchy of arithmetic operations in C [CLO 1] [PLO 2]

- A. / + * -
- B. * - / +
- C. + - / *
- D. * / + -

8. Program which is written originally by the programmer is classified as [CLO 1] [PLO 1]

- A. object code
- B. machine code
- C. source program
- D. interactive programs

9. Data types are differed on the basis of [CLO 1] [PLO 1]

- A. the way of storage
- B. the type of operations
- C. the type of operators used
- D. both a and b

10. Loop statement which is repeated for some given number of times is classified as [CLO 1]

- A. FOR loop
- B. GO loop
- C. REPEAT loop
- D. GO REPEAT loop

11. Type of statement written in sequence and is repeated until the specific condition met is classified as [CLO 1] [PLO 1]

- A. format
- B. loop
- C. case
- D. condition

12. Size of an array is declared by [CLO 1] [PLO 1]

- A. programmer

B. program users

C. software

D. declared automatically

13. Programming language 'FORTRAN' stands for [CLO 1] [PLO 1]

A. formula translator

B. formula translation

C. free translator

D. free translation

14. Functions that are used in the programs and are defined by the programmers are called [CLO 1] [PLO 2]

A. program layout

B. program procedure

C. built-in functions

D. user-defined function

15. An assembler translates [CLO 1] [PLO 1]

A. machine code into assembly code

B. assembly code into machine code

C. processing time into manual time

D. routine into subroutine

16. Name given by a programmer to any particular data is classified as [CLO 1] [PLO 1]

A. identifier

B. identification

C. exponent

D. mantissa

17. When variable used in program is whole number, the variable is stored as [CLO 1] [PLO 2]

A. fixed string

B. integers

C. negative whole numbers

D. positive whole numbers

18. In programming, programmers use comments to [CLO 1] [PLO 1]

A. highlight program modules

B. explain module functions

C. explain used variables

D. all of above

19. Variable which uses the same name in whole program and in its all routines thus best classified

as [CLO 1] [PLO 2]

A. middle variable

B. default variable

C. local variable

D. global variable

20. Statement which is used to make choice between two options and only option is to be performed

is written as [CLO 1] [PLO 2]

A. if statement

B. if else statement

C. then else statement

D. else one statement

21. The_____ statement is used to transfer the control to the end of statement block in a loop:

a. Continue [CLO 1] [PLO 2]

b. Break

c. Switch

d. Goto

22. Which of the following is not a valid variable name declaration? [CLO 1] [PLO 2]

a) int _a3;

b) int a_3;

c) int 3_a;

d) int _3a

23. All keywords in C are in [CLO 1] [PLO 2]

a) LowerCase letters

b) UpperCase letters

c) CamelCase letters

d) None of the mentioned

24. Which of the following is true for variable names in C? [CLO 1] [PLO 2]

a) They can contain alphanumeric characters as well as special characters

b) It is not an error to declare a variable to be one of the keywords(like goto, static)

c) Variable names cannot start with a digit

d) Variable can be of any length

25. The format identifier '%i' is also used for _____ data type? [CLO 1] [PLO 2]

a) char

b) int

c) float

d) double

26. Which of the following is a User-defined data type? [CLO 1] [PLO 3]

a) typedef int Boolean;

b) typedef enum {Mon, Tue, Wed, Thu, Fri} Workdays;

- c) struct {char name[10], int age};
- d) all of the mentioned

27. What is the output of this C code? [CLO 1] [PLO 3]

```
1. #include <stdio.h>
2. int main()
3. {
4.     signed char chr;
5.     chr = 128;
6.     printf("%d\n", chr);
7.     return 0;
8. }
```

- a) 128
- b) -128
- c) Depends on the compiler
- d) None of the mentioned

28. What is the output of this C code? [CLO 1] PLO 3]

```
1. #include <stdio.h>
2. int main()
3. {
4.     j = 10;
5.     printf("%d\n", j++);
6.     return 0;
7. }
```

- a) 10 b) 11
- c) Compile time error d) 0

Explanation : j is not belongs to any data type

29. The following code 'for(; ;)' represents an infinite loop.

It can be terminated by. [CLO 1] [PLO 3]

- a) break b) exit(0)

c) abort() d) all of the mentioned

30. The keyword 'break' cannot be simply used within: [CLO 1] [PLO 3]

- a) do-while b) if-else
- c) for d) while

31. Which keyword is used to come out of a loop only for that iteration? [CLO 1] [PLO 3]

- a) break b) continue
- c) return d) none of the mentioned

32. The first step in problem solving is ----- [CLO 1] [PLO 2]

- a) Understand the problem c) Identify the problem
- b) Developing algorithm/Flowchart d) Listing the possible outcome

33. The solutions which has series of actions to solve a problem are called ----- [CLO 1] [PLO 2]

- a) Heuristic solutions
- b) Algorithmic solutions

34. The solution can be reached by completing the actions in steps. These steps are called [CLO 1] [PLO 2]

- a) Sequence c) Algorithm
- b) Flowchart d) Steps

35. Solutions that cannot be reached through a direct set of steps are called ----- [CLO 1] [PLO 2]

- a) Algorithmic solutions b) Heuristic solutions

36. ----- means the outcome or the completed computer-assisted answer. [CLO 1] [PLO 1]

- a) Solution b) Result c) Program

37. The field of computer that deals with the problem of heuristic solution is called----- [CLO 1] [PLO 1]

- a) Artificial Intelligence
- b) Expert System c) Computer Management

38. ----- means the set of instructions that make up the solution after they

have been coded
into a programming language. [CLO 1] [PLO1]

a) solution b) result c) program d) error

39. ----- are organized facts. [CLO 1] [PLO1]

a) Data b) Information

40. A language applied for wide range of application is called as _____language[CLO 1][PLO 2]

a. Special purpose c)General purpose
b. Individual purpose d)Scientific purpose

41. Compiler converts the source code into _____ [CLO 1][PLO 4]

a. C code b. Byte code c. Object code d. Executable code

42. _____ converts source code to machine language one line at a time. [CLO 1][PLO 4]

a. Compiler b. Interpreter c. Assembler d.CPU

43. The program that converts high level language to a machine language is called _____

[CLO 1] [PLO 1]

a. Interpreter b. Linker c. Compiler d. Loader

44. ‘#’ symbol is known to be _____ [PLO 2] [CLO 1]

a. Linker b.Compiler c. Assembler d.Preprocessor directive

45. A _____ is a notational system for describing computations in both machine and human readable form [PLO 1] [CLO 1]

a. Programming language c.Machine language
b. High-level language d.Assembly language

46. ----- program converts assembly language into machine language [PLO 2] [CLO 1]

a. compiler b. interpreter c. assembler d. preprocessor directive

47. which symbol is been used for processing in flowchart? Ans: a [CLO 1] [PLO 3]

a) b) c) d)

48. which symbol is been used for input/output in flowchart? Ans: c [PLO 3] [CLO 1]

a) b) c) d)

49. which symbol is been used for decision making in flowchart? Ans: d [PLO 3] [CLO 1]

a) b) c) d)

50. which is used as start/stop symbol in flowchart? Ans:b [PLO 3] [CLO 1]

a) b) c) d)

51. ----- is a distinguishing characteristic of human excellence in every area of behavior

[CLO 1] [PLO 1]

a) creativity b) thinking c) visualization d) problem solving

52. .What is the value of x in this C code? [PLO 2] [CLO 1]

```
#include <stdio.h>
void main()
{
    int x = 5 * 9 / 3 + 9;
}
```

(a). 3.75 b) Depends on compiler c) 24 d) 3

53.A character variable can store how many characters at a time? [PLO 2][CLO 1]

(a) 1 character

- (b)8 characters
- (c)255 character

54.Which of the following is the correct way of writing comments? [PLO 2] [CLO 1]

- (a)*/comments/*
- (b)/*comment*/
- (c)**comment** (
- d){comment}

55. C programming language is [PLO 1] [CLO 1]

- (a)object oriented programming language
- (b)Procedure oriented programming language
- (c)function oriented programming language
- (d)None of the above

56.The memory space taken for a char type data is [PLO 2] [CLO 1]

- (a)2 bytes
- (b)4 bytes
- (c)8 bytes
- (d)1bytes

57 .The memory space taken for a int type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes

- (c) 8 bytes
- (d)10bytes

58.The memory space taken for a float type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes
- (c) 8 bytes
- (d)10bytes

59. What is the only function all programs must contain ? [PLO 2] [CLO 1]

- (a) start()
- (b) system()
- (c) main()
- (d) program

60. For 16-bit compiler allowable range for integer constants is _____? [PLO 2] [CLO 1]

- (a) -3.4e38 to 3.4e38
- (b) -32767 to 32768
- (c) -32668 to 32667
- (d) -32768 to 32767

61. Every statement in C program is to be terminated by a _____ [PLO 2] [CLO 1]

- (a) dot(.)
- (b) semi-colon(;)
- (c) colon(:)
- (d) Question mark(?)

62. The escape sequence „\b“ is a [PLO 2] [CLO 1]

- (a) back space
- (b) next line
- (c) tab
- (d) none of the above

63. The memory space taken for a long int type data is [PLO 2] [CLO 1]

- (a) 2 bytes
- (b) 4 bytes
- (c) 8 bytes
- (d) 10 bytes

64. Which of the following will not be valid expressions in C? [PLO 2] [CLO 1]

- (a) a=2+(b=5);
- (b) a=11%3
- (c) a=b=c=5

(d) $b+5=2$

65. The ----- operator is true only when both the operands are true. [PLO 2]
[CLO 1]

- a) &&
- b) ||
- c) !
- d) ?:

66. The ----- statement when executed in a switch statement causes immediate exit

from the structure. [CLO 1] [PLO 2]

- a) goto
- b) default
- c) break
- d) switch

66. The ternary conditional expression using the operator?: could be easily coded using ----

- statement [CLO 1][PLO 2]

- a) Nested if
- b) if-else
- c) if
- d) for

4 Marks :

Unit-1 CLO-1

1. Comment "C is mid level language"? [CLO-1][PLO 1]

2. What is problem solving? [CLO-1][PLO 1]

3. What are the six steps of problem solving? [CLO-1][PLO 1]

4. Discuss about how the problems can be solved with computers? [CLO-1][PLO 1]
5. What is a program? [CLO-1][PLO 2]
6. Why is problem analysis important? [CLO-1][PLO 2]
7. What are the tools of problem solving available? [CLO-1][PLO 2]
8. How do problem-solving tools help in leading to a solution? [CLO-1][PLO 2]
9. Why it is important to test a solution before coding it? [CLO-1][PLO 2]
10. What is an algorithm? Give the characteristics of the algorithm. [CLO-1][PLO 2]
11. What is a flowchart? Give the symbols/shapes used in the flowchart.

[CLO-1][PLO 3]
12. Define pseudocode and give its importance with an example. [CLO-1][PLO 2]
13. Discuss the difficulties with problem solving in detail. [CLO-1][PLO 2]
14. State the use of %d and %f .Write a printf statement in C using the above mentioned symbols? [CLO-1][PLO 4]
15. What is main difference between variable and constant? [CLO-1][PLO 3]
16. Explain bitwise left shift operator? [CLO-1][PLO 4]
17. Explain primary data types used in C? [CLO-1][PLO 3]
18. Difference between formatted & unformatted statement ? [CLO-1][PLO 3]
19. What is mean by storage class of variable? [CLO-1][PLO 3]
20. Explain with example ++i and i++. [CLO-1][PLO 4]
21. Refer all Elab programs. [CLO-1][PLO 3] [PLO 2][PLO 4]

12 Marks:

- 1) Explain the evolution of programming languages. [CLO-1][PLO 1]
- 2) Explain the various steps involved in problem solving with diagram. [CLO-1][PLO 1]
- 3) Draw the flowchart and write the algorithm and c code to find the sum and to reverse the digits of given five digit number. [CLO-1][PLO 1] [PL02]
- 4) Write an algorithm and draw a flow chart to find the factorial and Fibonacci series of given number. [CLO-1][PLO 2] [PLO 3]
- 5) Write a note on Algorithm, Flow chart and Pseudocode. [CLO-1][PLO 3] [PLO 2]
- 6) Explain the scope, lifetime of variable in C with example. [CLO-1][PLO 2][PL03]
- 7) Write down the algorithm to find the largest number among three given numbers and outline the steps in the algorithm with the inputs 5, 17, 3. [CLO-1][PLO 2][PL03]
- 8) Draw flowchart to compute the salary of an employee in a company. Assume that there are two types of employees in the company daily wages and regular. Salary is calculated as number of hours worked* wages per hour for daily wagers and basic pay +(% of DA * basic pay)/100 + HRA + medical allowance for regular employees. Sketch the flow of your design for a regular employee with basic pay = 5000, % of DA = 75 % and HRA = 500. (8)
[CLO-1][PLO 2][PL03][PL04]
- 9) Explain in details about operators with an example [CLO-1][PLO 1] [PLO 2][PL03]
- 10) Refer all elab Programs [CLO-1][PLO 3] [PLO 2][PLO 4]

UNIT - II

PART A

1. The ----- operator is true only when both the operands are true.
e) && b) || c) ! d) ?:

Answer: a

2. The ----- statement when executed in a switch statement causes immediate exit

from the structure.

a) goto b) default c) break d) switch

Answer: c

3. The ternary conditional expression using the operator?: could be easily coded using -----

- statement

a) Nested if b) if-else c) if d) for

Answer: b

4. What will be the output when the following segment is executed?

```
Char ch='a';
```

```
Switch(ch)
```

```
{
```

```
case 'a':
```

```
Printf("A");
```

```
case 'b':
```

```
Printf("B");
```

```
Default:
```

```
printf("C");
```

```
}
```

a)A b)B c)C d) a

Answer: a

5. What will be the output of the following segment when executed?

```
int x=10, y=20;
```

```
if((x<y)|| (x+5)>10)
```

```
printf("%d",x);
```

```
else
```

```
printf("%d",y);
```

a)10 b) 20 c) 15 d)5

Answer: a

6. The ----- statement is used to skip a part of the statements in a loop.

a) Continue b) break c) goto d)switch

Answer: a

7. A for loop with no test condition is known as ----- loop

a) Infinite b) time delay c) for d) Incrementing

Answer: a

8. The sentinel -controlled loop is also known as -----loop

a) Indefinite repetition loop c) time delay

b) Definite repetition loop d) infinite

Answer: a

9. In an exit controlled loop the body of the loop is always executed minimum number of ----

a) 1 time b)2 times c) 3 times d)n times

Answer:a

10. The while is an ----- loop statement.

a) Entry-controlled b)exit-controlled c)indefinite repetition

d)definite repetition

Answer: a

11. The -----specification is used to read or print integers

a)h b)l c)L d)c

Answer: a

12. To print the data left-justified, must use ----- in the field specification

a) - b) + c) / d) *

Answer: a

13. By default, the real numbers are printed with a precision of ----- decimal.

a)6 b)2 c)4 d)0

Answer:a

14. The expression !(x!=y) can be replaced by the expression-----

a) x!=y b) x==y c)x=!y d)!x=!y

Answer:c

15. In a counter controlled loop, variable known as -----is used to count the
loop

operations.

a) Counter b)sentinel c)i d)n

Answer: a

16. Which of the following special symbol allowed in a variable name?

(a) * (asterisk) (b)| (pipeline) (c) - (hyphen) (d) _ (underscore)

17 . Which of the following are invalid variable names?

a) Minimum b)n\$ c) Integer d) float

```
18. int a=10;
    ++a;
    a++;
    Printf("%d",a);
```

a)10 b) 11 c)12 d) 13

```
19. int a=11;
    a=a%2;
    a=a/2;
```

The value of a is,

a) 1, 1 b) 5,1 c) 1,5 d)5

20. Which of the following is not logical operator?

A. &

B. &&

C. ||

D. !

21 Which of the following cannot be checked in a switch-case statement?

A. Character

B. Integer

C. Float

D. enum

22. What is the output of this C code?

```
int main()
{
    int a = 0, i = 0, b;
    for (i = 0; i < 5; i++)
    {
        a++;
        continue;
    }
}
```

A. 2 B. 3

C. 4 D. 5

23. What is the output of this C code?

```
void main()
{
    int i = 0, j = 0;
    for (i = 0; i < 5; i++)
    {
        for (j = 0; j < 4; j++)
        {
            if (i > 1)
                break;
        }
        printf("Hi \n");
    }
}
```

A. Hi is printed 5 times B. Hi is printed 9 times

C. Hi is printed 7 times D. Hi is printed 4 times

24. What is the output of this C code?

```
void main()
{
    int i = 0;
    int j = 0;
    for (i = 0; i < 5; i++)
    {
        for (j = 0; j < 4; j++)
        {
            if (i > 1)
                continue;
            printf("Hi \n");
        }
    }
}
```

- A. Hi is printed 9 times B. Hi is printed 8 times
C. Hi is printed 7 times D. Hi is printed 6 times

25. What is the output of this C code?

```
void main()
{
    int i = 0;
    for (i = 0; i < 5; i++)
    {
        if (i < 4)
        {
            printf("Hello");
            break;
        }
    }
}
```

- A. Hello is printed 5 times B. Hello is printed 4 times
C. Hello D. Hello is printed 3 times

27. What is the output of this C code?

```
int main()
{
```

```

int i = 0;
char c = 'a';
while (i < 2){
i++;
switch (c) {
case 'a':
printf("%c ", c);
break;
break;
}
}

printf("after loop\n");
}

```

- A. a after loop B. a a after loop
C. after loop D. None of the mentioned

28. int main()
{
printf("before continue ");
continue;
printf("after continue\n");
}

- A. Before continue after continue B. Before continue
C. after continue D. Compile time error

29. What is the output of the code given below?

```

int main()
{
printf("%d ", 1);
goto l1;
printf("%d ", 2);
l1:goto l2;
printf("%d ", 3);
l2:printf("%d ", 4);
}

```

- A. 1 4 B. Compilation error

C. 1 2 4 D. 1 3 4

30. What is the output of code given below?

```
int main()
{
printf("%d ", 1);
11:12:
printf("%d ", 2);
printf("%d\n", 3);
}
```

A. Compilation error B. 1 2 3

C. 1 2 D. 1 3

31. What will happen if in a C program you assign a value to an array element whose subscript exceeds the size of array?

A. The element will be set to 0.

B. The compiler would report an error.

C. The program may crash if some important data gets overwritten.

D. The array size would appropriately grow.

32 .In C, if you pass an array as an argument to a function, what actually gets passed?

A. Value of elements in array

B. First element of the array

C. Base address of the array

D. Address of the last element of array

33. Result of a logical or relational expression in C is?

A. True or False

B. 0 or 1

C. 0 if expression is false and any positive number if expression is true

D. None of the mentioned

34. What will be the value of d in the following program?

```
int main()
{
    int a = 10, b = 5, c = 5;
    int d;
    d = b + c == a;
    printf("%d", d);
}
```

A. Syntax error

B. 1

C. 5

D. 10

35. What is the output of this C code?

```
int main()
{
    int a = 10, b = 5, c = 3;
    b != !a;
    c = !!a;
    printf("%d\t%d", b, c);
}
```

A. 5 1

B. 0 3

C. 5 3

D. 1 1

36. What is meaning of following declaration ?

```
int arr[20];
```

- a) Array of size 20 that can have integer address
- b) None of the above
- c) Integer array of size 20
- d) Array of size 20

37. In C Programming, If we need to store word "INDIA" then syntax is as below -

- a) `char name[6]={ 'I', 'N', 'D', 'I', 'A', '\0' }`
- b) `char name[6]={ "I", "N", "D", "I", "A" }`
- c) `char name[6]={ 'I', 'N', 'D', 'I', 'A' }`
- d) `char name[] ; name="INDIA"`

38. what is the way to initialize array?

- a) `int num[6] = {2,4,12,5,45,5};`
- b) `int n{} = {2,4,12,5,45,5};`
- c) `int n{6}={2,4,12};`
- d) `int n(6)={2,4,12,5,45,5};`

39. what will be the output of the program?

- a) 3, 2, 15
- b) 2, 3, 20
- c) 2, 1, 15

d) 1, 2, 5

Answer: Option A

Solution:

>> int a[5] = {5, 1, 15, 20, 25}; The variable arr is declared as an integer array with a size of 5 and it is initialized to

a[0] = 5, a[1] = 1, a[2] = 15, a[3] = 20, a[4] = 25.

>> int i, j, m; The variable i, j, m are declared as an integer type.

>> i = ++a[1]; becomes i = ++1; Hence i = 2 and a[1] = 2

>> j = a[1]++; becomes j = 2++; Hence j = 2 and a[1] = 3.

>> m = a[i++]; becomes m = a[2]; Hence m = 15 and i is incremented by 1(i++ means 2++ so i=3)

>> printf("%d, %d, %d", i, j, m); It prints the value of the variables i, j, m

Hence the output of the program is 3, 2, 15.

39. what will be the output of the program?

a) 5

b) 6

c) 9

d) Error

e) None of the above

Answer: Option C

Solution:

x[i] is equivalent to *(x + i),

so (buf + 1)[5] is *(buf + 1 + 5), i.e. buf[6].

40) An array elements are always stored in _____memory locations

- a) sequential
- b) Random
- c) Sequential and Random
- d) None of the above

41) Let x be an array. which of the following operations are illegal?

- a) ++x b) x+1 c) x++ d) x*2

options:

- a) I and II
- b) I,II and III
- c) II and III
- d) I, III and IV
- e) III and IV

Answer: Option D

Solution: `int x[10]; * x` will store the base address of array. *
Statement I, III and IV is invalid.

Statement I and III : `++x` and `x++` are throwing an error while compile (lvalue required as increment operand)

Since, x is storing in the address of the array which is static value which cannot be change by the operand.

Statement IV : `x*2` is also throw an error while compile (invalid operands to binary * (have 'int *' and 'int'))

Statement II : `x+1` is throw a warning: assignment makes integer from pointer without a cast [enabled by default]

42) what is the maximum number of dimensions an array in c may have?

- a) 2
- b) 8
- c) 20
- d) 50
- e) theoretically no limit. the only practical limits are memory size and compilers

43) size of the array need not be specified, when

- a) initialized is a part of definition
- b) it's a declaration
- c) it is a formal parameter
- d) All of these

44) what will be the output of the program?

- a) Equal
- b) Unequal
- c) Error

d) None of these

Answer: Option B

Solution:

Strings are compared using strcmp() function defined under string.h header file.

45)

Unit-2 PART -B

1. Give the general syntax of conditional operator?
2. Define operator in C. What role an operator plays in C program?
3. Differentiate between relational and logical operators used in C?
4. Give the types of decision making statements
5. What is the general form of switch statement?
6. Differentiate do... while.. and while... loop.
7. Determine the output of the following and justify your answer

```
int main()
{
    int ch; printf("enter a value between 1 to 2:");
    scanf("%d", &ch);
    switch (ch, ch -1){
    case 1: printf("1\n"); break;
    case 2: printf("2"); break;
    } return 0; }
```

8. Explain the limitations of arrays.
9. Give the steps of looping process.
10. Brief the use of continue, goto, break statement and give the syntax.
11. What are loop control structures? Explain for loop, while loop and do-while

loop with their syntax.

12. What do you mean by infinite loop? Give suitable of any infinite loop in a C program.

13. Explain break and continue statements with examples.

14. Determine the output of the following program justify your answer in a few words.

```
int main() {  
    float f = 1.0;  
    switch(f) {  
  
    case 1.0: printf("one"); break;  
    case 2.0: printf("two"); break;  
    default: printf("%f", f);  
  
    }  
}
```

15. Write a program to print series of number divisible by 3 from 1 to 100 using for loop.

16. Write a C program to print 1, 2, 4, 8, 16, 32, 64N using do...while loop (Read N from user).

17. What are Arrays in C programming? Give the importance of Array in C language

13. What are the rules to declare one dimensional array?

14. What do you mean by compile time initialization? Give suitable example of Compile time initialization of C Array.

15. Describe the array index out of bound error in context of C array program.

PART C

1) Lucy is celebrating her 15th birthday. Her father promised her that he will buy her a new computer on her birthday if she solves the question asked by him. He asks Lucy to find whether the year on which she had born is leap year or not. Help her to solve this puzzle so that she celebrates her

birthday happily. If her birth year is 2016 and it is a leap year display 2016 is a leap year.? Else display 2016 is not a leap year and check with other leap year conditions

2) Write a C program to print the multiplication table of an integer n upto m rows using a while loop

sample input and output

INPUT

5

4

OUTPUT

1*5=5

2*5=10

3*5=15

4*5=20

3) Write a program to generate a following @s triangle:

@ @ @ @ @

@ @ @ @

@ @ @

@ @

@

4) Write a program that determines a student's grade. The program will read three scores and determine the grade based on the following rules: score >=90% =>grade=A , score >= 70% and <90% => grade=B, score>=50% and <70% =>grade=C, score<50% =>grade=F

5) i)Why switch case is better than else..if... ladder. Justify.

ii)Write a program to read a value (1-7) and print the equivalent day of the week

(i.e. 1-SUN, 2-MON.....)

iii) Explain briefly about for loop and Nested for loop with suitable example

iv) Write a program to print the following pattern.

12345

1234

123

12

1

6.a. i) Write a C program to insert an element at a specified position in the array.

7)) Write a program to generate a following patterns

a)

b)

c)

d)

e)

8) write c Program for Student mark list generation by using control and looping statements

9) write c program for digit rotation. "For any positive integer, we define a digit rotation as either moving the first digit to the end of the number (left digit rotation), or the last digit to the front of the number (right digit rotation). For example, the number 12345 could be left digit rotated to 23451, or right digit rotated to 51234.

10) Write a program to find the sum of positive numbers in an array

11) write c program for palindrome strings , two strings A and B, each consisting of lower case alphabets.

12) Write a arithmetic operation menu driven program using while(1)

13) i) Write short notes on switch case.

ii) Program to create a simple calculator, Performing addition, subtraction, multiplication, division depending the input from user.

14) i)write a C program to read 10 nos. and reverse it using array

ii)Concatenate two arrays of length minimum 5 numbers.

UNIT III

PART A

1) Which of the following is not possible statically in C?

a) Jagged Array

- b) Rectangular Array
- c) Cuboidal Array
- d) Multidimensional Array

2) What is the output of this C code?

```
#include <stdio.h>
void main()

{
int a[2][3] = {1, 2, 3, 4, 5};
int i = 0, j = 0;
for (i = 0; i < 2; i++)
for (j = 0; j < 3; j++)
printf("%d", a[i][j]);
}
```

- a) 1 2 3 4 5 0
- b) 1 2 3 4 5 junk
- c) 1 2 3 4 5 5
- d) Run time error

3) Predict the output of below program:

```
#include <stdio.h>

int main()
{
    int arr[5];

    // Assume that base address of arr is 2000 and size of integer
    // is 32 bit
    arr++;
    printf("%u", arr);
}
```



```
    return 0;
}
```

- a) 2002
- b) 2004
- c) 2020
- d) lvalue required

4) Predict the output of below program:

```
#include <stdio.h>
```

```
int main()
{
    int arr[5];
    // Assume base address of arr is 2000 and size of integer is 32 bit
    printf("%u %u", arr + 1, &arr + 1);
    return 0;
}
```

- i. 2004 2020
- ii. 2004 2004
- iii. 2004 Garbage Value
- iv. The program fails to compile because address of operator cannot be used with array name

5) Size of the array need not be specified, when

- A. Initialization is a part of definition
- B. It is a declaratrion
- C. It is a formal parameter
- D. All of these

6) While passing an array as an actual argument, the function call must have the array name

A. with empty brackets

B.with its size

C.alone

D.none of the above

7) The parameter passing mechanism for an array is

A. call by value

B. call by value-result

C. call by reference

D. none of these

8) Under which of the following conditions, the size of an one-dimensional array need not be specified?

A. when initialization is a part of definition

B. when it is a declaration

C. when it is a formal parameter and an actual argument

D. All of the above

9) If a two dimensional array is used as a formal parameter, then

A. both the subscripts may be left empty

B.the first (row) subscript may be left empty

C. the first subscript must be left empty

D. both the subscripts must be left empty

11) Choose the statement that best defines an array

A. It is a collection of items that share a common name

B. It is a collection of items that share a common name and occupy consecutive memory location

C. It is a collection of items of the same type and storage class that share a common name and occupy consecutive memory locations

D. None of the above

12) Choose the correct statements

A. Array stores data of the same type

B. Array can be a part of a structure

C. Array of structure is allowed

D. All of the above

13) What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. int a[2][3] = {1, 2, 3, 4, 5};
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
```

a) 1 2 3 4 5 0

b) 1 2 3 4 5 junk

c) 1 2 3 4 5 5

d) Run time error

Answer: a

14) What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. int a[2][3] = {1, 2, 3, , 4, 5};
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
```

a) 1 2 3 junk 4 5

b) Compile time error

c) 1 2 3 0 4 5

d) 1 2 3 3 4 5

Answer: b

15. What is the output of this C code?

```
1. #include <stdio.h>
2. void f(int a[][3])
3. {
4. a[0][1] = 3;
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
10. void main()
11. {
12. int a[2][3] = {0};
13. f(a);
14. }
```

- a) 0 3 0 0 0 0
- b) Junk 3 junk junk junk junk
- c) Compile time error
- d) All junk values

Answer: a

16. What is the output of this C code?

```
1. #include <stdio.h>
2. void f(int a[][])

3. {
4. a[0][1] = 3;
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }

10. void main()
11. {
12. int a[2][3] = {0};
13. f(a);
14. }
```

- a) 0 3 0 0 0 0
- b) Junk 3 junk junk junk junk
- c) Compile time error
- d) All junk values

Answer: c

17. What is the output of this C code?

```
1. #include <stdio.h>
2. void f(int a[2][])
3. {
4. a[0][1] = 3;
5. int i = 0, j = 0;
6. for (i = 0; i < 2; i++)
7. for (j = 0; j < 3; j++)
8. printf("%d", a[i][j]);
9. }
10. void main()
11. {
12. int a[2][3] = {0};
13. f(a);
14. }
```

- a) 0 3 0 0 0 0
- b) Junk 3 junk junk junk junk
- c) Compile time error
- d) All junk values

Answer: c

18. Comment on the following statement:

```
int (*a)[7];
```

- a) An array "a" of pointers.
- b) A pointer "a" to an array.
- c) A ragged array.
- d) None of the mentioned

Answer: b

19. Comment on the 2 arrays regarding P and Q:

1. `int *a1[8];`
2. `int *(a3[8]);`
3. P. Array of pointers
4. Q. Pointer to an array

- a) a1 is P, a2 is Q
- b) a1 is P, a2 is P
- c) a1 is Q, a2 is P
- d) a1 is Q, a2 is Q

Answer: b

20. Which of the following is not possible statically in C?

- a) Jagged Array
- b) Rectangular Array
- c) Cuboidal Array
- d) Multidimensional Array

Answer: a

21. What is the correct syntax to send a 3-dimensional array as a parameter? (Assuming declaration `int a[5][4][3];`)

- a) `func(a);`

- b) func(&a);
- c) func(*a);
- d) func(**a);

Answer: a

22. Applications of multidimensional array are?

- a) Matrix-Multiplication
- b) Minimum Spanning Tree
- c) Finding connectivity between nodes
- d) All of the mentioned

Answer: d

23. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4.     int ary[2][3];
5.     foo(ary);
6. }
7. void foo(int *ary[])
8. {
9.     int i = 10, j = 2, k;
10.    ary[0] = &i;
11.    ary[1] = &j;
12.    *ary[0] = 2;
13.    for (k = 0; k < 2; k++)
14.        printf("%d\n", *ary[k]);
15. }
```

- a) 2 2

- b) Compile time error
- c) Undefined behaviour
- d) 10 2

Answer: a

24. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4.     int ary[2][3];
5.     foo(ary);
6. }
7. void foo(int (*ary)[3])
8. {
9.     int i = 10, j = 2, k;
10.    ary[0] = &i;
11.    ary[1] = &j;
12.    for (k = 0; k < 2; k++)
13.        printf("%d\n", *ary[k]);
14. }
```

- a) Compile time error
- b) 10 2
- c) Undefined behaviour
- d) segmentation fault/code crash

Answer: a

25. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. foo(ary);
5. }
6. void foo(int **ary)
7. {
8. int i = 10, k = 10, j = 2;
9. int *ary[2];
10. ary[0] = &i;
11. ary[1] = &j;
12. printf("%d\n", ary[0][1]);
13. }
```

- a) 10
- b) 2
- c) Compile time error
- d) Undefined behaviour

Answer: d

26. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. int ary[2][3][4], j = 20;
5. ary[0][0] = &j;
6. printf("%d\n", *ary[0][0]);
7. }
```

- a) Compile time error
- b) 20
- c) Address of j
- d) Undefined behaviour

Answer: a

27. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {

4. int ary[2][3];
5. ary[][] = {{1, 2, 3}, {4, 5, 6}};
6. printf("%d\n", ary[1][0]);
7. }
```

a) Compile time error

b) 4

c) 1

d) 2

Answer: a

28. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. void foo();
5. printf("1 ");
6. foo();
7. }
8. void foo()
9. {
10. printf("2 ");
11. }
```

- a) 1 2
- b) Compile time error
- c) 1 2 1 2
- d) Depends on the compiler

Answer: a

29. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. void foo(), f();
5. f();
6. }
7. void foo()
8. {
9. printf("2 ");
10. }
11. void f()
12. {
13. printf("1 ");

14. foo();
15. }
```

- a) Compile time error as foo is local to main
- b) 1 2
- c) 2 1
- d) Compile time error due to declaration of functions inside main

Answer: b

30. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4. void foo();
5. void f()
6. {
7. foo();
8. }
9. f();
10. }
11. void foo()
12. {
13. printf("2 ");
14. }
```

- a) 2 2
- b) 2
- c) Compile time error
- d) Depends on the compiler

Answer: d

Explanation: Even though the answer is 2, this code will compile fine only with gcc. GNU C supports nesting of functions in C as a language extension whereas standard C compiler doesn't.

31. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo();
3. int main()
4. {
5. void foo();
6. foo();
7. return 0;
8. }
9. void foo()
10. {
```

```
11. printf("2 ");  
12. }
```

- a) Compile time error
- b) 2
- c) Depends on the compiler
- d) Depends on the standard

Answer: b

32. What is the output of this C code?

```
1. #include <stdio.h>  
2. void foo();  
3. int main()  
4. {  
5. void foo(int);  
6. foo(1);  
7. return 0;  
8. }  
9. void foo(int i)  
10. {  
11. printf("2 ");  
12. }
```

- a) 2
- b) Compile time error
- c) Depends on the compiler
- d) Depends on the standard

Answer: a

33. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo();
3. int main()
4. {
5. void foo(int);
6. foo();
7. return 0;
8. }
9. void foo()
10. {
11. printf("2 ");
12. }
```

- a) 2
- b) Compile time error
- c) Depends on the compiler
- d) Depends on the standard

Answer: b

34. What is the output of this C code?

```
1. include <stdio.h>
2. void m()
3. {
4. printf("hi");
5. }
6. void main()
7. {
8. m();
9. }
```

- a) hi
- b) Run time error

c) Nothing

d) Varies

Answer: a

35. What is the output of this C code?

```
1. #include <stdio.h>
2. void m();
3. void n()
4. {
5. m();
6. }
7. void main()
8. {
9. void m()
10. {
11. printf("hi");
12. }
13. }
```

a) hi

b) Compile time error

c) Nothing

d) Varies

Answer: b

36. What is the return-type of the function sqrt()

a) int

b) float

c) double

d) depends on the data type of the parameter

Answer: c

37. Which of the following function declaration is illegal?

- a)

```
double func();  
int main(){  
    double func();
```
- b)

```
double func(){};  
int main(){
```
- c)

```
int main()  
{  
    double func();  
}  
double func(){//statements}
```
- d) None of the mentioned

Answer: d

38. What is the output of this code having void return-type function?

```
1. #include <stdio.h>  
2. void foo()  
3. {  
4.     return 1;  
5. }  
6. void main()  
7. {  
8.     int x = 0;  
9.     x = foo();  
10.    printf("%d", x);  
11. }
```

- a) 1
- b) 0
- c) Runtime error
- d) Compile time error

Answer: d

39. What will be the data type returned for the following function?

```
1. #include <stdio.h>
2. int func()
3. {
4. return (double)(char)5.0;
5. }
```

- a) char
- b) int
- c) double
- d) multiple type-casting in return is illegal

Answer: b

40. What is the problem in the following declarations?

```
int func(int);
double func(int);
int func(float);
```

- a) A function with same name cannot have different signatures
- b) A function with same name cannot have different return types
- c) A function with same name cannot have different number of parameters
- d) All of the mentioned

Answer: d

41. The output of the code below is

```
1. #include <stdio.h>
2. void main()
3. {
4. int k = m();
5. printf("%d", k);
6. }
7. void m()
8. {
9. printf("hello");
10. }
```

- a) hello 5
- b) Error
- c) Nothing
- d) Junk value

Answer: a

42. The output of the code below is

```
1. #include <stdio.h>
2. int *m()
3. {
4. int *p = 5;
5. return p;
6. }

7. void main()
8. {
9. int *k = m();
10. printf("%d", k);
11. }
```

- a) 5

- b) Junk value
- c) 0
- d) Error

Answer: a

43. The output of the code below is

```
1. #include <stdio.h>
2. int *m();
3. void main()
4. {
5.     int *k = m();
6.     printf("hello ");
7.     printf("%d", k[0]);
8. }
9. int *m()
10. {
11.     int a[2] = {5, 8};
12.     return a;
13. }
```

- a) hello 5 8
- b) hello 5
- c) hello followed by garbage value
- d) Compilation error

Answer: c

Part B

1. What is the need for user defined function?
2. Write a multi function program
3. List the Elements of user defined function
4. Give short note on 2-D array processing.
5. Write a C program to find the length of a string.
6. HELLO encode it as IFMMP using array
7. Show the difference between actual and formal parameter in function with piece of code
8. Compare user define function vs System define function
9. List categories of function
10. Explain the concept of function call by reference with a sample program.
11. Explain the concept of function call by value with a sample program
12. Explain about functions with example
13. Discuss Character Arrays
14. Predict output of the following program (CL03)

```
int main(){
    inta[][]={{1,2},{3,4}};

    int i,j;
    for (i=0;i<2;i++)
    for(j=0;j<2;j++)
    printf("%d",a[i][j]);
    return 0;
}
```

15. Why array index starts from zero?

16. Contrast function declaration vs function definition.

17. List the advantages of functions.

nd

18. Program to Find the 2 Largest Elements in an Array.

19. Write a program to convert the given string (srm university) Lower to upper case

20. Advantages and limitations of multi dimensional array initialization

21. Explain call by value and call by reference with an example

22. Common programming errors in 2D arrays

12. Explain String functions with example

PART C

1. Array construction for student mark list for 100 students. output need to display register number, marks of

five subjects , CGPA and PASS/Fail Status

2. Explain in details about String Functions: gets(), puts(), getchar(), putchar(), printf(), with an example

programs

3. i)Print the given pattern using 2-D array

1234567

23456

345

4

ii) Program to divide one array into two arrays

4. i) What are strings in C? Write a C program to read a string in lowercase and convert it to

uppercase.

ii)) Illustrate call by value and call by reference with example for each.

5. i) Find the factorial of 10 using function recursion.

ii) What is a string? Explain any 5 string functions.

6. Explain in details about String Functions: atoi, strlen, strcat, strcmp with an example

7. Functions declaration and definition, Types: Call by Value & Call by Reference with example programs

8. Explain and write c program for Function with and without Arguments and no Return Values

9. Explain and write c program for Passing Array to Functions with return type, Recursion Functions

10. Write c program for Matrix Multiplication using Multi-dimensional array

11. Program to Delete duplicate elements from an array

INPUT

1 3 4 5 3

OUTPUT

1 3 4 5

b. Write a program to perform Matrix addition and Multiplication using 2-D arrays

12. i) State the importance of functions. List out the different types of Function

ii) Write a swap function using call by value and call by reference

iii) Write a C program to concatenate two strings.

iv) Write a program to read two strings and compare them and print a message that the first string

is equal, less or greater than the second one accordingly.

13. Write c program for Matrix addition and Matrix Transpose using multi dimensional array

14. Given a number , find whether it is a power of 2 or not

15. Write a C program to swap elements in cyclic order using call by reference.

UNIT 4

1. What is the output of this C code?

```
1. #include <stdio.h>
```

```
2. void foo(int*);
```

```
3. int main()
```

```
4. {
```

```
5. int i = 10;
```

```
6. foo(&i++);
```

```
7. }
```

```
8. void foo(int *p)
```

```
9. {
```

```
10. printf("%d\n", *p);
```

```
11. }
```

a) 10

b) Some garbage value

c) Compile time error

d) Segmentation fault/code crash

Answer: c

2. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo(int*);

3. int main()
4. {

5. int i = 10, *p = &i;
6. foo(p++);

7. }
8. void foo(int *p)

9. {
10. printf("%d\n", *p);

11. }
```

- a) 10
- b) Some garbage value
- c) Compile time error
- d) Segmentation fault

Answer: a

3. What is the output of this C code?

```
1. #include <stdio.h>
2. void foo(float *);
```

```

3. int main()
4. {

5. int i = 10, *p = &i;
6. foo(&i);

7. }
8. void foo(float *p)

9. {
10. printf("%f\n", *p);

11. }

```

- a) 10.000000
- b) 0.000000
- c) Compile time error
- d) Undefined behaviour

Answer: b

4. What is the output of this C code?

```

1. #include <stdio.h>
2. int main()

3. {
4. int i = 97, *p = &i;

5. foo(&i);
6. printf("%d ", *p);

7. }
8. void foo(int *p)

9. {
10. int j = 2;

```

```
11. p = &j;  
12. printf("%d ", *p);  
  
13. }
```

- a) 2 97
- b) 2 2
- c) Compile time error
- d) Segmentation fault/code crash

Answer: a

5. What is the output of this C code?

```
1. #include <stdio.h>  
2. int main()  
  
3. {  
4. int i = 97, *p = &i;  
  
5. foo(&p);  
6. printf("%d ", *p);  
  
7. return 0;  
8. }  
  
9. void foo(int **p)  
10. {  
  
11. int j = 2;  
12. *p = &j;  
  
13. printf("%d ", **p);  
14. }
```

- a) 2 2
- b) 2 97

- c) Undefined behaviour
- d) Segmentation fault/code crash

Answer: a

6. What is the output of this C code?

```
1. #include <stdio.h>
2. int main()
3. {
4.     int i = 11;
5.     int *p = &i;
6.     foo(&p);
7.     printf("%d ", *p);
8. }
9. void foo(int *const *p)
10. {
11.     int j = 10;
12.     *p = &j;
13.     printf("%d ", **p);
14. }
```

- a) Compile time error
- b) 10 10
- c) Undefined behaviour
- d) 10 11

Answer: a

7. What is the output of this C code?

```
1. #include <stdio.h>

2. int main()
3. {

4.     int i = 10;
5.     int *p = &i;

6.     foo(&p);
7.     printf("%d ", *p);

8.     printf("%d ", *p);
9. }

10. void foo(int **const p)
11. {

12.     int j = 11;
13.     *p = &j;

14.     printf("%d ", **p);
15. }
```

- a) 11 11 11
- b) 11 11 Undefined-value
- c) Compile time error
- d) Segmentation fault/code-crash

Answer: b

8. What is the output of the code below?

```
1. #include <stdio.h>
2. int main()
```

```

3. {
4. int i = 10;

5. int *const p = &i;
6. foo(&p);

7. printf("%d\n", *p);
8. }

9. void foo(int **p)
10. {

11. int j = 11;
12. *p = &j;

13. printf("%d\n", **p);
14. }

```

- a) 11 11
- b) Undefined behaviour
- c) Compile time error
- d) Segmentation fault/code-crash

Answer: a

9. Which of the following are correct syntaxes to send an array as a parameter to function:

- a) func(&array);
- b) func(#array);
- c) func(*array);
- d) func(array[size]);

Answer: a.

11. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. int k = 5;

5. int *p = &k;
6. int **m = &p;

7. printf("%d%d%d\n", k, *p, **m);
8. }
```

- a) 5 5 5
- b) 5 5 junk value
- c) 5 junk junk
- d) Run time error

Answer: a

12. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
3. {
4. int k = 5;

5. int *p = &k;
6. int **m = &p;

7. printf("%d%d%d\n", k, *p, **p);
8. }
```

- a) 5 5 5

- b) 5 5 junk value
- c) 5 junk junk
- d) Compile time error

Answer: d.

13. What is the output of this C code?

```
1. #include <stdio.h>

2. void main()
3. {

4. int k = 5;
5. int *p = &k;

6. int **m = &p;
7. **m = 6;

8. printf("%d\n", k);
9. }
```

- a) 5
- b) Compile time error
- c) 6
- d) Junk

Answer: c

14. What is the output of this C code?

```
1. #include <stdio.h>
2. void main()
```



```
3. {  
4. int a[3] = {1, 2, 3};  
  
5. int *p = a;  
6. int *r = &p;  
  
7. printf("%d", (**r));  
8. }
```

- a) 1
- b) Compile time error
- c) Address of a
- d) Junk value

Answer: b

15. What is the output of this C code?

```
1. #include <stdio.h>  
  
2. void main()  
3. {  
  
4. int a[3] = {1, 2, 3};  
5. int *p = a;  
  
6. int **r = &p;  
7. printf("%p %p", *r, a);  
  
8. }
```

- a) Different address is printed
- b) 1 2
- c) Same address is printed.
- d) 1 1

View Answer

Answer: c

Explanation: None.

16. How many number of pointer (*) does C have against a pointer variable declaration?

- a) 7
- b) 127
- c) 255
- d) No limits.

Answer: d

17. What is the output of this C code?

```
1. #include <stdio.h>

2. int main()
3. {

4. int a = 1, b = 2, c = 3;
5. int *ptr1 = &a, *ptr2 = &b, *ptr3 = &c;

6. int **sptr = &ptr1; //-Ref
7. *sptr = ptr2;

8. }
```

- a) ptr1 points to a
- b) ptr1 points to b
- c) sptr points to ptr2
- d) None of the mentioned

Answer: b

18. What is the output of this C code?

```
1. #include <stdio.h>

2. void main()
3. {

4. int a[3] = {1, 2, 3};
5. int *p = a;

6. int **r = &p;
7. printf("%p %p", *r, a);

8. }
```

- a) Different address is printed
- b) 1 2
- c) Same address is printed.
- d) 1 1

Answer: c

PART B

- 1) short notes on Passing Array Element to Function with an example
- 2) Explain Formal and Actual Parameters
- 3) Give the Advantages of using Functions ,
- 4) example program for Processor Directives and #define Directives
- 5) write a program for Pointer Declaration and dereferencing,
- 6) write a program for pointers, Void Pointers and size of Void Pointers
7. Write algorithm and pseudo code for leap year calculation. [CLO 1]
8. Contrast the declaration of break and continue in while and for loop with justification

[CLO 2]

9. Limitations of two dimensional arrays.

PART C

- 1) i) c program to read array elements and print the values with their address
ii) what is Pointers and address operator, explain with an example of Size of Pointer Variable and Pointer, Operator
- 2) Explain in details with an example Pointer Declaration and dereferencing, pointers, Void Pointers and size of Void Pointers
- 3) write program for Arithmetic Operations, Incrementing Pointers
- 4) what is Constant Pointers, Pointers to array elements and strings
- 5) what is function pointer & Array of Function Pointers with an example
- 6) write the c program for Accessing Array of Function Pointers, Null Pointers
- 7) Justify and explain the different size of data types using pointer variables with example program.
- 8) i) write a c program to count vowels and consonants in a string using pointer
ii) Write call by reference and call by value with an example program

UNIT 5

Part A

1. Which of the following are themselves a collection of different data types?

- a) string
- b) structures
- c) char
- d) all of the mentioned

View Answer

Answer: b

Explanation: None.

2. User-defined data type can be derived by_____

- a) struct
- b) enum
- c) typedef
- d) all of the mentioned

Answer: d

3. Which operator connects the structure name to its member name?

- a) -
- b) <-
- c) .
- d) Both <- and .

Answer: c

4. Which of the following cannot be a structure member?

- a) Another structure
- b) Function
- c) Array
- d) None of the mentioned

Answer: b

5. Which of the following structure declaration will throw an error?

- a) struct temp{}s;
main(){}
- b) struct temp{};
struct temp s;
main(){}
- c) struct temp s;
struct temp{};
main(){}
- d) None of the mentioned

Answer: d

6. What is the output of this C code?

- 1. #include <stdio.h>
- 2. struct student
- 3. {
- 4. int no;

```
5. char name[20];

6. }
7. void main()

8. {
9. struct student s;

10. s.no = 8;
11. printf("hello");

12. }
```

- a) Compile time error
- b) Nothing
- c) hello
- d) Varies

Answer: a

7. What is the output of this C code?

```
1. #include <stdio.h>

2. struct student
3. {

4. int no = 5;
5. char name[20];

6. };
7. void main()

8. {
9. struct student s;

10. s.no = 8;
11. printf("hello");
```

12. }

a) Nothing

b) Compile time error

c) hello

d) Varies

Answer: b

8. What is the output of this C code?

1. #include <stdio.h>

2. struct student

3. {

4. int no;

5. char name[20];

6. };

7. void main()

8. {

9. student s;

10. s.no = 8;

11. printf("hello");

12. }

a) Nothing

b) hello

c) Compile time error

d) Varies

Answer: c

9. What is the output of this C code?

```
1. #include <stdio.h>

2. void main()
3. {

4. struct student
5. {

6. int no;
7. char name[20];

8. };
9. struct student s;

10. s.no = 8;
11. printf("%d", s.no);

12. }
```

- a) Nothing
- b) Compile time error
- c) Junk
- d) 8

Answer: d

10. Can the above code be compiled successfully?

```
1. #include <stdio.h>

2. struct p
3. {
```

```
4. int k;  
5. char c;  
  
6. float f;  
7. };  
  
8. int main()  
9. {  
  
10. struct p x = {.c = 97, .f = 3, .k = 1};  
11. printf("%f\n", x.f);  
  
12. }
```

- a) Yes
- b) No
- c) Depends on the standard
- d) Depends on the platform

Answer: c

11. What is the output of this C code?

```
1. #include <stdio.h>  
  
2. void main()  
3. {  
  
4. struct student  
5. {  
  
6. int no;  
7. char name[20];  
  
8. };  
9. struct student s;
```

```
10. no = 8;  
11. printf("%d", no);  
  
12. }
```

- a) Nothing
- b) Compile time error
- c) Junk
- d) 8

Answer: b

12. Number of bytes in memory taken by the below structure is

```
1. #include <stdio.h>  
  
2. struct test  
3. {  
  
4. int k;  
5. char c;  
  
6. };
```

- a) Multiple of integer size
- b) integer size+character size
- c) Depends on the platform
- d) Multiple of word size

Answer: a

13. What is the output of this C code?

```
1. #include <stdio.h>
2. struct
3. {
4. int k;

5. char c;
6. };

7. int main()
8. {

9. struct p;
10. p.k = 10;

11. printf("%d\n", p.k);
12. }
```

- a) Compile time error
- b) 10
- c) Undefined behaviour
- d) Segmentation fault

14. What is the output of this C code?

```
1. #include <stdio.h>

2. struct
3. {

4. int k;
5. char c;

6. } p;
7. int p = 10;

8. int main()
9. {
```

```
10. p.k = 10;
11. printf("%d %d\n", p.k, p);

12. }
```

- a) Compile time error
- b) 10 10
- c) Depends on the standard
- d) Depends on the compiler

Answer: a

15. What is the output of this C code?

```
1. #include <stdio.h>

2. struct p
3. {

4. int k;
5. char c;

6. };
7. int p = 10;

8. int main()
9. {

10. struct p x;
11. x.k = 10;

12. printf("%d %d\n", x.k, p);
13. }
```

- a) Compile time error
- b) 10 10
- c) Depends on the standard

d) Depends on the compiler

Answer: b

16. What is the output of this C code?

```
1. #include <stdio.h>
2. struct p
3. {
4. int k;

5. char c;
6. float f;

7. };
8. int p = 10;

9. int main()
10. {

11. struct p x = {1, 97};
12. printf("%f %d\n", x.f, p);

13. }
```

a) Compile time error

b) 0.000000 10

c) Somegarbage value 10

d) 0 10

Answer: b

17. What is the output of this C code(according to C99 standard)?

advertisement

```
1. #include <stdio.h>
2. struct p

3. {
4. int k;

5. char c;
6. float f;

7. };
8. int main()

9. {
10. struct p x = {.c = 97, .f = 3, .k = 1};

11. printf("%f\n", x.f);
12. }
```

- a) 3.000000
- b) Compile time error
- c) Undefined behaviour
- d) 1.000000

Answer: a

18. What is the output of this C code(according to C99 standard)?

```
1. #include <stdio.h>
2. struct p

3. {
4. int k;

5. char c;
6. float f;
```

```

7. };
8. int main()

9. {
10. struct p x = {.c = 97, .k = 1, 3};

11. printf("%f \n", x.f);
12. }

```

- a) 3.000000
- b) 0.000000
- c) Compile time error
- d) Undefined behaviour

Answer: b

19. What is the output of this C code(according to C99 standard)?

```

1. #include <stdio.h>
2. struct p

3. {
4. int k;

5. char c;
6. float f;

7. };
8. int main()

9. {
10. struct p x = {.c = 97};

11. printf("%f\n", x.f);
12. }

```

- a) 0.000000

- b) Somegarbagevalue
- c) Compile time error
- d) None of the mentioned

Answer: a

PART B

1. Write algorithm and pseudo code for leap year calculation.
2. Contrast the declaration of break and continue in while and for loop with justification
3. Limitations of two dimensional arrays.
4. Explain about nested pre-processor MACRO
5. Categorize the basic operations that can be performed on a file with suitable declarations.
6. Write a c program to demonstrate double pointer for accessing the value of another pointer
7. Explain about file operations and mode with syntax.
- 8) explain shortly about Initializing Structure, Declaring structure, variable,
- 9) give the example of Structure using typedef, Accessing members
- 10)what is Nested structure, Accessing elements in a structure array,
- 11)write c program for Array of structure, Accessing elements in a structure array
12. Explain about file operations and mode with syntax.
13. Give a brief note about preprocessor directives.
14. Differentiate between a union and a structure.
15. Write a program to read a file character by character, and display it simultaneously on the screen
- 16) write a program for Passing Array of structure to function,

- 17) write a program for Array of pointers to structures
18. Explain about nested pre-processor MACRO
19. Categorize the basic operations that can be performed on a file with suitable declarations.
20. Write a c program to demonstrate double pointer for accessing the value of another pointer

PART C

- 1) explain in details about Bit Manipulation to structure and Pointer, to structure, Union Basic and declaration with an example
- 2) explain in details about Accessing Union Members Pointers to Union, Dynamic memory allocation, malloc, realloc, free with an example
- 3) explain in details about Allocating Dynamic Array, Multidimensional array using dynamic memory allocation with an example
- 4) Explain in details about array of structures and accessing elements in a structure array.
- 5) Write a c program to insert a line at the end of text file
- 6) write a program to copy the content from one file to another file
- 7) explain and write the program for library management system using union
- 8) write a c program for file: opening, defining, closing, File Modes, File Types, Writing contents into a file
- 9) write a c program for Reading file contents, Appending an existing file
- 10) Differentiate between gets() and scanf()
- (11) Write a program to convert the given string "hello world" to "dlrow olleh".
12. Write a program that passes a pointer to a structure to a function
- 13 Briefly discuss about file operations in C

14) c program to read array elements and print the values with their address

32(a) Write a program that passes a pointer to a structure to a function

15) Justify and explain the different size of data types using pointer variables with example program.

[CLO 5]

(OR)

16) write a c program to count vowels and consonants in a string using pointer

32(b) Briefly discuss about file operations in C [CLO 5]

17) Write call by reference and call by value with an example program

18) a) i) Explain in details about array of structures and accessing elements in a structure array.

ii) Write a c program to insert a line at the end of text file

19) i) write a program to copy the content from one file to another file

ii) explain and write the program for library management system using union

SRM Institute of Science and Technology, Ramapuram
Department of Computer Science and Engineering

18CSS101J-Programming for Problem Solving

UNIT 1 MCQ

1. What is the value of variable z?

```
int x;  
int y;  
int z;  
x=3;  
y=4;  
z = ++x * y++;
```

- a) 9
- b) 12
- c) 16
- d) 20

Answer: c

2. To represent two different conditions which of the following is used.

- a) Rectangle
- b) Diamond
- c) Circle
- d) Parallelogram

Ans B

3. Give the output of the following program?

```
#include <stdio.h>
main()
{

int a=9, b=9;
a=b++;
b=a++;
b=++b;
printf("%d %d",a,b);

}
```

- a) 9,9
- b) 10,10
- c) 9,10
- d) 10,9

Answer: b

4. In the below mentioned operators, which of the following has the highest precedence?

- a) Unary +
- b) *
- c) >=
- d) ==

Answer: a

5. What is the output of the given code?

```
counter = 2
while counter < 68
puts counter
counter**=2
end
```

- a) 2 4 16 64
- b) 2 4 16
- c) 2 4 16 256
- d) None of the mentioned

Answer: b

6. Which of the following is a valid assignment operator?

- a) +=
- b) -=
- c) *=
- d) All of the mentioned

Answer: d

7. The output for the following C code is?

```
#include <stdio.h>
void main()
{

int x = 97;
int y = sizeof(x++);
printf("x is %d", x);
```

- ```
}
a) x is 97
b) x is 98
c) x is 99
d) Run time error
```

Answer: a

8. Give the correct answer for the following C code?

```
#include <stdio.h>
void main()

{
int x = 4, y, z;
y = --x;
z = x--;
printf("%d%d%d", x, y, z);
```

- ```
}
a) 3 2 3
b) 2 2 3
c) 3 2 2
d) 2 3 3
```

Answer: d

9. What will be the output of the following C code?

```
#include <stdio.h>
void main()
{

int x = 4;
int *p = &x;
int *k = p++;
int r = p - k;
printf("%d", r);

}
```

- a) 4
- b) 8
- c) 1
- d) Run time error

Answer: c

10. Among the following options, which is not a valid representation in bits?

- a) 8-bit
- b) 24-bit
- c) 32-bit
- d) 64-bit

Answer: b

11. Name the entities whose values can be changed?

- a) Constants
- b) Variables
- c) Modules
- d) Tokens

Answer: b

12. Which is not a basic data type ?

- a) float
- b) int
- c) real

- d) char

Answer: c

13. Name the correct default value of a Boolean type?

- a) 0
- b) 1
- c) True
- d) False
- e) -1

Answer: Option D

14. Which of the statements given below are correct?

- 1. We can assign values of any type to variables of type object.
- 2. When a variable of a value type is converted to object, it is said to be unboxed.
- 3. When a variable of type object is converted to a value type, it is said to be boxed.
- 4. Boolean variable cannot have a value of null.
- 5. When a value type is boxed, an entirely new object must be allocated and constructed.

- a) 2, 5
- b) 1, 5
- c) 3, 4
- d) 2, 3

Answer: Option B

15. Name the correct ways to set a value as 3.14 in a variable pi such that it cannot be modified?

- a) `float pi = 3.14F;`
- b) `#define pi 3.14F;`
- c) `const float pi = 3.14F;`
- d) `const float pi; pi = 3.14F;`
- e) `pi = 3.14F;`

Answer: C

16. why the variable names beginning with underscore is not encouraged?

- a) It is not standard form
- b) To avoid conflicts since assemblers and loaders use such names
- c) To avoid conflicts since library routines use such names
- d) To avoid conflicts with environment variables of an operating system

Ans:c

17. Mention the variable name which is not valid in C ?

- a) `int number;`
- b) `float rate;`
- c) `int variable_count;`
- d) `int $main;`

Ans:d

18. Which among the following options is true for the variable names in C?

- a) They can contain alphanumeric characters as well as special characters
- b) It is not an error to declare a variable to be one of the keywords (like goto, static)
- c) Variable names can't start with a digit
- d) Variable can be of any length

Ans:c

19. Give the output for the following code?

```
#include <stdio.h>
int main()
{
    int main = 5;
    printf("%d", main);
    return 0;
}
```

- a) compile-time error
- b) run-time error
- c) run without any error and prints 5
- d) experience infinite looping

Ans:c

20. Which of the following can never be a variable name in C?

- a) friend
- b) true
- c) volatile
- d) export

Ans: c

21. Mention the User-defined data type from the following?

- a) struct {char name[10], int age};
- b) typedef enum {Mon, Tue, Wed, Thu, Fri} Workdays;
- c) typedef int Boolean;
- d) all of the mentioned

Answer:d

22. What is short int in C programming?

- a) Basic datatype of C
- b) Qualifier
- c) short is the qualifier and int is the basic data type
- d) All of the mentioned

Ans:c

23. What is the output of this C code?

```
#include <stdio.h>
int main()
{
    signed char chr;

    chr = 128;
    printf("%d\n", chr);
    return 0;
}
```

- a) 128
- b) -128
- c) Depends on the compiler
- d) None of the mentioned

Ans:b

24. Which of the data types have size that is in variable?

- a) int
- b) struct
- c) float
- d) double

Ans:b

25. Mention the User-defined data type?

- a) typedef int Boolean;
- b) typedef enum {Mon, Tue, Wed, Thu, Fri} Workdays;
- c) struct {char name[10], int age};
- d) All of the mentioned

Answer: Option D

26. Output of this C code is ?

```
int main()
```

```

{
char chr;
chr = 128;
printf("%d\n", chr);
return 0;
}

```

- a) 128
- b) - 128
- c) Depends on the compiler
- d) None of the mentioned

Answer: Option B

27. What is the output for the C program ?

```

#include <stdio.h>
int main()
{

static int i=5;
if (--i){

printf("%d ",i);

main();
}

}

```

- a) 4 3 2 1
- b) 1 2 3 4
- c) 4 4 4 4
- d) 0 0 0 0

Answer: A

28. Which of the following is not a storage class in C?

- a) automatic storage class
- b) register storage class
- c) external storage class
- d) internal storage class

Ans: d

29. Scope of which of these in C is not local to the block in which it is defined

- a) automatic storage class
- b) register storage class
- c) static storage class
- d) external storage class

Ans d

30. Single line comments are denoted by ____.

- a) //
- b) \\\
- c) /*
- d) */

Answer: A

31 . How many comment lines are present in the below program -

```
main()
{
printf(\"Hello World in C //Comment \");
printf(\"Hello World in C \"); //Comment
//Comment printf(\"Hello World in C \");
}
```

- a) 1
- b) 0
- c) 2
- d) 3

Answer: C

32. Comments are case Sensitive !

- a) True
- b) False

Answer: B

33. What is the use of getchar()?

- a) The next input character each time it is called
- b) EOF when it encounters end of file
- c) The next input character each time it is called EOF when it encounters end of file
- d) None of the mentioned

Answer: c

34. Name the method which uses a list of well-defined instructions to complete a task

- a) Algorithm
- b) Flowchart
- c) Programs
- d) Functions

Ans C

35. Sequence logic will not be used while

- a) Accepting input from the user
- b) Giving output to the user
- c) Adding two numbers
- d) Comparing two sets of data

Ans B

36. Flowchart and algorithms are used for

- a) Better programming
- b) Easy understanding and debugging
- c) Efficient coding
- d) All the above

Ans d

37. What is the default initial value of static storage class?

- a) 0
- b) garbage value
- c) 1
- d) -1

Ans a

38. Resolving errors in a program is known as...

- a) Debugging
- b) Refixing
- c) Error Checking
- d) Problem Solving

Ans a

39. Many features of C were derived from an earlier language called ____.

- a) FORTRAN
- b) BASIC

- c) B
- d) PASCAL

Ans C

40. C Programming was created at ?

- a) MIT University
- b) AT&T Bell Laboratory
- c) L&T Laboratory
- d) Haward University

Ans B

41. What will be the output of the following pseudocode?

Integer a, b

Set a = 15, b = 7

a = a mod (a - 3)

b = b mod (b - 3)

a = a mod 1

b = b mod 1

Print a + b

- a) 15
- b) 7
- c) 2
- d) 0

[Note-mod finds the remainder after the division of one number by another. For example, the expression "5 mod 2" leaves a quotient of 2 and a remainder of 1]

Answer: d

42. What will be the output of the following pseudocode?

Integer a, b, c

Set b = 5, a = 2, c = 2

if(b>a && a>c && c>b)

b = a + 1

Else

a = b + 1

End if

Print a + b + c

[Note-&&: Logical AND - The logical AND operator (&&) returns the Boolean value true(or 1) if both operands---. If (x) gets executed if the value if(), i.e., x is not zero]

- a) 2
- b) 13

- c) 26
- d) 5

Ans:B

UNIT 2 MCQ

1. Does rational operators in the C language are calculated with the short circuit?

- a) True
- b) False
- c) Depends on the compiler
- d) Depends on the standard

Answer: a

2. What is the outcome of relational or logical operation in C?

- a) True or False
- b) 0 or 1
- c) 0 if an expression is false and any positive number if an expression is true
- d) None of the mentioned

Answer: b

3. What will be the output of the following C code?

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int a = 20, b = 10, c = 10;
```

```
int d;
```

```
d = b + c == a;
```

```
printf("%d", d);
```

```
}
```

a)Syntaxerror

b)1

c)5

d) 10

Answer: b

4. What will be the output of the following C program.

```
#include<stdio.h>

int main()

{ int a=-1, b=1, c, d;

c = !a && b;

d = !a || b;

printf("%d %d %d %d",a,b,c,d);

return 0;
}
```

- a) -1 1 1 0
- b) -1 1 0 0
- c) -1 1 1 1
- d) -1 1 0 1

Ans:- d

5. Which operator of the following is not a logical or relational operator?

- a) !=
- b) ==
- c) ||
- d) =

Answer: d

6. Relational operators will not be used in _____

- a) structure
- b) long
- c) strings
- d) float

Answer: a

7. Which are the operators used to relate the values of operands to create logical value in

C
language?

- a) Logical operator
- b) Relational operator
- c) Assignment operator
-) None of the above

Answer: b

8. In the following operators which one performs processes on data in binary level?

- a) Logical operator
- b) Bitwise operator
- c) Additional operators
- d) None of the above

Answer: b

9. How many number of operands needed to unary operator logical not (!)?

- a) 4
- b) 3
- c) 2
- d) 1

Answer: d

10. An expression comprises relational, arithmetic and assignment operators. If interpolations are not indicated, the order of valuation of the operators will be:

- a) Assignment, arithmetic, relational
- b) Relational, assignment, arithmetic
- c) Assignment, relational, arithmetic
- d) Arithmetic, relational, assignment

Answer: d

11. Which group has higher priority between (>,<=,<,<=) and (==,!)?

- a) (>, >=, <, <=) has lower priority (==, !=)
- b) (>, >=, <, <=) has higher priority (==, !=)
- c) (>, >=, <, <=) has equal priority with (==, !=)
- d) None of the above

Answer: b

12. The || and && operators

- A. Relate two Boolean values
- B. Relate two numeric values
- C. Combine two Boolean values
- D. Combine two numeric values

Answer: C

13. Increment operators need L-value Expression What will be the output of the following program?

```
#include<stdio.h>
int main()
{
    int i = 10;

    printf("%d", ++(-i));
    return 0;
}
```

- A) 11
- B) 10
- C) -9
- D) None - l-value required

Answer: D

14. What will be the output of the following C program?

```
#include <stdio.h>
int main()
{
    int a = 3, b = -8, c = 2;

    printf("%d", a % b / c);
}
```

```
return 0; }
```

- A) 1
- B) 15
- C) -9
- D) 12

Answer: A

15. What will be the output of the following C code?

```
#include <stdio.h>
```

```
void main()  
{ double x = 123828749.66;
```

```
int y = x;  
printf("%d\n", y);  
printf("%lf\n", y);}
```

- a) 0, 0.0
- b) 123828749, 123828749.66
- c) 12382874, 12382874.0
- d) 123828749, 0.000000

Answer: d

16. Which of the following selection statement test only equality?

- a) if
- b) switch
- c) if & switch
- d) none of the mentioned

Answer: b

17. Which of the following will execute the body of loop condition is false?

- a) do-while
- b) while
- c) for
- d) none of the mentioned

Answer: a

18. Which of this following statement is incorrect?

- a) switch statement is more efficient than a set of nested ifs
- b) two case constants in the same switch can have identical values
- c) switch statement can only test for equality, whereas if statement can evaluate any type of Boolean expression
- d) it is possible to create a nested switch statements

Answer: b

19. The break statement is used to quite from:

- a. DO loop
- b. FOR loop
- c. SWITCH STATEMENT
- d. All of above.

Answer: d

20. The advantage of a SWITCH statement over an ELSE-IF statement:

- a. A default condition can be used in the SWITCH.
- b. It is easier to understand.
- c. Several conditions can cause one set of statements to be executed in a SWITCH.
- d. Several diffeent statements can be executed in a SWITCH.

Answer: b

21. The comma (,) operator is primarily used in conjunction with:

- a. FOR statement.
- b. IF-ELSE statement.
- c. DO-While statement.
- d. All of above.

Answer: a

22. The GOTO statement is used:

- a. To permit two dissimilar expressions to appear in condition where only one expression would ordinarily use.
- b. to dismiss loops or to exit from a switch.
- c. an unconditional transfer of control to a named label.
- d. to carry out a logical test and then take one of two possible actions, depending upon the

outcome of a test.

Answer: c

23. What is an Array in C language?

- A) A group of elements of same data type.
- B) An array contains more than one element
- C) Array elements are stored in memory in continuous or contiguous locations.
- D) All the above.

Answer :d

24. An array Index starts with?

- A) -1
- B) 0
- C) 1
- D) 2

Answer :b

25. What is the output of c program?

```
Int main()  
{  
    Int a[]={5,2,3,4};  
    Int b[4]={10,6,7,8};  
    Printf("%d,%d", a[0], b[0]);  
}
```

- A) 5,10
- B) 2,6
- C) 0 0
- D) Compiler error

Answer : A

26. Which is correct format for initializing an array in C?

- a) `int arr[5] = (1,2,3,4,5);`
- b) `int arr(3) = {1,2,3,4,5};`
- c) `int arr[3] = {1,2,3,4,5};`
- d) `int arr(3) = (1,2,3,4,5);`

Answer: c

27. Which of the following concepts make extensive use of arrays?

- a) Binary trees

- b) Scheduling of processes
- c) Caching
- d) Spatial locality

Answer: d

28. What are the disadvantages of arrays?

- a) Data structure like queue or stack cannot be implemented
- b) There are chances of wastage of memory space if elements inserted in an array are lesser than the allocated size
- c) Index value of an array can be negative
- d) Elements are sequentially accessed

Answer: b

29. Array is accessed by _____ elements.

- a) randomly
- b) sequentially
- c) exponentially
- d) logarithmically

Answer: a

30. What is the output of C program with arrays?

```
int main()

{char str[]={"C","A","T","\0"}};

printf("%s",str);

return 0;}
```

- A) C
- B) CAT
- C) CAT\0
- D) Compiler error

Answer: D

31. What is the output of C program with string arrays?

```
int main()
```

```
{ char *p1 = "GOAT";  
  
char *p2;  
  
p2 = p1;  
  
p2="ANT";  
  
printf("%s", p1);  
  
}
```

- A) ANT
- B) GOAT
- C) G
- D) A

Answer: B

32. A one-dimensional array contains one-dimensional arrays is called

- A. Two-dimensional array
- B. Multi-casting array
- C. Multi-dimensional array
- D. Three-dimensional array

Answer: A

33. Objects in a sequence that have the same type, is called

- A. Arrays
- B. Operators
- C. Functions
- D. Stacks

Answer: A

34. In an array starting from the beginning and inspecting each element one after other for finding an object, until the object is found is called as

- A. Linear search algorithm
- B. Searching
- C. Inspecting
- D. All of them

Answer: A

35. Which of the following technique is used to find value in an array?

- A. Binary search algorithm
- B. Bubble sort
- C. Linear search algorithm
- D. All of them

Answer: d

36. Programmer's uses _____ for defining their own types.?

- A. Algorithms
- B. Operators
- C. Enumerations
- D. None of them

Answer: c

37. A 1-D array is considered as_____?

- A. Complex
- B. Sequential
- C. Linear
- D. Both C and B

Answer: d

38. Array comprising elements are numbered as 0,1,2,3? These numbers are called as

- A. Subscripts of the array
- B. Index values
- C. Members of an array
- D. Both A and B

Answer: d

39. Find the output of the code

```
#include<stdio.h>
void main()
{
int m[5]={10,20,30,40,50};
int a,b,c;
a=++m[1];
b=m[1]++
c=m[i++];

printf("%d,%d,%d",a,b,c);
}
```

- A. 21,20,30

- B. 20,21,30
- C. 22,21,30
- D. 20,21,40

Answer : C

40. What will be the resulting array after rotating `arr[]={2, 4, 6, 8, 10}` by 2?

- A. 4, 6, 8, 10, 2
- B. 10, 2, 4, 6, 8
- C. 8, 10, 2, 4, 6
- D. 6, 8, 10, 2, 4

Answer : D

41. Which one is used in arrays extensively

- A. Scheduling
- B. Caching
- C. Spatial Locality
- D. Binary trees

Answer : C

42. What is the size of `int arr[10]` if the `int` is of 4 bytes?

- A. 4
- B. 40
- C. 10
- D. 14

Answer : B

43. The resulting array after reversing `arr[]={21,23,25,27,29}` is

- A. 29,27,25,23,21
- B. 25,27,29,21,23
- C. 27,29,21,23,25
- D. 23,25,27,29,21

Answer : A

44. Maximum number of dimensions an array have

- A. 8
- B. 4
- C. No limit
- D. 50

Answer : C

45. What does `int arr[30]` mean?

- A. Array of size 30
- B. Integer Array of size 30
- C. Array of size 30 that have integer address
- D. None

Answer : A

UNIT 3 MCQ

1. ____ quotes is used for character array initialization.

- (A) Any
- (B) Double
- (C) No
- (D) Single

Answer: B

2. Data type for array in C belongs to-----

- (A) Primitive
- (B) User defined
- (C) Derived
- (D) Basic

Answer: C

3. To compare the two strings which of the following one is used

- (A) `Strcmp ()`
- (B) `Strcpy ()`
- (C) `Stcp()`
- (D) `Stcmp()`

Answer: A

4. ----- Array example is matrix

- (A) Single
- (B) 2-D
- (C) 3-D
- (D) Multi-dimensional

Answer: B

5. Array Size should be always -----

- (A) Positive
- (B) Negative

- (C) Whole number
- (D) Real number

Answer: A

6. The first subscript of 2D array represents the ----- size

- (A) Row
- (B) Column
- (C) Object
- (D) Diagonal

Answer: A

7. _____ array is an array in which each element is itself an array

- (A) 1-D
- (B) 2-D
- (C) 4-D
- (D) 5-D

Answer: B

8. ----- character type is used for indicating the terminator of the string.

- (A) Float
- (B) Int
- (C) Null
- (D) String

Answer: C

9. How many Array types available?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Answer: B

10. For write function, how many parameters required?

- (A) Two
- (B) Three
- (C) Four
- (D) Five

Answer: A

11. Arranging of elements in array with respect to ascending or descending is called ____

- (A) Merging
- (B) Filtering
- (C) Sorting
- (D) Ordering

Answer: C

12. The position of array elements always begins from ____

- (A) One
- (B) Two
- (C) Three
- (D) Zero

Answer: D

13. What will be the output of the following C code?

```
#include <stdio.h>

int main()

{

ARRAY1(ary);

}

void ARRAY1(int **ary)

{

int i = 10, k = 10, j = 2;

int *ary[2];

ary[0] = &i;

ary[1] = &j;

printf("%d\n", ary[0][1]);

}
```

a.22

b. Compile time error

c. Undefined behavior

d.102

Answer: c

14. What will be the output for the following program?

```
#include <stdio.h>

int main()

{

int ary[2][3];

ary[][] = {{1, 2, 3}, {4, 2, 6}};

printf("%d\n", ary[1][0]);

return 0;

}
```

- a. 4
- b. Compile time error
- c. 2
- d. 1

Answer: a

15. `int (*b)[6];`

- a. An array “b” of pointers
- b. A pointer “b” to an array
- c. A ragged array
- d. None of the mentioned

Answer: b

15. Statement on the following 2 arrays with respect to M and N.

```
int *x1[8];
```

```
int *(x2[8]);
```

M. Array of pointers

N. Pointer to an array

A) x1 is M x1 is N

B) x1 is M, x2 is M

C) x1 is N, x2 is M

D) x1 is N, x2 is N

Answer: b

16. Which of the following is not possible statically in C?

a. Jagged array

b. Rectangular array

c. Cuboidal array

d. Multidimensionl Array.

Answer:D

17. What will be the output of the following C code?

```
#include <stdio.h>
```

```
void main()
```

```
{
```

```
int a[2][3] = {1, 3, 5, 7, 9};
```

```
int i = 0, j = 0;
```

```
for (i = 0; i < 2; i++)
```

```
for (j = 0; j < 3; j++)
```

```
printf("%d", a[i][j]);
```

```
}
```

a.135790

b.13579junk

c.135799

d. Run time error

Answer: a

18. Write the output of the following C code?

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
char *S1 = "hello, world";
```

```
char *S2 = "hello, world";
```

```
if (strcmp(S1, S2))
```

```
printf("equal");
```

```
else
```

```
printf("unequal");
```

```
}
```

a.equal

b.Unequal

c.Compilation errot

d. Depends on the compiler

Answer: b

19. Write the output for the below code.

```
#include <stdio.h>

int main()

{

char *str = "hello, word\n";

printf("%d", strlen(str));

}
```

a. Compilation error

b. Undefined behavior

c.12

d.11

Answer: c

20. Always----- function adds null character

a. strcmp()

b. strcat()

c. strcpy()

d. strrev()

Answer: b

21.----- function is not declared in math.h ?

a) and()

b)pow()

c)exp()

d) acos()

Answer: a

22. Which one is correct for multidimensional array example.

- a. `int[][] code = {{1,2},{3,4,5}};`
- b. `int[2][] code = {{1,2},{3,4,5}};`
- c. `int[][] code={1,2,3,4,5};`
- d. all

Answer: A

23. The function function1 is called _____ if it calls the same function function1.

- A. indirect recursive
- B. direct recursive
- C. Both A and B
- D. None of the above

Answer : B

24. Iteration not requires more system memory than recursion.

- A. TRUE
- B. FALSE
- C. Can be true or false
- D. Can not Say

Answer : A

25. Write the output of the following C code?

```
#include<stdio.h>
```

```
main()
```

```
{
```

```
int n;
```

```

n=f1(3);

printf("%d",n);

}

f1(int x)

{

int b;

if(x==1)

return 1;

else

b=x*f1(x-1);

return b;

}

```

a.6

b.4

c.12

d.10

Answer: a

26. Find the error of following function give on compilation ?

```

f(int a,int b)
{
int b;
a = 20;
return a;
}

```

- a. Missing parentheses in return statement
- b. Function should be define as `int f(int a, int b)`
- c. Redclaration if b
- d. No error

Answer:c

27. It is essential to declare the sort of a function within the calling program on the off chance that the function

- a.Returns an integer
- b.Returns a non-integer value
- c.is not defined in the same file
- d.None of these.

Answer:b

28. What is meant by void?

- a. Representation of NULL data type
- b. Representation of void pointer
- c. Error
- d. None of above

Answer: A

29. If the two strings are not identical, then `strcmp()` function returns

- a. -1
- b. 0
- c. 1
- d. Yes

Answer: c

30.What is the Format specifier used to print a integer array in C Printf or Scanf function.?

A) %c

B) %C

C) %s

D) %d

Answer:D

31. Write the output for the below code.

```
int main()

{

char ary[]="SRM IST";

printf("%s",ary);

return 0;

}
```

A) S

B) SRM

C) SRMIST

D) SRM IST

Answer : D

32.How do you convert this char array to string.?

```
char str[]={ 'g','o','o','d'};
```

A) str[4] = 0;

B) str[4] = '\0'

C) str[]={ 'g','o','o','d','\0'};

D) All the options

Answer : D

33.Find the output?

```
int main()

{

char str[]={ "G", "O", "D", "\0" };

printf("%s",str);

return 0;

}
```

- A) Compiler error
- B) GOD
- C) GOD\0
- D)G

Answer : A

34.What is the output of C Program with arrays.?

```
int main()

{

char str[100];

scanf("%s", str);

printf("%s",str);

return 0;

}

//input: WEST INDIES
```

- A) WEST
- B) WEST INDIES

C)W

D) Compiler error

Answer : D

35. What is the output of C Program with functions.?

```
int main()

{

FUN();

printf("LTD ");

return 0;

}


void FUN()

{

printf("HDFC ");

}
```

A) HDFC LTD

B) LTD HDFC

C) LTD

D) Compiler error

Answer : D

36. In C program, Function return ----- number of value at a time.?

A) Only One Value

B) Maximum of two values

C) Maximum of three values

D) Maximum of 8 values

Answer: A

37. Select the types of function

A) Library Functions

B) User Defined Functions

C) Both Library and User Defined

D) None

Answer: C

38. What is the constrain for number of functions in a C Program.?

A) 12

B) 24

C) 56

D) No limit

Answer: D

39. What is the least number of functions to be present in a C Program?

A) 1

B) 3

C) 4

D) 2

Answer: A

40. Each C Program ought to contain which function.?

- A) main()
- B) show()
- C) scanf()
- D) printf()

Answer: A

UNIT 4 MCQ

1. How will you declare the function which is intended to receive an array as an argument
- a. return_type function(type arrayname[])
 - b. return_type function(type arrayname[SIZE])
 - c. return_type function(type *arrayname)
 - d. All of the above

Ans: d

2. To store the array returned from the function, we can define a _____ which points to that array.
- a. Structure
 - b. Array
 - c. Pointer
 - d. List

Ans: c

3. The difference between Actual Parameters and Formal Parameters is that Actual Parameters are the values that are _____ the function when it is invoked while Formal Parameters are the variables defined by the function that _____ values when the function is called.
- a. Passed to, receives
 - b. Received by, pass
 - c. Prints, process
 - d. Process, prints

Ans: a

4. The advantages of using functions are:
- a. Avoid repetition of codes.
 - b. Increases program readability.

- c. Divide a complex problem into simpler ones.
- d. All of the above

Ans: d

5. The C preprocessor is a _____ that is used automatically by the C compiler to transform your program before actual compilation.

- a. Macro processor
- b. Microprocessor
- c. Macro controller
- d. Micorcontroller

Ans: a

6. Proprocessordirecives are executed _____ compilation.

- a. Before
- b. After
- c. During
- d. None of the above

Ans: a

7. The _____ preprocessor directive is used to paste code of given file into current file.

- a. #define
- b. #include
- c. #ifdef
- d. #pragma

Ans: b

8. A macro is a _____ which is replaced by the value of macro. Macro is defined by _____ directive.

- a. segment of code, #pragma
- b. file, #define
- c. segment of code, #define
- d. none of the above

Ans: c

9. _____ is used to undefine a macro definition.

- a. #undef
- b. #unfed
- c. #defun
- d. #undef

Ans: d

10. What is the output of the following program?

```
#include<stdio.h>
#ifndef __MATH_H
#error First include then compile
#else
void main(){

float a;
a=sqrt(7);

printf("%f",a);
}
#endif
```

- a. 2.64575
- b. 49
- c. 0.7
- d. None of the above

Ans: a

11. The _____ directive is used by the compiler to offer machine or operating-system feature.

- a. #define
- b. #elif
- c. #include
- d. #pragma

Ans: d

12. Calculate the output of the following program?

```
#include<stdio.h>
#define Area(x) x*x
#define Costpaint(x,y,z) (z*y + Area (x))
void main()
{
    int A = 8, B= 6, C = 4;
    clrscr();
    printf("The area of square= %d\n", Area(A));
    printf("Cost of paint= %d\n", Costpaint(A,B,C));
}
```

- a. 64
- b. 88
- c. 32
- d. 76

Ans: b

13. Identify the value that gets printed in the following program

```
#include <stdio.h>
int main()
{

int a=10; //variable declaration
int *p; //pointer variable declaration
p=&a; //store address of variable a in pointer p

printf("Address stored in a variable p is:%x\n",p); //accessing the address
printf("Value stored in a variable p is:%d\n",**p); //accessing the value

return 0;
}
```

- a. 10 10
- b. 10, 60ff08
- c. 60ffd, 10
- d. None of the above

Ans: a

14. _____ operator can be used to evaluate size of a variable/pointer in C.

- a. size()
- b. eval()
- c. sizeof()
- d. None of the above

Ans: c

15. A pointer to void means a _____ pointer that can point to any data type.

- a. Specific
- b. Generic
- c. Exact
- d. Null

Ans: b

16. Select the possible arithmetic operations are applicable on the pointer in C language:

- a. Increment, Decrement
- b. Addition, Subtraction
- c. Comparison
- d. All of the above

Ans: d

17. What is the output of the following program:

```
#include<stdio.h>
void main ()
{
    int arr[5] = {1, 2, 3, 4, 5};
    int *p = arr;
    int i;
    printf("printing array elements...\n");
    for(i = 0; i < 5; i++)
    {
        printf("%d ", *(p+i));
    }
}
```

- a. 1, 2, 3, 4, 5
- b. 5, 4, 3, 2, 1
- c. 1, 2, 3, 5, 8
- d. None of the above

Ans: a

18. Identify the illegal pointer arithmetic operations:

- a. Address + Address
- b. Address * Address
- c. Address % Address
- d. All of the above

Ans: d

19. What is the output of the given program:

```
#include<stdio.h>
int addition ();
```

```

int main ()
{
    int result;
    int (*ptr)();
    ptr = &addition;
    result = (*ptr)();
    printf("The sum is %d",result);
}
int addition()
{
    int a=5, b=2;
    return a+b;
}

```

a. 10

b. 3

c. 7

d. 25

Ans: c

20. A _____ pointer in C cannot change the address of the variable to which it is pointing

a. Fixed

b. Null

c. Void

d. Constant

Ans: d

21. An array of pointers to strings is an array of character pointers where each pointer points

to the _____ of the string or the _____ of the string.

a. first character, base address

b. last character, last address

c. middle character, middle address

d. None of the above

Ans: a

22. A function pointer points to _____, not _____.

a. data, code

b. code, data

c. type, const

d. None of the above

Ans: b

23. Identify the correct function pointer declaration:

- a. int * foo(int)
- b. int (*foo)(int);
- c. int (int)(*foo)
- d. None of the above

Ans: b

24. What is the output of the given program:

```
#include <stdio.h>
void Hi_function (int times); /* function */
int main() {
void (*function_ptr)(int); /* function pointer Declaration */

function_ptr = Hi_function; /* pointer assignment */
function_ptr (3); /* function call */
return 0;}

void Hi_function (int times) {
int k;
for (k = 0; k < times; k++) printf("Hi");}
```

- a. Hi HiHiHiHiHi
- b. Hi HiHiHiHi
- c. Hi HiHiHi
- d. Hi HiHi

Ans. d

25. What is the output of the given program:

```
#include <stdio.h>
void* cube (const void* num);
int main() {

int x, cube_int;
x = 4;

cube_int = cube (&x);
printf("%d cubed is %d\n", x, cube_int);
return 0;}

void* cube (const void *num) {
int result;
result = (*(int *)num) * (*(int *)num) * (*(int *)num);
return result;}
```

- a. 2 cubed is 9

- b. 3 cubed is 28
- c. 4 cubed is 64
- d. 5 cubed is 26

Ans. c

26. What is the output of the given program:

```
#include<stdio.h>
main (){

int a[3] = {10,20,30};
int *p[3],i;
for (i=0; i<3; i++)

p[i] = &a[i]; //initializing base address of array
printf ("elements of the array are")

for (i=0; i<3; i++)
printf ("%d \t", *p[i]); //printing array of pointers
getch();
}
```

- a. elements of the array are 10 20 30
- b. elements of the array are 9 19 29
- c. elements of the array are 11 21 31
- d. None of the above

Ans. a

27. What is the output of the following program:

```
#include<stdio.h>
main (){

int a = 10;
int *p;
int **q;
p = &a;
q = &p;

printf("%d",a);
printf("%d", *p);
printf("%d", **q);
}
```

- a. 10 10 11
- b. 10 11 12
- c. 10 10 10
- d. None of the above

Ans. d

28. Choose the correct way of declaring pointer array:

- a. datatype pointername [size];
- b. datatype *pointername [size];
- c. datatype[size]pointername;
- d. None of the above

Ans. b

29. A _____ is a pointer that does not point to any memory location.

- a. Null pointer
- b. Void pointer
- c. Array pointer
- d. String pointer

Ans. a

30. The null pointer basically stores the ____ value.

- a. Void
- b. Int
- c. Char
- d. Null

Ans. d

31. _____ is used to initialize a pointer variable when the pointer does not point to a valid memory address.

- a. Void
- b. Int
- c. Char
- d. Null

Ans. d

32. What is the output of the following program:

```
#include <stdio.h>
int main()
{
    int *ptr;
    printf("Address: %d", ptr);

    printf("Value: %d", *ptr);
    return 0;
}
```

- a. Produces output
- b. Program crashes
- c. Show error

d. None of the above

Ans. b

33. What is the output of the following program:

```
#include <stdio.h>
int main()
{
    int *ptr;
    ptr=(int*)malloc(4*sizeof(int));
    if(ptr==NULL)
    {
        printf("Memory is not allocated");
    }
    else
    {
        printf("Memory is allocated");
    }
    return 0;
}
```

a. Memory is not allocated

b. Memory is allocated

c. Shows compiler error

d. Program crashes

Ans. b

34. What is the output of the following program:

```
#include <stdio.h>
#include <conio.h>
#define NUMBER 0
void main() {
    #if (NUMBER==0)
    printf("Value of Number is: %d",NUMBER);
    #endif
    getch();
}
```

a. Value of Number is: 10

b. Value of Number is: 11

c. Value of Number is: 0

d. Value of Number is 9

Ans. c

35. The #error preprocessor directive indicates error. The compiler gives fatal error if #error directive is found and skips further compilation process.

- a. #ifdef, continue
- b. #error, continue
- c. #ifdef, skips
- d. #error, skips

Ans. d

36. What is the output of the following program:

```
#include <stdio.h>
int main()
{
char *cities[] = {"Iran", "Iraq"};
int i;
for(i = 0; i < 2; i++)
printf("%s\n", cities[i]);
return 0;
}
```

- a. Iraq, Iran
- b. India, Iran
- c. Iran, Iraq
- d. Iraq, India

Ans. c

37. What is the output of the following program:

```
#include <stdio.h>
#include <string.h>
void function(char**);
int main()
{
char *str = "Pointer-to-string";
int i, j = strlen(str);
for(i = 0; i < j; i++)
printf("%c", *str++);
return 0;
}
```

- a. gnirts-ot-retnioP
- b. Pointers to string

- c. Pointers-to-string
- d. All of the above

Ans. c

38. Choose the correct way that express the Pointer and array elements

- a. a[i]
- b. i[a]

- c. *(a + i)
- d. All of the above

Ans. d

39. What is the output of the following program:

```
#include<stdio.h>
int main(){
int i = 3;
int *j;
int **k;
j = &i;
k = &j;
k++;
printf("%d ",**k);
return 0;
```

- ```
}
```
- a. Garbage Value
  - b. Compilation error
  - c. Runtime error
  - d. Linker error

Ans. c

40. What is the output of the following program:

```
#include<stdio.h>
int main(){
int i = 3;
int *j;
j = &i;
j++;
printf("%d ",*j);
return 0;
```

- ```
}
```
- a. Garbage Value
 - b. Compilation error
 - c. Runtime error
 - d. Linker error

Ans.a

UNIT 5 MCQ

1. What is the output of the program?

```

#include <stdio.h>
struct student
{

char *name;
};

struct student s;
struct student fun(void)

{
s.name = "newton";
printf("%s\n", s.name);
s.name = "alan";

return s;
}

void main()
{

struct student m = fun();
printf("%s\n", m.name);

m.name = "turing";
printf("%s\n", s.name);

}

```

- a) newtonalanalan
- b) run tiime error
- c) alanalan newton
- d) no error

Answer: a

2. What will be the output of the program?

```

#include <stdio.h>

```

```

struct student
{
char *name;
};

```

```

void main()

```

```

{

struct student s, m;
s.name = "st";
m = s;
printf("%s%s", s.name, m.name);

}

```

a) Compile time error

b) no error

c) runtime error

d) stst

Answer: D

3. Which return-type cannot be used for a function?

a) main

b)int

c) void

d) none of the mentioned

Answer:D

4. What will be the output of the program?

```
#include <stdio.h>
```

```
struct temp
```

```
{
int a;
} s;
```

```
voidfunc(struct temp s)
```

```
{

s.a = 10;
printf("%d\t", s.a);
```

```
}
main()
```

```
{
func(s);
```

```
printf("%d\t", s.a);
```

```
}
```

a) 0

b) 10 20

c) 10 0

d) (Garbage Value) 10

Answer: C

5. Which of the following is not possible??

a) s1 = s2;

b) s1 != s2;

c) (*s1).number = 50;

d) None of the mentioned

Answer: D

6. Which operation is illegal in structures?

a) Typecasting of structure

b) none

c) Dynamic allocation of memory for structure

d) All

Answer: A

7. "s.t.b = 10" indicates _____

a) Syntax Error

b) Structure

c) float data type

d) variable name

Answer: B

8. What will be the output of the program?

```
#include <stdio.h>
```

```
struct student
```

```
{
```

```

char *name;
};

struct student fun(void)
{

struct student s;
s.name = "alan";

return s;
}

void main()
{

struct student m = fun();
s.name = "turing";

printf("%s", m.name);
}

```

- a) alan
- b) Runtime error
- c) Compile time error
- d) Noth

Answer:C

9. What will be the output of the program?

```

#include <stdio.h>
struct point

{
int x;
int y;

};

int main()

{
struct point p = {1};
struct point p1 = {1};
if(p == p1)
printf("equal\n");

```

```
else
printf("not equal\n");

}
```

- a) Compile time error
- b) Runtime error
- c) depends on the standard
- d) equal

Answer: a

10. What will be the output of the program?

```
#include <stdio.h>
```

```
struct point
{

int x;
int y;

};
struct notpoint

{
int x;
int y;

};
struct point foo();
int main()

{
struct point p = {1};
struct notpoint p1 = {2, 3};

p1 = foo();
printf("%d\n", p1.x);

}
struct point foo()

{
struct point temp = {1, 2};
return temp;
```



```
}
```

- a) Compile time error
- b) 1
- c) Runtime error
- d) Undefined behaviour

Answer: a

11. What will be the output of the program?

```
#include <stdio.h>
```

```
struct point
```

```
{
```

```
int x;
```

```
int y;
```

```
};
```

```
struct notpoint
```

```
{
```

```
int x;
```

```
int y;
```

```
};
```

```
int main()
```

```
{
```

```
struct point p = {1};
```

```
struct notpoint p1 = p;
```

```
printf("%d\n", p1.x);
```

```
}
```

- a) Compile time error
- b) 1
- c) 0
- d) Runtime error

Answer: a

12. What will be the output of the program?

```

#include <stdio.h>

struct point
{

int x;
int y;

};
structnotpoint

{
int x;
int y;

};

void foo(struct point);
int main()

{
structnotpoint p1 = {1, 2};
foo(p1);

}
void foo(struct point p)

{
printf("%d\n", p.x);

}

```

- a) Compile time error
- b) 1
- c) 0
- d) Runtime error

Answer: a

13. What will be the output of the program?

```

#include <stdio.h>

struct point
{

```

```

int x;
int y;

};
void foo(struct point*);
int main()

{
struct point p1 = {1, 2};
foo(&p1);

}
void foo(struct point *p)

{
printf("%d\n", *p.x++);

}

```

- a) Compile time error
- b) Segmentation fault
- c) code crash
- d) Runtime error

Answer: a

14. What will be the output of the program?

```
#include <stdio.h>
```

```
struct point
```

```

{
int x;
int y;

};
void foo(struct point*);
int main()

{
struct point p1 = {1, 2};
foo(&p1);

```

```

}
void foo(struct point *p)

{
printf("%d\n", *p->x++);

}

```

- a) Compile time error
- b) Runtime error
- c) Segmentation fault
- d) 5

Answer: a

15. What will be the output of the program?

```
#include <stdio.h>
```

```

struct student fun(void)
{

```

```

    struct student
    {

```

```

        char *name;
    };

```

```

    struct student s;
    s.name = "alan";

```

```

    return s;
}

```

```

void main()
{

```

```

    struct student m = fun();
    printf("%s", m.name);

```

```

}

```

- a) Compile time error
- b) Runtime error
- c) Nothing
- d) code crash

Answer: a

16. What will be the output of the program?

```
#include <stdio.h>
struct student

{char *name;
};

struct student fun(void)
{

struct student s;
s.name = "alan";

return s;
}

void main()
{

struct student m = fun();
printf("%s", m.name);

}
```

- a) Compile time error
- b) alan
- c) Run time error
- d) code crash

Answer: b

17. What is the correct syntax to access the member of the ith structure?
Assuming: struct temp

```
{
int b;
}s[50];
a) s.b.[i];
b) b[i].s;
c) b[i];
d) s[i].b;
```

Answer: d

18. What is the output of the program.

```
#include <stdio.h>
```

```
struct temp
```

```
{
```

```
int a;
```

```
int b;
```

```
int c;
```

```
};
```

```
main()
```

```
{
```

```
struct temp p[] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
```

```
}
```

- a) No Compile time error, generates an array of structure of size 3
- b) Compile time error, generates an array of structure of size 9
- c) no Compile time error, illegal declaration of a multidimensional array
- d) illegal assignment to members of structure

Answer: a

19. Which of the following uses structure?

- a) Array
- b) Binary Tree
- c) linked list
- d) All

Answer: D

20. What is the correct syntax to declare a function foo() which receives an array of structure in function?

- a) void foo(struct *var);
- b) void foo(struct.var);
- c) void(struct(var));
- d) none

Answer: a

21. What is the output of the program(size of int be 4)

```
#include <stdio.h>
```

```
struct temp
```

```
{
```

```
int a;
```

```
int b;
```

```
int c;
```

```
} p[] = {0};
```

```
main()
```

```
{
```

```
printf("%d", sizeof(p));
```

```
}
```

a) 8

b) 12

c) 30

d) Can't be estimated

Answer: b

22. What is be the output of the program?

```
#include <stdio.h>
```

```
struct student
```

```
{
```

```
char *name;
```

```
};
```

```
struct student s[2];
```

```
void main()
```

```
{
```

```
s[0].name = "alan";
```

```
s[1] = s[0];
```

```
printf("%s%s", s[0].name, s[1].name);
```

```
s[1].name = "turing";
```

```
printf("%s%s", s[0].name, s[1].name);
```

```
}
```

- a) alanalanalanturing
- b) alanturingturing
- c) alanturingturing
- d) run time error

Answer: a

23. What is the output of the program?

```
#include <stdio.h>
```

```
struct student
```

```
{
```

```
char *name;
```

```
};
```

```
struct student s[2], r[2];
```

```
void main()
```

```
{
```

```
s[0].name = "alan";
```

```
s[1] = s[0];
```

```
r = s;
```

```
printf("%s%s", r[0].name, r[1].name);
```

```
}
```

- a) Runtime error
- b) Compile time error
- c) Varies
- d) None

Answer: b

24. What is the output of the program?

```
#include <stdio.h>
```

```
struct student
```

```
{
```

```
char *name;
```



```
};
void main()

{
struct student s[2], r[2];
s[1] = s[0] = "alan";
printf("%s%s", s[0].name, s[1].name);

}
```

- a) alanalan
- b) None
- c) Compile time error
- d) Runtime error

Answer: c

25. What is be the output of the program?

```
#include <stdio.h>
```

```
struct student
{
};

void main()
{

struct student s[2];
printf("%d", sizeof(s));

}
```

- a) 1
- b) 8
- c) 10
- d) 0

Answer: d

26. What is the output of the program?

```
#include <stdio.h>
```

```
struct point
{
```

```

int x;
int y;

};
void foo(struct point*);
int main()

{
struct point p1[] = {1, 2, 3, 4};

foo(p1);
}

void foo(struct point p[])
{

printf("%d\n", p[1].x);
}

```

- a) Compile time error
- b) 3
- c) 12
- d) 4

Answer: b

27. What is the output of the program?

```

#include <stdio.h>
struct point

{
int x;
int y;

};
void foo(struct point*);
int main()

{
struct point p1[] = {1, 2, 3, 4};
foo(p1);

```

```

}
void foo(struct point p[])

{
printf("%d\n", p->x);

}

```

- a) 1
- b) 2
- c) 4
- d) Run time error

Answer: a

28. What is the output of the program?

```
#include <stdio.h>
```

```

struct point
{

int x;
int y;

};
void foo(struct point*);
int main()

{
struct point p1[] = {1, 2, 3, 4};
foo(p1);

}
void foo(struct point p[])

{
printf("%d %d\n", p->x, ++p->x);

}

```

- a) 2 4
- b) 2 2
- c) Compile time error
- d) Undefined

Answer: b

29. What is the output of the program?

```
#include <stdio.h>
```

```
struct point
```

```
{
```

```
int x;
```

```
int y;
```

```
} p[] = {1, 2, 3, 4, 5};
```

```
void foo(struct point*);
```

```
int main()
```

```
{
```

```
foo(p);
```

```
}
```

```
void foo(struct point p[])
```

```
{
```

```
printf("%d %d\n", p->x, p[2].y);
```

```
}
```

a) 1 0

b) Run time error

c) garbage value

d) Undefined

Answer: a

30. What is the output of the program?

```
#include <stdio.h>
```

```
struct point
```

```
{
```

```
int x;
```

```
int y;
```

```
};
```

```

void foo(struct point*);
int main()

{
    struct point p1[] = {1, 2, 3, 4, 5};
    foo(p1);

}
void foo(struct point p[])

{
    printf("%d %d\n", p->x, p[3].y);

}

```

- a) Run time error
- b) 2 0
- c) 1 some garbage value
- d) None

Answer: c

31. What is the output of the program?

```
#include <stdio.h>
```

```

struct point
{

int x;
int y;

};
void foo(struct point*);
int main()

{
    struct point p1[] = {1, 2, 3, 4, 5};
    foo(p1);

}
void foo(struct point p[])

{
    printf("%d %d\n", p->x, (p + 2).y);
}

```

```
}
```

- a) Compile time error
- b) 2 0
- c) garbagevalue
- d) Undefined

Answer: a

32. What is the output of the program?

```
#include <stdio.h>
```

```
struct point
```

```
{
```

```
int x;
```

```
int y;
```

```
};
```

```
void foo(struct point*);
```

```
int main()
```

```
{
```

```
struct point p1[] = {1, 2, 3, 4, 5};
```

```
foo(p1);
```

```
}
```

```
void foo(struct point p[])
```

```
{
```

```
printf("%d %d\n", p->x, (p + 2)->y);
```

```
}
```

- a) Runtime error
- b) 1 0
- c) garbagevalue
- d) undefined

Answer: b

33. What is the output of the program?

```
#include <stdio.h>
```

```

struct student
{

char *c;
};

void main()
{

struct student s[2];
printf("%d", sizeof(s));

```

- }
- a) 2
- b) 3
- c) 1
- d) 8

Answer: d

34. What is the output of the program?

```

#include <stdio.h>
struct p

{
int x;
char y;

};

int main()

{
struct p p1[] = {1, 92, 3, 94, 5, 96};
struct p *ptr1 = p1;
int x = (sizeof(p1) / 3);
if (x == sizeof(int) + sizeof(char))
printf("%d\n", ptr1->x);
else
printf("falsen");

}

```

- a) Compile time error
- b) Runtime error

- c) Undefined
- d) false

Answer: d

35. What is the output of the program?

```
#include <stdio.h>
```

```
struct p
{

int x;
char y;

};

int main()

{
struct p p1[] = {1, 92, 3, 94, 5, 96};
struct p *ptr1 = p1;
int x = (sizeof(p1) / sizeof(ptr1));
if (x == 1)
printf("%d\n", ptr1->x);
else
printf("false\n");

}
```

- a) Compile time error
- b) Runtime error
- c) false
- d) Undefined

Answer: c

36. What is the output of the program?

```
#include <stdio.h>
```

```
struct p
{

int x;
char y;
```



```

};
typedef struct p* q*;
int main()

{
struct p p1[] = {1, 92, 3, 94, 5, 96};

q ptr1 = p1;
printf("%d\n", ptr1->x);

}

```

- a) Compile time error
- b) Runtime error
- c) Undefined
- d) Segmentation fault

Answer: a

37. What is the output of the program?

```
#include <stdio.h>
```

```

struct p
{

int x;
char y;

};
void foo(struct p* );
int main()

{
typedef struct p* q;
struct p p1[] = {1, 92, 3, 94, 5, 96};
foo(p1);

}
void foo(struct p* p1)

{
q ptr1 = p1;

printf("%d\n", ptr1->x);
}

```

- a) Compile time error
- b) Runtime error
- c) Segmentation fault

d) Undefined

Answer: a.

38. Which is an incorrect syntax for pointer to structure?

(Assuming structtemp{int b;}*my_struct;)

- a) *my_struct.b = 10;
- b) (*my_struct).b = 10;
- c) my_struct->b = 10;
- d) Both *my_struct.b = 10; and (*my_struct).b = 10;

Answer: a

39. Which is an incorrect syntax to pass by reference a member of a structure?

(Assume: structtemp{int a;}s;)

- a) func(&s.a);
- b) func(&(s).a);
- c) func(&(s.a));
- d) none of the mentioned

Answer: d

40. What is the output of the program?

```
#include <stdio.h>
```

```
struct temp  
{
```

```
int a;  
} s;
```

```
void change(struct temp);  
main()
```

```
{  
s.a = 10;  
change(s);  
printf("%d\n", s.a);
```

```
}  
void change(struct temp s)  
  
{  
s.a = 1;  
  
}
```

- a) Output will be 20
- b) Output will be 10
- c) Output varies
- d) Compile time error

Answer: b